DevOrSysAdminScripts

Laurent Frédéric Bernard François Lyaudet
https://lyaudet.eu/laurent/
Laurent.Lyaudet@gmail.com

November 11, 2026

Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

Current version: 2026-11-11
Current number of commits: 1022
Current git SHA1: 3c1fbe3e69f5d123919339c7341622e64e77e144
Code lines: 14995 total lines, 13712 not empty lines, 1283 empty lines.
Buildtarget: 1USLetter

Contents

1 Files trees **3**

2 Notes on licenses and dependencies **6**

3 Listing of files **9**
3.1 archive_libre_office_documents.exec.php . . . 9
3.2 build_and_checks_dependencies/build_and_checks.exec.sh.tpl . . . 10
3.3 build_and_checks_dependencies/build_dependencies_notes.exec.php . . . 13
3.4 build_and_checks_dependencies/build_dependencies_notes.libr.php . . . 14
3.5 build_and_checks_dependencies/build_file_from_printable_file__standard.exec.sh . . . 19
3.6 build_and_checks_dependencies/build_md_from_printable_md.exec.sh . . . 21
3.7 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh . . . 23
3.8 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh . . . 25
3.9 build_and_checks_dependencies/check_URLs.libr.sh . . . 27
3.10 build_and_checks_dependencies/common_build_and_checks.exec.sh . . . 29
3.11 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl . . . 49
3.12 build_and_checks_dependencies/common_options.libr.sh . . . 67
3.13 build_and_checks_dependencies/comparisons.libr.sh . . . 77
3.14 build_and_checks_dependencies/create_PDF.exec.sh . . . 81
3.15 build_and_checks_dependencies/generate_from_template.libr.sh . . . 93
3.16 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh . . . 95
3.17 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh . . . 97
3.18 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh . . . 99
3.19 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl . . . 107
3.20 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl . . . 107
3.21 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl . . . 108
3.22 build_and_checks_dependencies/lines_counts.libr.sh . . . 108
3.23 build_and_checks_dependencies/lines_filters.libr.sh . . . 112
3.24 build_and_checks_dependencies/lines_maps.libr.sh . . . 122
3.25 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh . . . 123
3.26 build_and_checks_dependencies/phpmd_baseline.xslt . . . 125
3.27 build_and_checks_dependencies/python_black_complement.libr.sh . . . 126

3.28 build_and_checks_dependencies/python_isort_complement.libr.sh . . . 128
3.29 build_and_checks_dependencies/repository_data.libr.sh . . . 129
3.30 build_and_checks_dependencies/shell_checks_complement.libr.sh . . . 131
3.31 build_and_checks_dependencies/split_score.exec.php . . . 133
3.32 build_and_checks_dependencies/split_score.libr.php . . . 135
3.33 build_and_checks_dependencies/strings_functions.libr.sh . . . 139
3.34 build_and_checks_dependencies/too_long_code_lines.libr.sh . . . 162
3.35 build_and_checks_dependencies/update_common_build_and_checks.exec.sh . . . 164
3.36 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh . . . 167
3.37 build_and_checks_variables/.gitignore . . . 170
3.38 build_and_checks_variables/bandit.ini . . . 170
3.39 build_and_checks_variables/bandit_baseline.json . . . 170
3.40 build_and_checks_variables/black.toml . . . 170
3.41 build_and_checks_variables/dependencies_data.json . . . 170
3.42 build_and_checks_variables/dependencies_notes.txt . . . 173
3.43 build_and_checks_variables/DevOrSysAdminScripts.html.tpl . . . 175
3.44 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl . . . 180
3.45 build_and_checks_variables/files_names_listing.txt . . . 183
3.46 build_and_checks_variables/licenses_notes.txt . . . 185
3.47 build_and_checks_variables/phpcs_ruleset.xml . . . 186
3.48 build_and_checks_variables/phpmd_baseline.xml . . . 193
3.49 build_and_checks_variables/phpstan_baseline.neon.tpl . . . 197
3.50 build_and_checks_variables/phpstan_config.neon.tpl . . . 198
3.51 build_and_checks_variables/pylintrc.toml . . . 198
3.52 build_and_checks_variables/repository_data.txt . . . 199
3.53 build_and_checks_variables/repository_data.txt.tpl . . . 202
3.54 build_and_checks_variables/ruff.toml . . . 203
3.55 build_and_checks_variables/shellcheck.ini . . . 204
3.56 build_and_checks_variables/shellcheck_notes.txt . . . 204
3.57 build_and_checks.exec.sh . . . 205
3.58 COPYING . . . 207
3.59 COPYING.LESSER . . . 219
3.60 files.libr.php . . . 222
3.61 git_helpers/git_checks.libr.sh . . . 235
3.62 git_helpers/post-commit . . . 238
3.63 git_helpers/post_commit_renaming_comment.libr.sh . . . 239
3.64 git_helpers/pre-commit . . . 246
3.65 IMAP_transfert.exec.py . . . 247
3.66 LiensUtiles.txt . . . 252
3.67 miscellaneous/funny.sh . . . 253
3.68 miscellaneous/grep_benchmark.sh . . . 253
3.69 miscellaneous/spampty_data.txt . . . 255
3.70 miscellaneous/string_escaping.html . . . 255
3.71 miscellaneous/string_escaping.xhtml . . . 256
3.72 README.md.tpl . . . 256
3.73 string_escaping.libr.php . . . 258
3.74 unicode.libr.php . . . 265
3.75 unicode_exceptions.libr.php . . . 281
3.76 upgrade_build_and_checks.exec.sh . . . 283
3.77 wget_sha512.libr.sh . . . 286

1 Files trees

```
.
├── archive_libre_office_documents.exec.php
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.libr.php
│   ├── build_file_from_printable_file__standard.exec.sh
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── phpmd_baseline.xslt
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpcs_ruleset.xml
│   ├── phpmd_baseline.xml
│   ├── phpstan_baseline.neon
```

```
│   ├── phpstan_baseline.neon.tpl
│   ├── phpstan_config.neon
│   ├── phpstan_config.neon.tpl
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts-A4.pdf
├── DevOrSysAdminScripts-Book.pdf
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── DevOrSysAdminScripts-USLetter.pdf
├── files.libr.php
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── .gitignore
├── IMAP_transfert.exec.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── funny.sh
│   ├── grep_benchmark.sh
│   ├── spampty_data.txt
│   ├── string_escaping.html
│   ├── string_escaping.xhtml
│   └──
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode_exceptions.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 90 files

[ 12K 2026-11-11 15:51]  ./
├── [1.8K 2026-08-14 17:47]  archive_libre_office_documents.exec.php*
├── [4.0K 2026-11-11 15:51]  build_and_checks_dependencies/
│   ├── [4.4K 2026-05-12 19:33]  build_and_checks.exec.sh.tpl*
│   ├── [1.5K 2026-08-14 17:46]  build_dependencies_notes.exec.php*
│   ├── [ 10K 2026-08-14 15:46]  build_dependencies_notes.libr.php
│   ├── [2.6K 2026-08-13 12:28]
│   │ build_file_from_printable_file__standard.exec.sh*
│   ├── [4.2K 2026-08-12 19:31]  build_md_from_printable_md.exec.sh*
```

```
│   ├── [3.1K 2026-05-12 19:34]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.3K 2026-05-12 19:37]  check_shell_scripts_indentation.libr.sh
│   ├── [4.0K 2026-08-12 19:14]  check_URLs.libr.sh
│   ├── [ 43K 2026-11-11 15:41]  common_build_and_checks.exec.sh*
│   ├── [ 38K 2026-09-18 22:14]  common_build_and_checks.exec.sh.tpl
│   ├── [ 19K 2026-05-12 19:37]  common_options.libr.sh
│   ├── [6.2K 2026-05-12 19:37]  comparisons.libr.sh
│   ├── [ 23K 2026-11-11 15:40]  create_PDF.exec.sh*
│   ├── [3.7K 2026-05-12 19:32]  generate_from_template.libr.sh
│   ├── [2.1K 2026-08-12 19:16]  get_common_text_glob_patterns.libr.sh
│   ├── [3.3K 2026-05-12 19:37]  grammar_and_spelling_check.libr.sh
│   ├── [4.0K 2026-08-13 13:00]  licenses_templates/
│   │   ├── [ 16K 2026-08-13 13:00]  build_licenses_templates.exec.sh*
│   │   ├── [ 720 2026-05-12 19:16]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 699 2026-05-12 19:17]  license_file_header_GPLv3+.tpl
│   │   └── [ 720 2026-05-12 19:18]  license_file_header_LGPLv3+.tpl
│   ├── [6.0K 2026-05-13 17:48]  lines_counts.libr.sh
│   ├── [ 15K 2026-05-13 15:25]  lines_filters.libr.sh
│   ├── [3.0K 2026-05-12 19:37]  lines_maps.libr.sh
│   ├── [3.3K 2026-05-12 19:37]  overwrite_if_not_equal.libr.sh
│   ├── [1.6K 2026-08-14 14:40]  phpmd_baseline.xslt
│   ├── [2.4K 2026-05-12 19:37]  python_black_complement.libr.sh
│   ├── [3.1K 2026-05-13 15:29]  python_isort_complement.libr.sh
│   ├── [2.9K 2026-05-13 15:08]  repository_data.libr.sh
│   ├── [3.9K 2026-05-12 19:37]  shell_checks_complement.libr.sh
│   ├── [2.8K 2026-08-14 17:45]  split_score.exec.php*
│   ├── [5.9K 2026-08-14 14:56]  split_score.libr.php
│   ├── [ 47K 2026-05-13 16:32]  strings_functions.libr.sh
│   ├── [4.8K 2026-08-12 19:25]  too_long_code_lines.libr.sh
│   ├── [4.5K 2026-08-14 14:33]  update_common_build_and_checks.exec.sh*
│   └── [6.0K 2026-05-12 19:37]
    update_or_check_files_names_listing.exec.sh*
├── [4.6K 2026-11-11 15:41]  build_and_checks.exec.sh*
├── [4.0K 2026-11-11 15:51]  build_and_checks_variables/
│   ├── [  15 2026-01-05 20:00]  bandit_baseline.json
│   ├── [  25 2026-01-05 18:49]  bandit.ini
│   ├── [  30 2026-02-28 20:30]  black.toml
│   ├── [5.1K 2026-04-17 16:50]  dependencies_data.json
│   ├── [4.3K 2026-04-17 16:57]  dependencies_notes.txt
│   ├── [5.7K 2026-05-12 19:31]  DevOrSysAdminScripts.html.tpl
│   ├── [ 38K 2026-11-11 15:41]  DevOrSysAdminScripts.tex
│   ├── [4.8K 2026-11-11 15:50]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.3K 2026-08-14 17:42]  files_names_listing.txt
│   ├── [  23 2026-03-24 19:33]  .gitignore
│   ├── [1.4K 2026-04-17 16:49]  licenses_notes.txt
│   ├── [ 16K 2026-08-14 17:54]  phpcs_ruleset.xml
│   ├── [5.2K 2026-08-14 14:53]  phpmd_baseline.xml
│   ├── [1.6K 2026-08-14 18:06]  phpstan_baseline.neon
│   ├── [1.7K 2026-08-14 18:06]  phpstan_baseline.neon.tpl
│   ├── [ 236 2026-08-13 12:29]  phpstan_config.neon
│   ├── [ 244 2026-08-12 19:23]  phpstan_config.neon.tpl
│   ├── [ 678 2026-02-24 22:38]  pylintrc.toml
│   ├── [5.8K 2026-11-11 14:52]  repository_data.txt
│   ├── [ 963 2026-11-11 15:43]  repository_data.txt.tpl
│   ├── [1.5K 2026-08-13 16:02]  ruff.toml
│   ├── [ 166 2026-03-24 19:33]  shellcheck.ini
```

```
│   └── [3.6K 2026-05-13 17:18]  shellcheck_notes.txt
├── [ 34K 2024-05-06 00:49]  COPYING
├── [7.5K 2025-12-27 21:48]  COPYING.LESSER
├── [468K 2026-11-11 15:42]  DevOrSysAdminScripts-A4.pdf
├── [468K 2026-11-11 15:42]  DevOrSysAdminScripts-Book.pdf
├── [575K 2026-11-11 15:41]  DevOrSysAdminScripts.html
├── [466K 2026-09-18 22:18]  DevOrSysAdminScripts.pdf
├── [468K 2026-11-11 15:42]  DevOrSysAdminScripts-USLetter.pdf
├── [ 21K 2026-08-14 18:01]  files.libr.php
├── [4.0K 2026-05-13 17:28]  git_helpers/
│   ├── [3.8K 2026-05-13 17:22]  git_checks.libr.sh
│   ├── [2.1K 2026-05-12 19:31]  post-commit*
│   ├── [ 15K 2026-05-13 17:28]  post_commit_renaming_comment.libr.sh
│   └── [1.8K 2026-05-12 19:31]  pre-commit*
├── [   8 2026-03-26 10:37]  .gitignore
├── [8.4K 2026-05-12 19:31]  IMAP_transfert.exec.py
├── [1.1K 2026-03-31 19:59]  LiensUtiles.txt
├── [ 24K 2026-08-13 18:22]  miscellaneous/
│   ├── [   0 2026-03-04 09:13]
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── [1.3K 2026-05-12 19:37]  funny.sh
│   ├── [2.4K 2026-08-13 18:22]  grep_benchmark.sh
│   ├── [  94 2026-03-03 19:56]  spampty_data.txt
│   ├── [ 461 2026-05-13 18:15]  string_escaping.html
│   ├── [ 653 2026-05-12 14:38]  string_escaping.xhtml
│   └── [   0 2026-03-04 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.1K 2026-05-12 12:17]  README.html
├── [4.3K 2026-03-22 21:29]  README.md
├── [4.3K 2026-03-22 21:29]  README.md.tpl
├── [7.5K 2026-08-14 17:03]  string_escaping.libr.php
├── [2.5K 2026-08-13 18:54]  unicode_exceptions.libr.php
├── [ 31K 2026-08-14 17:04]  unicode.libr.php
├── [6.4K 2026-05-13 18:46]  upgrade_build_and_checks.exec.sh*
└── [2.5K 2026-05-12 19:37]  wget_sha512.libr.sh

6 directories, 90 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;

- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
`https://www.gnu.org/software/sed/`,
`https://savannah.gnu.org/projects/sed/`.

ShellCheck can be found at the following URLs:
`https://www.shellcheck.net/`,
`https://github.com/koalaman/shellcheck`.

GNU Wget can be found at the following URLs:
`https://www.gnu.org/software/wget/`,
`https://savannah.gnu.org/projects/wget/`.

more of util-linux can be found at the following URLs:
`https://www.kernel.org/pub/linux/utils/util-linux/`,
`https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/`,
`https://github.com/util-linux/util-linux` official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.
pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.
tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.
git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.
pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.
notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.
black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.
isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.
mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.
phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/,
https://docs.phpdoc.org/,
https://github.com/phpdocumentor/phpdocumentor.
pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).
ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.
xmllint (libxml2) can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home,
https://gitlab.gnome.org/GNOME/libxml2.
bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.
phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.
php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.
tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.
pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.
python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.
If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 archive_libre_office_documents.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
```

```
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'files.libr.php';

if(PHP_SAPI !== 'cli'){
  fwrite(
    STDERR,
    "archive_libre_office_documents.exec.php must be run in a shell.\n",
  );
  die(1);
}

if(count($argv) < 3 || count($argv) > 4){
  fwrite(
    STDERR,
    "split_score.exec.php must be given 2 or 3 arguments.\n",
  );
  die(1);
}

DOSASfiles\archive_directory_into_another(
  '/home/'.$argv[1].'/.config/libreoffice/4/user/backup/',
  '/home/'.$argv[2].'/Documents/LibreOfficeArchives/',
  true,
  true,
  // Following argument is if you want same relative path and same content
  // at the same time.
  // This is in case you want to track when the file goes back to some
  // old state.
  isset($argv[3]) ? (bool) (int) $argv[3] : false,
  true,
  true,
  '\\.~\\d+~',
);
?>
```

3.2 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
```

```
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir --parents "${LFBFL_subdir}/licenses_templates"
  local LFBFL_variables_directory
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  readonly LFBFL_variables_directory
  mkdir --parents -- "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
```

```
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  local LFBFL_script_download_URL
  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
  LFBFL_script_download_URL+="${LFBFL_common_file_name}"
  readonly LFBFL_script_download_URL
  declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.3 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'string_escaping.libr.php';
require_once 'build_dependencies_notes.libr.php';

if(PHP_SAPI !== 'cli'){
  fwrite(
    STDERR,
    "build_dependencies_notes.exec.php must be run in a shell.\n",
  );
  die(1);
}

if(count($argv) !== 2){
  fwrite(
    STDERR,
    "split_score.exec.php must be given exactly 1 argument.\n",
  );
  die(1);
}
```

```

\DOSAS_dependencies_notes\build_dependencies_notes($argv[1]);
?>
```

3.4 build_and_checks_dependencies/build_dependencies_notes.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
2026-03-26T11:59+01:00: This file was renamed from
"build_dependencies_notes.exec.php"
to
"build_dependencies_notes.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

// require_once 'string_escaping.libr.php';
namespace DOSAS_dependencies_notes;

use Exception;
use string_escaping;

/**
This function looks at the dependencies data found in a work directory,
and generate three files: a txt file, a TeX file (.sub),
and an HTML file (.sub) with the data in a nicer format.

```

```
@param string $s_work_directory The directory where the work must be done.

@throws \Exception When the content of one of the files needed as input
                   cannot be loaded into a string.
                   Or that some part of the structure from the JSON
                   isn't valid.

@return void
*/
function build_dependencies_notes(string $s_work_directory) : void {
  $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
  $s_subdir2 = $s_subdir1.'/temp';
  $s_data_file_path = $s_subdir1.'/dependencies_data.json';
  $s_data = file_get_contents($s_data_file_path);
  if($s_data === false){
    throw new Exception(
      'Unexpected failure when fetching content of dependencies_data.json'
    );
  }
  $arr_data = json_decode($s_data, true);
  if(!is_array($arr_data)){
    throw new Exception(
      "dependencies_data.json doesn't contain an array."
    );
  }
  $s_data_file_path2 = $s_subdir1.'/repository_data.txt';
  $s_data2 = file_get_contents($s_data_file_path2);
  if($s_data2 === false){
    throw new Exception(
      'Unexpected failure when fetching content of repository_data.txt'
    );
  }
  $i_max_line_length = 70;
  foreach(explode("\n", $s_data2) as $s_line){
    if(str_starts_with($s_line, 'max_line_length=')){
      $i_max_line_length = (
        (int) substr($s_line, strlen('max_line_length='))
      );
      break;
    }
  }

  $s_result_content1 = '';
  $s_result_content2 = '';
  $s_result_content3 = '';

  foreach($arr_data as $arr_dependency){
    if(!is_array($arr_dependency)){
      throw new Exception("Dependency isn't an array.");
    }
    if(!isset($arr_dependency['URLs'])){
      throw new Exception("Dependency doesn't have an URLs array.");
    }
    if(!is_array($arr_dependency['URLs'])){
      throw new Exception("URLs isn't an array.");
    }
```

```
108     if(!isset($arr_dependency['dependency_name'])){
109       throw new Exception("Dependency doesn't have a name.");
110     }
111     if(!is_string($arr_dependency['dependency_name'])){
112       throw new Exception("Dependency name isn't a string.");
113     }
114
115     $s_URL_word = 'URL';
116     if(count($arr_dependency['URLs']) > 1){
117       $s_URL_word .= 's';
118     }
119
120     $s_result_content3 .= "<p>\n";
121
122     $s_dependency1 = $arr_dependency['dependency_name'];
123     $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
124     $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);
125
126     $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
127     $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
128     $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
129     $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";
130
131     for(
132       $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
133       $i <= $i_max;
134       ++$i
135     ){
136       $mixed_URL = $arr_dependency['URLs'][$i];
137       $s_comment = '';
138       $s_URL = '';
139       if(is_array($mixed_URL)){
140         if(!isset($mixed_URL['URL'])){
141           throw new Exception("Dependency URL doesn't have an URL.");
142         }
143         if(!isset($mixed_URL['comment'])){
144           throw new Exception("Dependency URL doesn't have a comment.");
145         }
146         $s_URL = $mixed_URL['URL'];
147         $s_comment = $mixed_URL['comment'];
148       }
149       else{
150         $s_URL = $mixed_URL;
151       }
152       if(!is_string($s_URL)){
153         throw new Exception("URL isn't a string.");
154       }
155       if(!is_string($s_comment)){
156         throw new Exception("Comment isn't a string.");
157       }
158
159       $i_URL_length = strlen($s_URL);
160       $i_comment_length = strlen($s_comment);
161
162       $s_result_content1 .= $s_URL;
163       $i_previous_line_length = $i_URL_length;
```

```
164         if($i_comment_length > 0){
165           if(
166             $i_previous_line_length + $i_comment_length + 1
167             > $i_max_line_length
168           ){
169             $s_result_content1 .= "\n";
170           }
171           else{
172             $s_result_content1 .= ' ';
173           }
174           $s_result_content1 .= $s_comment;
175         }
176         $s_result_content1 .= "\n";
177
178         if($i_URL_length > 60){
179           $s_result_content2 .= (
180             "{\\small\n"
181             .'  \url{'.$s_URL."}\n"
182             .'}'
183           );
184           $i_previous_line_length = 1;
185         }
186         else{
187           $s_result_content2 .= '\url{'.$s_URL.'}';
188           $i_previous_line_length = $i_URL_length + 6;
189         }
190         if($i === $i_max){
191           ++$i_previous_line_length;
192         }
193         else{
194           $i_previous_line_length += 9;
195         }
196         $s_comment2 = string_escaping\TeX\escape_text($s_comment);
197         $i_comment_length2 = strlen($s_comment2);
198         if($i_comment_length2 > 0){
199           if(
200             $i_previous_line_length + $i_comment_length2 + 1
201             > $i_max_line_length
202           ){
203             $s_result_content2 .= "\n";
204           }
205           else{
206             $s_result_content2 .= ' ';
207           }
208           $s_result_content2 .= $s_comment2;
209         }
210         if($i === $i_max){
211           $s_result_content2 .= ".\n";
212         }
213         else{
214           $s_result_content2 .= ",\\newline\n";
215         }
216
217         $s_result_content3 .= (
218           "<a\n"
219           .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
```

```
        ."><span style=\"font-family:monospace\"\n"
        .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
        .'></a>'
      );
      if($s_comment !== ''){
        $s_result_content3 .= (
          ' '.string_escaping\HTML\escape_text($s_comment)
        );
      }
      if($i === $i_max){
        $s_result_content3 .= ".\n";
      }
      else{
        $s_result_content3 .= ",<br>\n";
      }
    }//end for urls

    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }//end foreach($arr_data as $arr_dependency)

  // Add a "footer".
  $s_closing_sentence = (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content1 .= $s_closing_sentence;
  $s_result_content2 .= $s_closing_sentence;
  $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";

  // We overwrite results if needed.
  $s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
  $s_existing_result_content1 = '';
  if(file_exists($s_result_path1)){
    $s_existing_result_content1 = file_get_contents($s_result_path1);
  }
  if($s_result_content1 !== $s_existing_result_content1){
    file_put_contents($s_result_path1, $s_result_content1);
  }
  /*
  Example of generated content of dependencies_notes.txt:
  ------------------------------------------------------------------------
  more of util-linux can be found at the following URLs:
  https://www.kernel.org/pub/linux/utils/util-linux/
  https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
  https://github.com/util-linux/util-linux official?
  ------------------------------------------------------------------------
  */
  $s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
  $s_existing_result_content2 = '';
  if(file_exists($s_result_path2)){
    $s_existing_result_content2 = file_get_contents($s_result_path2);
  }
  if($s_result_content2 !== $s_existing_result_content2){
    file_put_contents($s_result_path2, $s_result_content2);
```

```
  }
  /*
  Example of generated content of dependencies_notes.tex.sub:
  ------------------------------------------------------------------------
  more of util-linux can be found at the following URLs:\newline
  \url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
  {\small
    \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
  },\newline
  \url{https://github.com/util-linux/util-linux} official?.
  ------------------------------------------------------------------------
  */
  $s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
  $s_existing_result_content3 = '';
  if(file_exists($s_result_path3)){
    $s_existing_result_content3 = file_get_contents($s_result_path3);
  }
  if($s_result_content3 !== $s_existing_result_content3){
    file_put_contents($s_result_path3, $s_result_content3);
  }
  /*
  Example of generated content of dependencies_notes.html.sub:
  <p>
  more of util-linux can be found at the following URLs:<br>
  <a
  href="https://www.kernel.org/pub/linux/utils/util-linux/"
  ><span style="font-family:monospace"
  >https://www.kernel.org/pub/linux/utils/util-linux/</span
  ></a>,<br>
  <a
  href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
  ><span style="font-family:monospace"
  >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
  ></a>,<br>
  <a
  href="https://github.com/util-linux/util-linux"
  ><span style="font-family:monospace"
  >https://github.com/util-linux/util-linux</span
  ></a> official?.
  </p>
  */
}//end build_dependencies_notes()
?>
```

3.5 build_and_checks_dependencies/build_file_from_printable_file__standard.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
```

```
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_file_from_printable_file__standard(){
  # $1=base_name
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --remove-leading-whitespaces-after-escaped-line-return
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -i LFBFL_remove_leading_whitespaces_after_escaped_line_return=0
  get_some_flag LFBFL_remove_leading_whitespaces_after_escaped_line_return\
    --remove-leading-whitespaces-after-escaped-line-return 1 "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name
  get_some_option LFBFL_base_name --base-name '' '' 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    if [[ "${LFBFL_base_name}" == "" ]]; then
      return 2
    fi
    printf "Searching file to build from printable file: %s.\n"\
      "${LFBFL_base_name}"
  fi

  local LFBFL_s_expression='s/\\\n//Mg'
  if [[
    LFBFL_remove_leading_whitespaces_after_escaped_line_return -eq 1
  ]]; then
    LFBFL_s_expression='s/\\\n *//Mg'
  fi
  if [[ -f "${LFBFL_base_name}.tpl" ]]; then
    sed --regexp-extended --null-data --expression="${LFBFL_s_expression}"\
      -- "${LFBFL_base_name}.tpl"\
```

```
      > "${LFBFL_base_name}.temp"
    overwrite_if_not_equal "${LFBFL_base_name}" "${LFBFL_base_name}.temp"
  else
    printf "No file %s.tpl\n" "${LFBFL_base_name}"
    return 3
  fi

  if [[ ! -f "${LFBFL_base_name}" ]]; then
    printf "No file %s\n" "${LFBFL_base_name}"
    return 4
  fi
}

build_file_from_printable_file__standard "$@"
```

3.6 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
```

```
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name
  get_some_option LFBFL_base_name --base-name README README 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 && "${LFBFL_base_name}" != "README" ]]; then
    printf "Searching md file: %s.\n" "${LFBFL_base_name}"
  fi

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence, it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed --regexp-extended --null-data\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      -- "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 2
  fi

  pandoc --from=markdown --to=html --standalone\
    --shift-heading-level-by=-1 --output="${LFBFL_base_name}.html.temp"\
    -- "${LFBFL_base_name}.md"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
```

```
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed --in-place --regexp-extended\
    --expression='s~(  <meta .*) />~\1>~g'\
    --expression='s~<html .*>~<html lang="en">~'\
    --expression='/    code\{white-space: pre-wrap;\}/d'\
    -- "${LFBFL_base_name}.html.temp"
  sed --in-place --regexp-extended --null-data\
    --expression='s~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
    --expression="${LFBFL_prewrap}"\
    -- "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.7 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
```

```
36    # $1=file_path
37    #
38    # declare -r LFBFL_file_path=$(basename -- "$1")
39    # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
40    #   return 0
41    # fi
42    # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
43    # ------------------------------------------------------------------
44    # is of course slower than
45    # ------------------------------------------------------------------
46    # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
47    # ------------------------------------------------------------------
48    # is of course slower than
49    # ------------------------------------------------------------------
50    declare -r LFBFL_head=$(head --lines=1 -- "$1")
51    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
52      || printf "%s:File has wrong shell script beginning.\n" "$1"
53  }
54
55  check_shell_scripts_beginnings(){
56    # Options:
57    #   --verbose
58    #   --work-directory=""
59    declare -i LFBFL_i_verbose=0
60    get_verbose_option "$@"
61    local LFBFL_work_directory=""
62    get_work_directory_option "$@"
63    declare -a LFBFL_return_traps_stack
64    init_return_trap
65    pushd_to_work_directory --trap-popd
66    can_continue_after_enhanced_pushd || return 1
67
68    # shopt -s globstar
69    # for LFBFL_file_path in **/*.sh; do
70    local LFBFL_file_path
71    local LFBFL_head
72    declare -r LFBFL_s_files_paths=$(
73      find . -type f -name "*.sh" -printf '%P\n'\
74      | relevant_find
75    )
76    if [[ -z "${LFBFL_s_files_paths}" ]]; then
77      [[ LFBFL_i_verbose -eq 1 ]]\
78        && printf "check_shell_scripts_beginnings: No file found.\n"
79      return
80    fi
81    declare -a LFBFL_arr_files_paths
82    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
83    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
84      [[ LFBFL_i_verbose -eq 1 ]]\
85        && printf "%s:Checking shell script beginning.\n"\
86            "${LFBFL_file_path}"
87      check_one_shell_script_beginning "${LFBFL_file_path}"
88    done
89  }
```

3.8 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  # $1=file_path
  declare -r LFBFL_NULL_PREVIOUS_LINE="--"
  declare -i LFBFL_i_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_i_line_length
  declare -i LFBFL_i_line_end_length
  declare -i LFBFL_i_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_i_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '

  # This grep would have been enough without some sed commands.
  declare -r LFBFL_s_lines=$(
    grep --before-context=1 --extended-regexp --line-number\
      --regexp='^(  )* ([^ ]|$)' --with-filename -- "$1"
  )
  if [[ -z "${LFBFL_s_lines}" ]]; then
    return
  fi
```

```
56
57    declare -a LFBFL_arr_lines
58    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
59    readonly LFBFL_arr_lines
60
61    for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
62      if [[ "${LFBFL_some_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]]; then
63        # When switching between code fragments, we reinit.
64        # It cannot happen for other grep output lines with line-number.
65        LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
66        continue
67      fi
68      # Note how it's funny that this loop always work because we check
69      # shell scripts beginnings in another script and thus we always
70      # have a line before.
71      if [[ "${LFBFL_previous_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]];
72      then
73        LFBFL_previous_line="${LFBFL_some_line}"
74        continue
75      fi
76      # Remove the name of the file and then the line number.
77      LFBFL_line_end=${LFBFL_some_line#*:}
78      LFBFL_line_end=${LFBFL_line_end#*:}
79      # But for the previous line '-' or ':' is used instead of ':'
80      # and filenames with '-' are much more frequent than with ':'...
81      LFBFL_i_line_length=${#LFBFL_some_line}
82      LFBFL_i_line_end_length=${#LFBFL_line_end}
83      LFBFL_i_line_start_length=$((
84        LFBFL_i_line_length - LFBFL_i_line_end_length
85      ))
86      LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
87      # Checking if increasing number of digits in the line numbers
88      # (99 -> 100).
89      LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
90      # Nice puzzle below :) Look for comments above :)
91      case ${LFBFL_substring1} in
92        -) ((LFBFL_i_offset += 1));;
93        :) ((LFBFL_i_offset += 1));;
94        [0-9]) ((LFBFL_i_offset += 2));;
95        *) printf ":-(\n";;
96      esac
97      LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
98      if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
99        # Line starts with spacing and -e
100       # Keep max spaces
101       LFBFL_spacing=${LFBFL_line_end%%-*}
102       LFBFL_i_offset=${#LFBFL_spacing}
103       LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
104       LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
105       if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
106         printf "%s\n" "${LFBFL_some_line}"
107         LFBFL_i_file_with_error=1
108       fi
109     else
110       printf "%s\n" "${LFBFL_some_line}"
111       LFBFL_i_file_with_error=1
```

```
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.9 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
```

```
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\0'\
```

```
      | xargs -0 grep --with-filename -- 'http:'\
      | grep --invert-match "| xargs -0 grep --with-filename -- 'htt"\
      | grep --invert-match --perl-regexp\
        "(['\"])http(:[^'\"]*)\\1.*(['\"])https\\2\\3"\
      | grep --invert-match\
        --regexp='http://www.w3.org/1999/xhtml'\
        --regexp="http://www.w3.org/1999/XSL/Transform"\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd'\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd'\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd'
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep --quiet "${LFBFL_substitution}" -- "${LFBFL_file_path}";
        then
          sed --in-place\
          --expression="s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            -- "${LFBFL_file_path}"
        fi
      done
    done
  done
}
```

3.10 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
```

```
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  declare -a LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    "--work-directory=${LFBFL_work_directory}"
  )

  source ./wget_sha512.libr.sh

  local LFBFL_subdir="build_and_checks_dependencies"
  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
```

```
68  declare -r LFBFL_subdir3="build_and_checks_variables"
69  local LFBFL_file_name
70  local LFBFL_script_download_URL
71  local LFBFL_file_path
72  local LFBFL_correct_sha512
73
74  wrapped_wget_sha512(){
75    wget_sha512 "${LFBFL_file_path}"\
76      "${LFBFL_script_download_URL}"\
77      "${LFBFL_correct_sha512}"\
78      "${LFBFL_verbose}"\
79      "${LFBFL_check_download}"
80  }
81
82  LFBFL_file_name="build_dependencies_notes.exec.php"
83  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
84  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
85  LFBFL_correct_sha512='3d881a757c69fbcc02cbc5a568c508cfa6249c1330ea6'
86  LFBFL_correct_sha512+='7eeca6d284950b3ca2ced7b4acce4142cdf90c5aaa88'
87  LFBFL_correct_sha512+='5bb3ee77f3a32ae0d4b412173cec0d944278ab0'
88  wrapped_wget_sha512
89  chmod +x "./${LFBFL_file_path}"
90
91  LFBFL_file_name="build_dependencies_notes.libr.php"
92  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
93  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
94  LFBFL_correct_sha512='6203cb1c31b656004ec8467d4f84288f342fcc7f5c236'
95  LFBFL_correct_sha512+='270e819b986fe967aa6abe101188a7fc1a55e2f4c9b5'
96  LFBFL_correct_sha512+='9a6dde6c9aa125b385d0ce85e02b1de69ce1084'
97  wrapped_wget_sha512
98
99  LFBFL_file_name="build_file_from_printable_file__standard.exec.sh"
100 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
101 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
102 LFBFL_correct_sha512='9c5b11280019b4bbc679bddb8eb5c66d9ef715d04c8c1'
103 LFBFL_correct_sha512+='9e2cbd55e08ad38169b82aa2de5fe59f46053552a358'
104 LFBFL_correct_sha512+='d92f3554cfc8ce7d930055b400ec1e6ee5e3069'
105 wrapped_wget_sha512
106 chmod +x "./${LFBFL_file_path}"
107
108 LFBFL_file_name="build_md_from_printable_md.exec.sh"
109 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
110 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
111 LFBFL_correct_sha512='a39101cb7c0a7a743386b0c8745071cc15f8fd93be3ac'
112 LFBFL_correct_sha512+='94bd5d69860836e4ae94b08a3219c2542f9d6501f0b4'
113 LFBFL_correct_sha512+='d4b1bc40700afcb1b555bbbe25de6485f95c361'
114 wrapped_wget_sha512
115 chmod +x "./${LFBFL_file_path}"
116
117 LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
118 LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
119 LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
120 LFBFL_correct_sha512='0b3e861820e09148a141142c8bb5549e7fa29d906dda9'
121 LFBFL_correct_sha512+='5e0fe44132610910687e6f767f1050646abe1ea262b8'
122 LFBFL_correct_sha512+='37b3a5d5e79497d0cdc42e43d3c74967f266abb'
123 wrapped_wget_sha512
```

```
124
125     LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
126     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
127     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
128     LFBFL_correct_sha512='3596fddc4b20ce353facac3e355e38fadb5593caee3c8'
129     LFBFL_correct_sha512+='05464aedd513d34860a79b296efa648dae18717ae4c0'
130     LFBFL_correct_sha512+='24a6df11eb0549c4797643427c49c4f80947840'
131     wrapped_wget_sha512
132
133     LFBFL_file_name="check_URLs.libr.sh"
134     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
135     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
136     LFBFL_correct_sha512='d3ddcd56520d1786cd5afb8fd0f7ee278ed7307a6f5d2'
137     LFBFL_correct_sha512+='1f948d0f2f6f2ed5210072d923f2036643c7181012fc'
138     LFBFL_correct_sha512+='15d7921f5617105012a14b8adc9acc76d2b50cb'
139     wrapped_wget_sha512
140
141     LFBFL_file_name="common_options.libr.sh"
142     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
143     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
144     LFBFL_correct_sha512='3cfaf6cea99f67ade931cf21e9f49658c421e182dc4ac'
145     LFBFL_correct_sha512+='211be526f9d99b2e57a09844991cf0d2e6d5e49c19e2'
146     LFBFL_correct_sha512+='a63bf9e8f1e3fe999542e9a4e45e5196b375fa9'
147     wrapped_wget_sha512
148
149     LFBFL_file_name="comparisons.libr.sh"
150     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
151     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
152     LFBFL_correct_sha512='f146beca0d7b4046c140f3b2292cec4c31515cf8c7dbe'
153     LFBFL_correct_sha512+='7e9b07fc6dae32513aff6692c6286fa31055caff1a20'
154     LFBFL_correct_sha512+='9a6a7384568fea0b50b3cbb2608a3ed9e6e5125'
155     wrapped_wget_sha512
156
157     LFBFL_file_name="create_PDF.exec.sh"
158     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
159     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
160     LFBFL_correct_sha512='588f29f1de07d9980a14259a710cea5f5093102e0ff8c'
161     LFBFL_correct_sha512+='366e71c45bc383f5040954a858ec0c84ba712c51ee01'
162     LFBFL_correct_sha512+='cb2c5ced6354103dfc9ea7e735c70b205cafbe6'
163     wrapped_wget_sha512
164     chmod +x "./${LFBFL_file_path}"
165
166     LFBFL_file_name="generate_from_template.libr.sh"
167     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
168     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
169     LFBFL_correct_sha512='5f322f27bd20d7042c729f4cb5d8f14f587b5b415c77e'
170     LFBFL_correct_sha512+='4a224902d8d4cdeb1483639ae85d81873c9f20c175bb'
171     LFBFL_correct_sha512+='4de76a6eaa3dbcbf8a300368e65bbd5303afd3d'
172     wrapped_wget_sha512
173
174     LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
175     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
176     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
177     LFBFL_correct_sha512='0945c9896ce9d26637a8ba5c08a1fdad102b725de26ce'
178     LFBFL_correct_sha512+='088e519a3696925465007ad0b460ae7433174d8af848'
179     LFBFL_correct_sha512+='b77dfdff0da78534453e90303d3bd458bf378e9'
```

```
    wrapped_wget_sha512

    LFBFL_file_name="grammar_and_spelling_check.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='88cc4f031c05d576a74ea7b14e79077f06c53b3b8a294'
    LFBFL_correct_sha512+='e9b8ddcfe884db5e2cb5cebde09a8d9aea7980fb595c'
    LFBFL_correct_sha512+='85c562462fb67e10503a471ee46da89c8f73b47'
    wrapped_wget_sha512

    # /licenses_templates/ ------------------------------------------
    local LFBFL_dependencies_URL2
    LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
    readonly LFBFL_dependencies_URL2
    LFBFL_file_name="build_licenses_templates.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    LFBFL_correct_sha512='9b06e0631a64496c0e412ef412fb676bff0ace10cdbf7'
    LFBFL_correct_sha512+='a6b92b7cb087049f8d2ae38261b316f74eedb5a9d2e3'
    LFBFL_correct_sha512+='4825396438729e70ce82e38df766df7fb416b5b'
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="license_file_header_AGPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    LFBFL_correct_sha512='5dc0dc20914c57feee2e26c2d329f020f221357b11b0d'
    LFBFL_correct_sha512+='0ff6add69941ad953f8356b68be32cd60df8c56e91a2'
    LFBFL_correct_sha512+='716c9cbf70ab8698e74e6aacc9c5439a93cc2bc'
    wrapped_wget_sha512

    LFBFL_file_name="license_file_header_GPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    LFBFL_correct_sha512='2d4b31697e78ba36ee0e3bd2b5393a57e15b5f5e62b2e'
    LFBFL_correct_sha512+='802b812dcb087588f63cb92a851b21180be24bf5dee4'
    LFBFL_correct_sha512+='eff7927244cde817e964512eaa1fb51d9064c01'
    wrapped_wget_sha512

    LFBFL_file_name="license_file_header_LGPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    LFBFL_correct_sha512='43976b6dc9a54fcea0440cfaf87530dade73efe22b5ae'
    LFBFL_correct_sha512+='a968453c9c5ac1fdc4d3798fdd445fb38a53f997da82'
    LFBFL_correct_sha512+='7c30be6b7ad42d5e56ae90aac5095b9526b203d'
    wrapped_wget_sha512
    # ---------------------------------------------------------------

    LFBFL_file_name="lines_counts.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='d131e24ad596988a6953b9362b13e5d17330f3bd8013b'
    LFBFL_correct_sha512+='03fc2acf74a6b931660a619149cce59d1fa3b075c11c'
    LFBFL_correct_sha512+='0a5e9c85e847b437d7bc3b58f034c00df3c5702'
    wrapped_wget_sha512

```

```
    LFBFL_file_name="lines_filters.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='d7b9d7456d18965b7910516cf80f96b5a89b76c59f315'
    LFBFL_correct_sha512+='f7558487f5f37afc8d0af13d4f11928f7c2bec8a03a1'
    LFBFL_correct_sha512+='7b49a9a5f71945d2a6fff955ce98518f5305234'
    wrapped_wget_sha512

    LFBFL_file_name="lines_maps.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='ccb9115be582a5b39790725dc96a628beb21b0c9951e7'
    LFBFL_correct_sha512+='03a7a7bc8ab2736e3940ed3df6a2fcb04e75350c10d7'
    LFBFL_correct_sha512+='3e92bcfada424b4049cc6141d41bf7901c0525b'
    wrapped_wget_sha512

    LFBFL_file_name="overwrite_if_not_equal.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='378f03b8ba77430e05db4ef5d5926803b466a87e1a2ec'
    LFBFL_correct_sha512+='eef0b51036f25460a93a018bd86d3f90146b4dfec9ab'
    LFBFL_correct_sha512+='15c5d1d57a526296bbdc4a0c03650088f3e9653'
    wrapped_wget_sha512

    LFBFL_file_name="phpmd_baseline.xslt"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='88f39ea6952acebdad60c267e0f49a0489b486f7ead53'
    LFBFL_correct_sha512+='4c6ccf32e9f2bfa86f92d772c2fa59fecef00d6c9e6c'
    LFBFL_correct_sha512+='4d31cf074dda4943070db069c64581a18bfe724'
    wrapped_wget_sha512

    LFBFL_file_name="python_black_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='4f08461329f1b78b58e54f675fb433aec363de2c96a91'
    LFBFL_correct_sha512+='34ba5e96f3c77b93c9d7d6e28771a0b886501c286457'
    LFBFL_correct_sha512+='a4a410389556f180f2dc8f66d7956bf5dd5acf2'
    wrapped_wget_sha512

    LFBFL_file_name="python_isort_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='f91f3236ff9720412b31ec9fec2a5a5b3ef2424931815'
    LFBFL_correct_sha512+='c0e6d1bd9821e86ef80a13a27a3f6d831327dc4b9698'
    LFBFL_correct_sha512+='5935a75793c87bdf735c4ae2a94327289d1c2ed'
    wrapped_wget_sha512

    LFBFL_file_name="repository_data.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='8148fae16b5853b5ed6ca141b73b32f6d8695d8d34a54'
    LFBFL_correct_sha512+='73ad8e8cc0dab61d92e87fd925b6f25c0632f70b2fee'
    LFBFL_correct_sha512+='78ee53b0e3af6aa25b3cc58d8769292c3e4defd'
    wrapped_wget_sha512

```

```
292     LFBFL_file_name="shell_checks_complement.libr.sh"
293     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
294     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
295     LFBFL_correct_sha512='143575d86b5fc0fd78e575b6e46054b0c8facaf2c6df6'
296     LFBFL_correct_sha512+='07dc596b6db0494c2b88211d5e0e011dee9abe70a2fc'
297     LFBFL_correct_sha512+='b8f2bc04eb5cd50b248b0afe423103ee1cd568b'
298     wrapped_wget_sha512
299
300     LFBFL_file_name="split_score.exec.php"
301     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
302     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
303     LFBFL_correct_sha512='2845905b970c2dae5cf4a072df3e00670607f576d88db'
304     LFBFL_correct_sha512+='22176e57f47081d487eaf0d5d006950550932bae2686'
305     LFBFL_correct_sha512+='9ad1933bb2adcaf2719469fad83ea55e53e154e'
306     wrapped_wget_sha512
307     chmod +x "./${LFBFL_file_path}"
308
309     LFBFL_file_name="split_score.libr.php"
310     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
311     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
312     LFBFL_correct_sha512='a66cf904ba51bf9180c887d67f81b8d64eb0ac9b95a3c'
313     LFBFL_correct_sha512+='696e51e050743956029d2d247e2252e2763ceffbd201'
314     LFBFL_correct_sha512+='459ac09ed5ed53e2af2436ed3610fe56a88d384'
315     wrapped_wget_sha512
316
317     LFBFL_file_name="strings_functions.libr.sh"
318     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
319     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
320     LFBFL_correct_sha512='526bdaeca9ce0732ad429c16410909f9f8014eef9e031'
321     LFBFL_correct_sha512+='837c428786b19bc31fea3eff6c8d93eea260b9076b67'
322     LFBFL_correct_sha512+='71d21220528e70aa08a1a2fb7579926db662ff2'
323     wrapped_wget_sha512
324
325     LFBFL_file_name="too_long_code_lines.libr.sh"
326     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
327     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
328     LFBFL_correct_sha512='0571332f70cd5700f35ba5eb804bbe20a58d01ca33635'
329     LFBFL_correct_sha512+='c36f40a8aa24ddc7926ee66d6372191b91e6b2ffcac3'
330     LFBFL_correct_sha512+='9d4e89ef8a1e0c32cbae6c2e1e69437a65d3efc'
331     wrapped_wget_sha512
332
333     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
334     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
335     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
336     LFBFL_correct_sha512='188fa5dbcab63be8e0f5b3c8997e099c7259f7c804433'
337     LFBFL_correct_sha512+='581c2c05a7d9c7bd3d82bc424e2021afa90cb9e9d6a5'
338     LFBFL_correct_sha512+='f84f549ed692c18b6351983c31b875846d7e1af'
339     wrapped_wget_sha512
340     chmod +x "./${LFBFL_file_path}"
341
342     # shellcheck source=check_shell_scripts_beginnings.libr.sh
343     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
344     # shellcheck source=check_shell_scripts_indentation.libr.sh
345     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
346     # shellcheck source=check_URLs.libr.sh
347     source "./${LFBFL_subdir}/check_URLs.libr.sh"
```

```
348     # shellcheck source=common_options.libr.sh
349     source "./${LFBFL_subdir}/common_options.libr.sh"
350     # shellcheck source=comparisons.libr.sh
351     source "./${LFBFL_subdir}/comparisons.libr.sh"
352     # shellcheck source=generate_from_template.libr.sh
353     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
354     # shellcheck source=get_common_text_glob_patterns.libr.sh
355     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
356     # shellcheck source=grammar_and_spelling_check.libr.sh
357     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
358     # shellcheck source=lines_counts.libr.sh
359     source "./${LFBFL_subdir}/lines_counts.libr.sh"
360     # shellcheck source=lines_filters.libr.sh
361     source "./${LFBFL_subdir}/lines_filters.libr.sh"
362     # shellcheck source=lines_maps.libr.sh
363     source "./${LFBFL_subdir}/lines_maps.libr.sh"
364     # shellcheck source=overwrite_if_not_equal.libr.sh
365     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
366     # shellcheck source=python_black_complement.libr.sh
367     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
368     # shellcheck source=python_isort_complement.libr.sh
369     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
370     # shellcheck source=shell_checks_complement.libr.sh
371     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
372     # shellcheck source=strings_functions.libr.sh
373     source "./${LFBFL_subdir}/strings_functions.libr.sh"
374     # shellcheck source=too_long_code_lines.libr.sh
375     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
376
377     enhanced_set_bash_option extglob
378     # shellcheck source=repository_data.libr.sh
379     source "./${LFBFL_subdir}/repository_data.libr.sh"
380     # shellcheck disable=SC2154,SC2309
381     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
382       && enhanced_unset_bash_option extglob
383
384     declare -a LFBFL_return_traps_stack
385     init_return_trap
386
387     enhanced_set_shell_option pipefail --trap-unset
388
389     declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
390     keep_options LFBFL_some_common_options2 --verbose
391
392     printf "Building license headers\n"
393     "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
394       "${LFBFL_some_common_options[@]}"
395
396     printf "Building README.md\n"
397     "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
398       --work-directory="${LFBFL_work_directory}"\
399       --base-name=README\
400       "${LFBFL_verbose}"
401
402     printf "Building other MarkDown files\n"
403     local LFBFL_some_directory
```

```
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
      LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building other standard files\n"
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.neon.tpl"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
      LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.tpl}
      "./${LFBFL_subdir}/build_file_from_printable_file__standard.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"\
        --remove-leading-whitespaces-after-escaped-line-return
    done
  fi

  printf "Building dependencies_notes\n"
  # https://github.com/php/php-src/issues/21538
  local LFBFL_annoying_warning
  LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
  LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
  LFBFL_annoying_warning=" by settings in"
  readonly LFBFL_annoying_warning
  if [[ -e "${LFBFL_subdir3}/dependencies_data.json" ]]; then
    php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
      -- "${LFBFL_work_directory}"\
      2>&1\
      | grep --invert-match "${LFBFL_annoying_warning}"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file dependencies_data.json.\n"
```

```
460   fi
461
462   declare -i LFBFL_i_directory_changed
463   pushd_to_work_directory
464   LFBFL_i_directory_changed=$?
465   can_continue_after_enhanced_pushd || return 1
466
467   local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
468
469   declare -i LFBFL_i_max_line_length
470   grep_variable "${LFBFL_data_file_path}" max_line_length\
471     --result-variable-prefix=LFBFL_i_
472
473   LFBFL_some_common_options+=(
474     "--max-line-length=${LFBFL_i_max_line_length}"
475   )
476   LFBFL_some_common_options2+=(
477     "--max-line-length=${LFBFL_i_max_line_length}"
478   )
479
480   declare -i LFBFL_i_upgrade_venvs=0
481   local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
482   readonly LFBFL_upgrade_venvs_ts_file
483   declare -i LFBFL_i_upgrade_venvs_ts
484   declare -i LFBFL_i_current_ts
485   local LFBFL_upgrade_venvs_answer
486
487   local LFBFL_upgrade_venvs_time_interval_in_seconds=""
488   get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
489     "${LFBFL_some_common_options2[@]}"
490
491   if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
492     LFBFL_i_upgrade_venvs_ts=$(
493       stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
494     )
495     LFBFL_i_current_ts=$(date +%s)
496     ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
497     ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
498     if [[ LFBFL_i_current_ts -gt 0 ]]; then
499       LFBFL_i_upgrade_venvs=1
500     fi
501   else
502     LFBFL_i_upgrade_venvs=1
503   fi
504   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
505     local LFBFL_upgrade_venvs=""
506     grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
507       --result-variable-prefix=LFBFL_
508     if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
509       read -r -n 1 -t 10\
510         -p "Upgrade venvs and composer global? [Y/n]"\
511         LFBFL_upgrade_venvs_answer
512       if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
513         LFBFL_i_upgrade_venvs=0
514       fi
515     fi
```

```
  fi

  printf "Analyzing too long lines\n"
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  printf "Running shellcheck\n"
  if [[ -e "${LFBFL_subdir3}/shellcheck.ini" ]]; then
    LFBFL_s_files_paths=$(
      find . -type f -name "*.sh"\
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_paths}" ]]; then
      mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
      for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
        shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
          -- "${LFBFL_file_path}"
      done
    fi
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file shellcheck.ini.\n"
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }
```

```
572
573   source_without_return_trap(){
574     declare -i LFBFL_i_no_return_traps=1
575     # shellcheck disable=SC1090,SC1091
576     source -- "$1"
577   }
578
579   printf "Running isort\n"
580   local LFBFL_isort_venv=""
581   grep_variable "${LFBFL_data_file_path}" isort_venv\
582     --result-variable-prefix=LFBFL_\
583     --replace-line-returns-by=""
584   if [[ -n "${LFBFL_isort_venv}" ]]; then
585     maybe_deactivate
586     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
587   fi
588   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
589     pip install --upgrade isort
590   fi
591   isort --profile=black .
592   if [[ -n "${LFBFL_isort_venv}" ]]; then
593     deactivate
594   fi
595   printf "Running python_isort_complement\n"
596   python_isort_complement "${LFBFL_some_common_options2[@]}"
597
598   printf "Running black\n"
599   # First, we update the configuration file with max_line_length.
600   if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
601     local LFBFL_update_max_length="s/^line-length = [0-9]*$"
602     LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
603     sed --regexp-extended --expression="${LFBFL_update_max_length}"\
604       "${LFBFL_subdir3}/black.toml"\
605       > "${LFBFL_subdir3}/black.toml.temp"
606     overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
607       "${LFBFL_subdir3}/black.toml.temp"
608   elif [[ LFBFL_i_verbose -eq 1 ]]; then
609     printf "No file black.toml.\n"
610   fi
611
612   local LFBFL_black_venv=""
613   grep_variable "${LFBFL_data_file_path}" black_venv\
614     --result-variable-prefix=LFBFL_\
615     --replace-line-returns-by=""
616   if [[ -n "${LFBFL_black_venv}" ]]; then
617     maybe_deactivate
618     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
619   fi
620   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
621     pip install --upgrade black
622   fi
623   if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
624     black --config="${LFBFL_subdir3}/black.toml" .
625   fi
626   if [[ -n "${LFBFL_black_venv}" ]]; then
627     deactivate
```

```
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_path}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
        mypy -- "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_path}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  if [[ -e "${LFBFL_subdir3}/bandit.ini" ]]; then
    bandit --ini="${LFBFL_subdir3}/bandit.ini"\
      --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
```

```
      --recursive .
    # Saving new baseline in temp if necessary.
    bandit --ini="${LFBFL_subdir3}/bandit.ini"\
      --format=json\
      --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
      --recursive .
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file bandit.ini.\n"
  fi
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_path}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  if [[ -e "${LFBFL_subdir3}/pylintrc.toml" ]]; then
    pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file pylintrc.toml.\n"
  fi
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_path}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  if [[ -e "${LFBFL_subdir3}/ruff.toml" ]]; then
    ruff check --config="${LFBFL_subdir3}/ruff.toml"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file ruff.toml.\n"
  fi
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

```

```
740   printf -- "---PHP---\n"
741   printf "Running PHP native linter\n"
742   LFBFL_s_files_paths=$(
743     find . -type f -name "*.php"\
744     | relevant_find
745   )
746   if [[ -n "${LFBFL_s_files_paths}" ]]; then
747     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
748     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
749       php --syntax-check -f="${LFBFL_file_path}"\
750         2>&1\
751         | grep --invert-match "${LFBFL_annoying_warning}"
752     done
753   fi
754
755   printf "Running PHP_CodeSniffer\n"
756   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
757     composer global require squizlabs/php_codesniffer
758   fi
759   if [[ -e "${LFBFL_subdir3}/phpcs_ruleset.xml" ]]; then
760     phpcs --report=code\
761       --standard="${LFBFL_subdir3}/phpcs_ruleset.xml"
762   elif [[ LFBFL_i_verbose -eq 1 ]]; then
763     printf "No file phpcs_ruleset.xml.\n"
764   fi
765
766   printf "Running PHPMD\n"
767   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
768     composer global require phpmd/phpmd
769   fi
770
771   declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
772   local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
773   LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
774   readonly LFBFL_temp_phpmd_baseline
775
776   local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
777   LFBFL_phpmd_rulesets+="design,naming,unusedcode"
778   readonly LFBFL_phpmd_rulesets
779
780   phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
781     . text "${LFBFL_phpmd_rulesets}"
782   # Saving new baseline in temp if necessary.
783   rm "${LFBFL_temp_phpmd_baseline}"
784   phpmd --color --generate-baseline\
785     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
786     . text "${LFBFL_phpmd_rulesets}"
787   # Remove all superfluous whitespaces.
788   # sed --in-place --regexp-extended --null-data\
789   #   --expression='s/\n +//Mg'\
790   #   "${LFBFL_temp_phpmd_baseline}"
791   xsltproc --output "${LFBFL_temp_phpmd_baseline}"\
792     build_and_checks_dependencies/phpmd_baseline.xslt\
793     "${LFBFL_temp_phpmd_baseline}"
794   # sed --in-place --expression='s/" file="/"\n    file="/g'\
795   #   "${LFBFL_temp_phpmd_baseline}"
```

```
  # sed --in-place --regexp-extended\
  #   --expression='s~(file=".*)/>~\1\n  />~g'\
  #   "${LFBFL_temp_phpmd_baseline}"
  sed --in-place\
    --expression='s/\([^"]*\("[^"]*"\)*\) rule="/\1\n    rule="/'\
    --expression='s/\([^"]*\("[^"]*"\)*\) file="/\1\n    file="/'\
    --expression='s/\([^"]*\("[^"]*"\)*\) method="/\1\n    method="/'\
    --expression='s~/>~\n  />~'\
    "${LFBFL_temp_phpmd_baseline}"
  diff --suppress-common-lines\
    "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
    | grep --invert-match\
      --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
  # See miscellaneous/grep_benchmark.

  printf "Running PHPSTAN\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpstan/phpstan
    composer global require phpstan/phpstan-strict-rules
  fi

  local LFBFL_temp_phpstan_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpstan_baseline+="phpstan_baseline.neon"
  readonly LFBFL_temp_phpstan_baseline
  if [[ -e "${LFBFL_subdir3}/phpstan_config.neon" ]]; then
    phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"
    # Saving new baseline in temp if necessary.
    rm "${LFBFL_temp_phpstan_baseline}"
    phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"\
      --generate-baseline="${LFBFL_temp_phpstan_baseline}"\
      --allow-empty-baseline
    sed --in-place --expression='s/\t/ /g' "${LFBFL_temp_phpstan_baseline}"
    cat "${LFBFL_temp_phpstan_baseline}"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file phpstan_config.neon.\n"
  fi

  printf "Running phpDocumentor\n"
  # shellcheck disable=SC2312
  docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  local LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd -- "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"
```

```
852
853     [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
854
855     printf "Checking listed files\n"
856     "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
857       "${LFBFL_some_common_options[@]}"
858
859     printf "Creating the PDF file of the listing of the source code\n"
860     "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
861
862     pushd_to_work_directory --trap-popd
863     can_continue_after_enhanced_pushd || return 1
864
865     if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
866       "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
867     fi
868
869     printf "Running xmllint\n"
870     # In some cases, we need to handle a cache of common DTDs by hand.
871     # XHTML DTDs validation can yield errors:
872     # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
873     # If you use Ubuntu/Debian, install w3c-sgml-lib instead.
874     LFBFL_file_name="xhtml1-strict.dtd"
875     LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
876     LFBFL_script_download_URL="${LFBFL_a}"
877     LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
878     LFBFL_correct_sha512='fe11db06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
879     LFBFL_correct_sha512+='d8352bf19947168c9bcf1ae212eca035824293abb1f9'
880     LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
881     wrapped_wget_sha512
882
883     LFBFL_file_name="xhtml1-transitional.dtd"
884     LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
885     LFBFL_script_download_URL="${LFBFL_a}"
886     LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
887     LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
888     LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
889     LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
890     wrapped_wget_sha512
891
892     LFBFL_file_name="xhtml1-frameset.dtd"
893     LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
894     LFBFL_script_download_URL="${LFBFL_a}"
895     LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
896     LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
897     LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
898     LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
899     wrapped_wget_sha512
900
901     LFBFL_file_name="xhtml11.dtd"
902     LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
903     LFBFL_script_download_URL="${LFBFL_a}"
904     LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
905     LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
906     LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
907     LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
```

```
908   wrapped_wget_sha512
909
910   LFBFL_file_name="xhtml-lat1.ent"
911   LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
912   LFBFL_script_download_URL="${LFBFL_a}"
913   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
914   LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
915   LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
916   LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
917   wrapped_wget_sha512
918
919   LFBFL_file_name="xhtml-special.ent"
920   LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
921   LFBFL_script_download_URL="${LFBFL_a}"
922   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
923   LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
924   LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
925   LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
926   wrapped_wget_sha512
927
928   LFBFL_file_name="xhtml-symbol.ent"
929   LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
930   LFBFL_script_download_URL="${LFBFL_a}"
931   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
932   LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
933   LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
934   LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
935   wrapped_wget_sha512
936
937   declare -r LFBFL_s_files_for_xmllint=$(
938     find . -type f -iregex ".*\.\(xml\|xhtml\)"\
939     | relevant_find\
940     | sort
941   )
942   if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
943     LFBFL_xmllint_use_downloaded_DTDs=""
944     grep_variable "${LFBFL_data_file_path}" xmllint_use_downloaded_DTDs\
945       --result-variable-prefix=LFBFL_
946
947     declare -a LFBFL_arr_files_for_xmllint
948     mapfile -t LFBFL_arr_files_for_xmllint\
949       <<< "${LFBFL_s_files_for_xmllint}"
950     readonly LFBFL_arr_files_for_xmllint
951     local LFBFL_xmllint_files_without_DTD=""
952     grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
953       xmllint_files_without_DTD\
954       --result-variable-prefix=LFBFL_
955     declare -a LFBFL_arr_xmllint_files_without_DTD
956     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
957       mapfile -t LFBFL_arr_xmllint_files_without_DTD\
958         <<< "${LFBFL_xmllint_files_without_DTD}"
959     fi
960     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
961       [[ LFBFL_i_verbose -eq 1 ]]\
962         && printf "xmllint on %s\n" "${LFBFL_file_path}"
963       # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
```

```
964       # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
965       # xmllint --pedantic --noout -- "${LFBFL_file_path}"
966       xmllint --pedantic --noout "${LFBFL_file_path}"
967     done
968
969     local LFBFL_file_path2
970     for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
971       for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
972       do
973         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
974           LFBFL_file_path=""
975         fi
976       done
977       if [[ "${LFBFL_file_path}" == "" ]]; then
978         continue
979       fi
980       [[ LFBFL_i_verbose -eq 1 ]]\
981         && printf "xmllint on %s\n" "${LFBFL_file_path}"
982       if [[
983         "${LFBFL_file_path}" == *.xhtml
984         && "${LFBFL_xmllint_use_downloaded_DTDs}" == "true"
985       ]]; then
986         if grep --fixed-strings\
987           "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
988           "${LFBFL_file_path}"
989         then
990           xmllint --pedantic --noout\
991             --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
992             "${LFBFL_file_path}"
993         elif grep --fixed-strings\
994           "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
995           "${LFBFL_file_path}"
996         then
997           xmllint --pedantic --noout\
998             --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
999             "${LFBFL_file_path}"
1000        elif grep --fixed-strings\
1001          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
1002          "${LFBFL_file_path}"
1003        then
1004          xmllint --pedantic --noout\
1005            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
1006            "${LFBFL_file_path}"
1007        elif grep --fixed-strings\
1008          "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
1009          "${LFBFL_file_path}"
1010        then
1011          xmllint --pedantic --noout\
1012            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
1013            "${LFBFL_file_path}"
1014        else
1015          printf "Unknown XHTML DTD.\n"
1016        fi
1017      else
1018        xmllint --pedantic --noout --valid "${LFBFL_file_path}"
1019      fi
```

```
  done
fi
# ----------------------------------------------------------------------

# ----------------------------------------------------------------------
# It is at the end that all the HTML files are generated and that
# we can check there are no mistakes.
printf "Running Nu W3C HTML CSS and SVG validator\n"
local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
grep_variable "${LFBFL_data_file_path}"\
  upgrade_vnu_jar_time_interval_in_seconds\
  --result-variable-prefix=LFBFL_

if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
then
  LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
fi

declare -i LFBFL_i_vnu_jar_ts
local LFBFL_vnu_jar_path=""
grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
  LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
  LFBFL_i_current_ts=$(date +%s)
  ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
  ((
    LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
  ))
  if [[ LFBFL_i_current_ts -gt 0 ]]; then
    printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
    printf "Go get the latest version here:\n"
    printf "https://github.com/validator/validator/releases\n"
  fi
  # vnu has false positives for xhtml.
  local LFBFL_use_vnu_for_xhtml=""
  grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
    --result-variable-prefix=LFBFL_
  local LFBFL_find_regexp
  if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
    LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
  else
    LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
  fi
  declare -r LFBFL_s_files_for_Nu=$(
    find . -type f -iregex "${LFBFL_find_regexp}"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
    declare -a LFBFL_arr_files_for_Nu
    mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
    readonly LFBFL_arr_files_for_Nu
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
  fi
```

```
  fi
  # ----------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.11 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
```

```
  LFBFL_verbose="--verbose"
  LFBFL_i_verbose=1
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download=y* ]]; then
  LFBFL_check_download="--check-download=y"
fi
readonly LFBFL_check_download

declare -a LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  "--work-directory=${LFBFL_work_directory}"
)

source ./wget_sha512.libr.sh

local LFBFL_subdir="build_and_checks_dependencies"
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_subdir3="build_and_checks_variables"
local LFBFL_file_name
local LFBFL_script_download_URL
local LFBFL_file_path
local LFBFL_correct_sha512

wrapped_wget_sha512(){
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

LFBFL_file_name="build_dependencies_notes.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.exec.php@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.libr.php@
wrapped_wget_sha512

LFBFL_file_name="build_file_from_printable_file__standard.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_file_from_printable_file__standard.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

```

```
    LFBFL_file_name="build_md_from_printable_md.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_build_md_from_printable_md.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_check_shell_scripts_beginnings.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_check_shell_scripts_indentation.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="check_URLs.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_check_URLs.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="common_options.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_common_options.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="comparisons.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_comparisons.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="create_PDF.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_create_PDF.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="generate_from_template.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_generate_from_template.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_get_common_text_glob_patterns.libr.sh@
    wrapped_wget_sha512

```

```
158     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
159     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
160     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
161     @sha512_grammar_and_spelling_check.libr.sh@
162     wrapped_wget_sha512
163
164     # /licenses_templates/ ------------------------------------------
165     local LFBFL_dependencies_URL2
166     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
167     readonly LFBFL_dependencies_URL2
168     LFBFL_file_name="build_licenses_templates.exec.sh"
169     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
170     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
171     @sha512_build_licenses_templates.exec.sh@
172     wrapped_wget_sha512
173     chmod +x "./${LFBFL_file_path}"
174
175     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
176     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
177     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
178     @sha512_license_file_header_AGPLv3+.tpl@
179     wrapped_wget_sha512
180
181     LFBFL_file_name="license_file_header_GPLv3+.tpl"
182     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
183     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
184     @sha512_license_file_header_GPLv3+.tpl@
185     wrapped_wget_sha512
186
187     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
188     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
189     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
190     @sha512_license_file_header_LGPLv3+.tpl@
191     wrapped_wget_sha512
192     # ---------------------------------------------------------------
193
194     LFBFL_file_name="lines_counts.libr.sh"
195     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
196     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
197     @sha512_lines_counts.libr.sh@
198     wrapped_wget_sha512
199
200     LFBFL_file_name="lines_filters.libr.sh"
201     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
202     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
203     @sha512_lines_filters.libr.sh@
204     wrapped_wget_sha512
205
206     LFBFL_file_name="lines_maps.libr.sh"
207     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
208     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
209     @sha512_lines_maps.libr.sh@
210     wrapped_wget_sha512
211
212     LFBFL_file_name="overwrite_if_not_equal.libr.sh"
213     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_overwrite_if_not_equal.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="phpmd_baseline.xslt"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_phpmd_baseline.xslt@
    wrapped_wget_sha512

    LFBFL_file_name="python_black_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_python_black_complement.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="python_isort_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_python_isort_complement.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="repository_data.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_repository_data.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="shell_checks_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_shell_checks_complement.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="split_score.exec.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_split_score.exec.php@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="split_score.libr.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_split_score.libr.php@
    wrapped_wget_sha512

    LFBFL_file_name="strings_functions.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_strings_functions.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="too_long_code_lines.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
    @sha512_too_long_code_lines.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_update_or_check_files_names_listing.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    # shellcheck source=check_shell_scripts_beginnings.libr.sh
    source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
    # shellcheck source=check_shell_scripts_indentation.libr.sh
    source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
    # shellcheck source=check_URLs.libr.sh
    source "./${LFBFL_subdir}/check_URLs.libr.sh"
    # shellcheck source=common_options.libr.sh
    source "./${LFBFL_subdir}/common_options.libr.sh"
    # shellcheck source=comparisons.libr.sh
    source "./${LFBFL_subdir}/comparisons.libr.sh"
    # shellcheck source=generate_from_template.libr.sh
    source "./${LFBFL_subdir}/generate_from_template.libr.sh"
    # shellcheck source=get_common_text_glob_patterns.libr.sh
    source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
    # shellcheck source=grammar_and_spelling_check.libr.sh
    source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
    # shellcheck source=lines_counts.libr.sh
    source "./${LFBFL_subdir}/lines_counts.libr.sh"
    # shellcheck source=lines_filters.libr.sh
    source "./${LFBFL_subdir}/lines_filters.libr.sh"
    # shellcheck source=lines_maps.libr.sh
    source "./${LFBFL_subdir}/lines_maps.libr.sh"
    # shellcheck source=overwrite_if_not_equal.libr.sh
    source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
    # shellcheck source=python_black_complement.libr.sh
    source "./${LFBFL_subdir}/python_black_complement.libr.sh"
    # shellcheck source=python_isort_complement.libr.sh
    source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
    # shellcheck source=shell_checks_complement.libr.sh
    source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
    # shellcheck source=strings_functions.libr.sh
    source "./${LFBFL_subdir}/strings_functions.libr.sh"
    # shellcheck source=too_long_code_lines.libr.sh
    source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

    enhanced_set_bash_option extglob
    # shellcheck source=repository_data.libr.sh
    source "./${LFBFL_subdir}/repository_data.libr.sh"
    # shellcheck disable=SC2154,SC2309
    [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
      && enhanced_unset_bash_option extglob

    declare -a LFBFL_return_traps_stack
    init_return_trap

    enhanced_set_shell_option pipefail --trap-unset
```

```
326
327     declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
328     keep_options LFBFL_some_common_options2 --verbose
329
330     printf "Building license headers\n"
331     "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
332       "${LFBFL_some_common_options[@]}"
333
334     printf "Building README.md\n"
335     "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
336       --work-directory="${LFBFL_work_directory}"\
337       --base-name=README\
338       "${LFBFL_verbose}"
339
340     printf "Building other MarkDown files\n"
341     local LFBFL_some_directory
342     declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
343     local LFBFL_s_files_paths
344     LFBFL_s_files_paths=$(
345       find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
346       | relevant_find
347     )
348     declare -a LFBFL_arr_files_paths
349     if [[ -n "${LFBFL_s_files_paths}" ]]; then
350       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
351       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
352         [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
353         printf "Found template %s.\n" "${LFBFL_file_path}"
354         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
355         LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
356         LFBFL_file_name=${LFBFL_file_name%.md.tpl}
357         "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
358           --work-directory="${LFBFL_some_directory}"\
359           --base-name="${LFBFL_file_name}"\
360           "${LFBFL_verbose}"
361       done
362     fi
363
364     printf "Building other standard files\n"
365     LFBFL_s_files_paths=$(
366       find "${LFBFL_work_directory}" -type f -name "*.neon.tpl"\
367       | relevant_find
368     )
369     if [[ -n "${LFBFL_s_files_paths}" ]]; then
370       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
371       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
372         printf "Found template %s.\n" "${LFBFL_file_path}"
373         LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
374         LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
375         LFBFL_file_name=${LFBFL_file_name%.tpl}
376         "./${LFBFL_subdir}/build_file_from_printable_file__standard.exec.sh"\
377           --work-directory="${LFBFL_some_directory}"\
378           --base-name="${LFBFL_file_name}"\
379           "${LFBFL_verbose}"\
380           --remove-leading-whitespaces-after-escaped-line-return
381       done
```

```
382   fi
383
384   printf "Building dependencies_notes\n"
385   # https://github.com/php/php-src/issues/21538
386   local LFBFL_annoying_warning
387   LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
388   LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
389   LFBFL_annoying_warning=" by settings in"
390   readonly LFBFL_annoying_warning
391   if [[ -e "${LFBFL_subdir3}/dependencies_data.json" ]]; then
392     php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
393       -- "${LFBFL_work_directory}"\
394       2>&1\
395       | grep --invert-match "${LFBFL_annoying_warning}"
396   elif [[ LFBFL_i_verbose -eq 1 ]]; then
397     printf "No file dependencies_data.json.\n"
398   fi
399
400   declare -i LFBFL_i_directory_changed
401   pushd_to_work_directory
402   LFBFL_i_directory_changed=$?
403   can_continue_after_enhanced_pushd || return 1
404
405   local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
406
407   declare -i LFBFL_i_max_line_length
408   grep_variable "${LFBFL_data_file_path}" max_line_length\
409     --result-variable-prefix=LFBFL_i_
410
411   LFBFL_some_common_options+=(
412     "--max-line-length=${LFBFL_i_max_line_length}"
413   )
414   LFBFL_some_common_options2+=(
415     "--max-line-length=${LFBFL_i_max_line_length}"
416   )
417
418   declare -i LFBFL_i_upgrade_venvs=0
419   local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
420   readonly LFBFL_upgrade_venvs_ts_file
421   declare -i LFBFL_i_upgrade_venvs_ts
422   declare -i LFBFL_i_current_ts
423   local LFBFL_upgrade_venvs_answer
424
425   local LFBFL_upgrade_venvs_time_interval_in_seconds=""
426   get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
427     "${LFBFL_some_common_options2[@]}"
428
429   if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
430     LFBFL_i_upgrade_venvs_ts=$(
431       stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
432     )
433     LFBFL_i_current_ts=$(date +%s)
434     ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
435     ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
436     if [[ LFBFL_i_current_ts -gt 0 ]]; then
437       LFBFL_i_upgrade_venvs=1
```

```
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Analyzing too long lines\n"
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└┼│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  printf "Running shellcheck\n"
  if [[ -e "${LFBFL_subdir3}/shellcheck.ini" ]]; then
    LFBFL_s_files_paths=$(
      find . -type f -name "*.sh"\
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_paths}" ]]; then
      mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
      for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
        shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
          -- "${LFBFL_file_path}"
      done
    fi
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file shellcheck.ini.\n"
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

```

```
  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }

  source_without_return_trap(){
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source -- "$1"
  }

  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_path}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
    local LFBFL_update_max_length="s/^line-length = [0-9]*$"
    LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
    sed --regexp-extended --expression="${LFBFL_update_max_length}"\
      "${LFBFL_subdir3}/black.toml"\
      > "${LFBFL_subdir3}/black.toml.temp"
    overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
      "${LFBFL_subdir3}/black.toml.temp"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file black.toml.\n"
  fi

```

```
  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_path}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
    black --config="${LFBFL_subdir3}/black.toml" .
  fi
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_path}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
        mypy -- "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi
```

```
606
607   printf "Running bandit\n"
608   LFBFL_bandit_venv=""
609   grep_variable "${LFBFL_data_file_path}" bandit_venv\
610     --result-variable-prefix=LFBFL_\
611     --replace-line-returns-by=""
612   if [[ -n "${LFBFL_bandit_venv}" ]]; then
613     maybe_deactivate
614     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
615   fi
616   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
617     pip install --upgrade bandit
618   fi
619   if [[ -e "${LFBFL_subdir3}/bandit.ini" ]]; then
620     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
621       --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
622       --recursive .
623     # Saving new baseline in temp if necessary.
624     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
625       --format=json\
626       --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
627       --recursive .
628   elif [[ LFBFL_i_verbose -eq 1 ]]; then
629     printf "No file bandit.ini.\n"
630   fi
631   if [[ -n "${LFBFL_bandit_venv}" ]]; then
632     deactivate
633   fi
634
635   printf "Running pylint\n"
636   LFBFL_pylint_venv=""
637   grep_variable "${LFBFL_data_file_path}" pylint_venv\
638     --result-variable-prefix=LFBFL_\
639     --replace-line-returns-by=""
640   if [[ -n "${LFBFL_pylint_venv}" ]]; then
641     maybe_deactivate
642     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
643   fi
644   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
645     pip install --upgrade pylint
646   fi
647   if [[ -e "${LFBFL_subdir3}/pylintrc.toml" ]]; then
648     pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
649   elif [[ LFBFL_i_verbose -eq 1 ]]; then
650     printf "No file pylintrc.toml.\n"
651   fi
652   if [[ -n "${LFBFL_pylint_venv}" ]]; then
653     deactivate
654   fi
655
656   printf "Running ruff\n"
657   LFBFL_ruff_venv=""
658   grep_variable "${LFBFL_data_file_path}" ruff_venv\
659     --result-variable-prefix=LFBFL_\
660     --replace-line-returns-by=""
661   if [[ -n "${LFBFL_ruff_venv}" ]]; then
```

```
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  if [[ -e "${LFBFL_subdir3}/ruff.toml" ]]; then
    ruff check --config="${LFBFL_subdir3}/ruff.toml"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file ruff.toml.\n"
  fi
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHP native linter\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.php"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      php --syntax-check -f="${LFBFL_file_path}"\
        2>&1\
        | grep --invert-match "${LFBFL_annoying_warning}"
    done
  fi

  printf "Running PHP_CodeSniffer\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require squizlabs/php_codesniffer
  fi
  if [[ -e "${LFBFL_subdir3}/phpcs_ruleset.xml" ]]; then
    phpcs --report=code\
      --standard="${LFBFL_subdir3}/phpcs_ruleset.xml"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file phpcs_ruleset.xml.\n"
  fi

  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
  local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

```

```
718   phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
719     . text "${LFBFL_phpmd_rulesets}"
720   # Saving new baseline in temp if necessary.
721   rm "${LFBFL_temp_phpmd_baseline}"
722   phpmd --color --generate-baseline\
723     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
724     . text "${LFBFL_phpmd_rulesets}"
725   # Remove all superfluous whitespaces.
726   # sed --in-place --regexp-extended --null-data\
727   #   --expression='s/\n +//Mg'\
728   #   "${LFBFL_temp_phpmd_baseline}"
729   xsltproc --output "${LFBFL_temp_phpmd_baseline}"\
730     build_and_checks_dependencies/phpmd_baseline.xslt\
731     "${LFBFL_temp_phpmd_baseline}"
732   # sed --in-place --expression='s/" file="/"\n    file="/g'\
733   #   "${LFBFL_temp_phpmd_baseline}"
734   # sed --in-place --regexp-extended\
735   #   --expression='s~(file=".*)/>~\1\n  />~g'\
736   #   "${LFBFL_temp_phpmd_baseline}"
737   sed --in-place\
738     --expression='s/\([^"]*\("[^"]*"\)*\) rule="/\1\n    rule="/'\
739     --expression='s/\([^"]*\("[^"]*"\)*\) file="/\1\n    file="/'\
740     --expression='s/\([^"]*\("[^"]*"\)*\) method="/\1\n    method="/'\
741     --expression='s~/>~\n  />~'\
742     "${LFBFL_temp_phpmd_baseline}"
743   diff --suppress-common-lines\
744     "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
745     | grep --invert-match\
746       --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
747   # See miscellaneous/grep_benchmark.
748
749   printf "Running PHPSTAN\n"
750   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
751     composer global require phpstan/phpstan
752     composer global require phpstan/phpstan-strict-rules
753   fi
754
755   local LFBFL_temp_phpstan_baseline="${LFBFL_subdir3}/temp/"
756   LFBFL_temp_phpstan_baseline+="phpstan_baseline.neon"
757   readonly LFBFL_temp_phpstan_baseline
758   if [[ -e "${LFBFL_subdir3}/phpstan_config.neon" ]]; then
759     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"
760     # Saving new baseline in temp if necessary.
761     rm "${LFBFL_temp_phpstan_baseline}"
762     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"\
763       --generate-baseline="${LFBFL_temp_phpstan_baseline}"\
764       --allow-empty-baseline
765     sed --in-place --expression='s/\t/ /g' "${LFBFL_temp_phpstan_baseline}"
766     cat "${LFBFL_temp_phpstan_baseline}"
767   elif [[ LFBFL_i_verbose -eq 1 ]]; then
768     printf "No file phpstan_config.neon.\n"
769   fi
770
771   printf "Running phpDocumentor\n"
772   # shellcheck disable=SC2312
773   docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
```

```
774   printf -- "---PHP end---\n"
775
776   printf -- "---JS---\n"
777   local LFBFL_npm_lint_directories=""
778   grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
779     --result-variable-prefix=LFBFL_
780   if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
781     printf "Running ESLint\n"
782     local LFBFL_JS_directory
783     declare -a LFBFL_arr_paths
784     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
785     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
786       (cd -- "${LFBFL_JS_directory}" && npm run lint)
787     done
788   fi
789   printf -- "---JS end---\n"
790
791   [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
792
793   printf "Checking listed files\n"
794   "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
795     "${LFBFL_some_common_options[@]}"
796
797   printf "Creating the PDF file of the listing of the source code\n"
798   "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
799
800   pushd_to_work_directory --trap-popd
801   can_continue_after_enhanced_pushd || return 1
802
803   if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
804     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
805   fi
806
807   printf "Running xmllint\n"
808   # In some cases, we need to handle a cache of common DTDs by hand.
809   # XHTML DTDs validation can yield errors:
810   # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
811   # If you use Ubuntu/Debian, install w3c-sgml-lib instead.
812   LFBFL_file_name="xhtml1-strict.dtd"
813   LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
814   LFBFL_script_download_URL="${LFBFL_a}"
815   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
816   LFBFL_correct_sha512='fe11db06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
817   LFBFL_correct_sha512+='d8352bf19947168c9bcf1ae212eca035824293abb1f9'
818   LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
819   wrapped_wget_sha512
820
821   LFBFL_file_name="xhtml1-transitional.dtd"
822   LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
823   LFBFL_script_download_URL="${LFBFL_a}"
824   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
825   LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
826   LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
827   LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
828   wrapped_wget_sha512
829
```

```
  LFBFL_file_name="xhtml1-frameset.dtd"
  LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
  LFBFL_script_download_URL="${LFBFL_a}"
  LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
  LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
  LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
  LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
  wrapped_wget_sha512

  LFBFL_file_name="xhtml11.dtd"
  LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
  LFBFL_script_download_URL="${LFBFL_a}"
  LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
  LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
  LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
  LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
  wrapped_wget_sha512

  LFBFL_file_name="xhtml-lat1.ent"
  LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
  LFBFL_script_download_URL="${LFBFL_a}"
  LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
  LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
  LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
  LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
  wrapped_wget_sha512

  LFBFL_file_name="xhtml-special.ent"
  LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
  LFBFL_script_download_URL="${LFBFL_a}"
  LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
  LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
  LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
  LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
  wrapped_wget_sha512

  LFBFL_file_name="xhtml-symbol.ent"
  LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
  LFBFL_script_download_URL="${LFBFL_a}"
  LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
  LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
  LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
  LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
  wrapped_wget_sha512

  declare -r LFBFL_s_files_for_xmllint=$(
    find . -type f -iregex ".*\.\(xml\|xhtml\)"\
    | relevant_find\
    | sort
  )
  if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
    LFBFL_xmllint_use_downloaded_DTDs=""
    grep_variable "${LFBFL_data_file_path}" xmllint_use_downloaded_DTDs\
      --result-variable-prefix=LFBFL_

    declare -a LFBFL_arr_files_for_xmllint
```

```
886     mapfile -t LFBFL_arr_files_for_xmllint\
887       <<< "${LFBFL_s_files_for_xmllint}"
888     readonly LFBFL_arr_files_for_xmllint
889     local LFBFL_xmllint_files_without_DTD=""
890     grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
891       xmllint_files_without_DTD\
892       --result-variable-prefix=LFBFL_
893     declare -a LFBFL_arr_xmllint_files_without_DTD
894     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
895       mapfile -t LFBFL_arr_xmllint_files_without_DTD\
896         <<< "${LFBFL_xmllint_files_without_DTD}"
897     fi
898     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
899       [[ LFBFL_i_verbose -eq 1 ]]\
900         && printf "xmllint on %s\n" "${LFBFL_file_path}"
901       # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
902       # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
903       # xmllint --pedantic --noout -- "${LFBFL_file_path}"
904       xmllint --pedantic --noout "${LFBFL_file_path}"
905     done
906
907     local LFBFL_file_path2
908     for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
909       for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
910       do
911         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
912           LFBFL_file_path=""
913         fi
914       done
915       if [[ "${LFBFL_file_path}" == "" ]]; then
916         continue
917       fi
918       [[ LFBFL_i_verbose -eq 1 ]]\
919         && printf "xmllint on %s\n" "${LFBFL_file_path}"
920       if [[
921         "${LFBFL_file_path}" == *.xhtml
922         && "${LFBFL_xmllint_use_downloaded_DTDs}" == "true"
923       ]]; then
924         if grep --fixed-strings\
925           "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
926           "${LFBFL_file_path}"
927         then
928           xmllint --pedantic --noout\
929             --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
930             "${LFBFL_file_path}"
931         elif grep --fixed-strings\
932           "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
933           "${LFBFL_file_path}"
934         then
935           xmllint --pedantic --noout\
936             --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
937             "${LFBFL_file_path}"
938         elif grep --fixed-strings\
939           "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
940           "${LFBFL_file_path}"
941         then
```

```
          xmllint --pedantic --noout\
            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
            "${LFBFL_file_path}"
        elif grep --fixed-strings\
          "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
          "${LFBFL_file_path}"
        then
          xmllint --pedantic --noout\
            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
            "${LFBFL_file_path}"
        else
          printf "Unknown XHTML DTD.\n"
        fi
      else
        xmllint --pedantic --noout --valid "${LFBFL_file_path}"
      fi
    done
  fi
  # ---------------------------------------------------------------------

  # ---------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml.
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
```

```
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_s_files_for_Nu=$(
      find . -type f -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
      declare -a LFBFL_arr_files_for_Nu
      mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
      readonly LFBFL_arr_files_for_Nu
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
    fi
  fi
  # ----------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.12 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
```

```
30    if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
31      LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
32    fi
33    if [[ "${FUNCNAME[$1]}" != "" ]]; then
34      LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
35    fi
36    readonly LFBFL_where_was_i
37    # Comment out line above in case someone would use distinct
38    # offsets in the same function.
39  }
40
41  # RETURN traps handling
42  # Add the following lines at the start of a function.
43  #   declare -a LFBFL_return_traps_stack
44  #   init_return_trap
45
46  init_return_trap(){
47    local LFBFL_previous_return_trap
48    local LFBFL_previous_return_trap2
49    LFBFL_previous_return_trap=$(trap -p RETURN)
50    # shellcheck disable=SC2154,SC2309
51    if [[ LFBFL_i_verbose -eq 1 ]]; then
52      local LFBFL_where_was_i
53      get_where_was_i 2
54      printf "%s previous_return_trap is: %s.\n"\
55        "${LFBFL_where_was_i}"\
56        "${LFBFL_previous_return_trap}"
57    fi
58    if [[ -z "${LFBFL_previous_return_trap}" ]]; then
59      LFBFL_previous_return_trap="trap - RETURN"
60    fi
61    readonly LFBFL_previous_return_trap
62    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
63    LFBFL_previous_return_trap2="${LFBFL_function_depth}:"
64    LFBFL_previous_return_trap2+="${LFBFL_previous_return_trap}"
65    readonly LFBFL_previous_return_trap2
66    LFBFL_return_traps_stack+=("${LFBFL_previous_return_trap2}")
67    trap 'execute_return_traps' RETURN
68  }
69
70  execute_return_traps(){
71    if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
72      return
73    fi
74    # -----------------------------------------------------------------------
75    # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
76    if [[ "${FUNCNAME[1]}" == "source" ]]; then
77      # Not Working
78      return
79    fi
80    if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
81      return
82    fi
83    # -----------------------------------------------------------------------
84    local LFBFL_some_return_trap
85    local LFBFL_where_was_i
```

```
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ ${#LFBFL_return_traps_stack[@]} -gt 0 ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    # shellcheck disable=SC2154,SC2309
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ ${#LFBFL_return_traps_stack[@]} -gt 1 ]]; then
        printf "%s executing some return trap: %s.\n"\
          "${LFBFL_where_was_i}"\
          "${LFBFL_some_return_trap}"
      else
        printf "%s setting back to previous_return_trap: %s.\n"\
          "${LFBFL_where_was_i}"\
          "${LFBFL_some_return_trap}"
      fi
    fi
    eval -- "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
}


is_top_dirstack_directory(){
  # Altough "${DIRSTACK[0]}" may seem to be already a realpath,
  # possible use of symbolic links makes this function necessary.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needed to be "realpathed" already
  # but I removed this micro-optimization.
  # A compiled language could have a "realpathed property" on
  # "path object/string" to optimize when possible.
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath -- "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  local LFBFL_real_path2
  LFBFL_real_path2=$(realpath -- "$1")
  readonly LFBFL_real_path2
  [[ "${LFBFL_real_path}" == "${LFBFL_real_path2}" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory (needed to be sanitized with realpath before call)
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  # $1 can be any path now, see my comment on the micro-optimization in
```

```
142   # is_top_dirstack_directory() just above.
143   # shellcheck disable=SC2154,SC2309
144   [[ LFBFL_i_verbose -eq 1 ]]\
145     && printf "enhanced_pushd requested: %s.\n" "$1"
146   # Value 0 will always be overwritten, it is superfluous.
147   declare -gi i_enhanced_pushd_result=0
148   if [[ -z "$1" ]]; then
149     i_enhanced_pushd_result=110
150     # shellcheck disable=SC2248
151     return ${i_enhanced_pushd_result}
152   fi
153   local LFBFL_real_path
154   LFBFL_real_path=$(realpath -- "$1")
155   readonly LFBFL_real_path
156   # shellcheck disable=SC2154,SC2309
157   [[ LFBFL_i_verbose -eq 1 ]]\
158     && printf "enhanced_pushd passed validation 1.\n"
159   is_top_dirstack_directory "${LFBFL_real_path}" && {
160     i_enhanced_pushd_result=111
161     # shellcheck disable=SC2248
162     return ${i_enhanced_pushd_result}
163   }
164   # shellcheck disable=SC2154,SC2309
165   [[ LFBFL_i_verbose -eq 1 ]]\
166     && printf "enhanced_pushd passed validation 2.\n"
167
168   pushd -- "${LFBFL_real_path}" || {
169     i_enhanced_pushd_result=$?
170     local LFBFL_where_was_i
171     get_where_was_i "$2"
172     printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3"\
173       "${LFBFL_real_path}"
174     # shellcheck disable=SC2248
175     return ${i_enhanced_pushd_result}
176   }
177   # shellcheck disable=SC2154,SC2309
178   [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed.\n"
179   # shellcheck disable=SC2248
180   return ${i_enhanced_pushd_result}
181 }
182
183 enhanced_popd(){
184   # Auxiliary function needed below.
185   # See usages below if you want to reuse it.
186   # $1=to_directory
187   # $2=offset_for_where_was_i
188   # shellcheck disable=SC2154,SC2309
189   [[ LFBFL_i_verbose -eq 1 ]]\
190     && printf "enhanced_popd requested: %s.\n" "$1"
191   if [[ -z "$1" ]]; then
192     return 0
193   fi
194
195   declare -i LFBFL_i_popd_result
196   popd || {
197     LFBFL_i_popd_result=$?
```

```
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed.\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[
    i_enhanced_pushd_result -eq 0
    || i_enhanced_pushd_result -eq 110
    || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_some_flag(){
  # $1=variable_name
  # $2=some_flag_long_name
  # $3=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  # This test:
  # if [[ "$*" == *--verbose* ]]; then
  # is faster than below.
  # But it is not bug free in a calling function,
  # and even incompatible with the current function where
```

```
254   # the flag_long_name is a mandatory argument.
255   printf -v "$1" "%d" 0
256   declare -i LFBFL_i
257   local LFBFL_argument
258   for ((LFBFL_i = 4; LFBFL_i <= $#; ++LFBFL_i)); do
259     LFBFL_argument="${!LFBFL_i}"
260     if [[ "${LFBFL_argument}" == "$2" ]]; then
261       printf -v "$1" "%d" 1
262       break
263     fi
264   done
265   if [[ $3 -eq 1 ]]; then
266     readonly "$1"
267   fi
268 }
269
270 get_verbose_option(){
271   # This command is to be called in another one with same arguments.
272   # Options:
273   #   --verbose
274   get_some_flag LFBFL_i_verbose --verbose 1 "$@"
275   # shellcheck disable=SC2154,SC2309
276   if [[ LFBFL_i_verbose -eq 1 ]]; then
277     local LFBFL_where_was_i
278     get_where_was_i 2
279     printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
280   fi
281 }
282
283
284 get_some_option(){
285   # $1=variable_name
286   # $2=some_option_long_name
287   # $3=option_default_value_if_flag_without_option_value_or_empty_value
288   # $4=option_default_value_if_nothing
289   # $5=read_only 0 or 1 to make the variable readonly
290   # This command is to be called in another one with same arguments on top
291   # of the four previous.
292   # If you truly have a need to distinguish:
293   # option_default_value_if_flag_without_option_value --thing
294   # option_default_value_if_flag_with_empty_option_value --thing= or =''
295   # then think twice before modifying this function.
296   printf -v "$1" "%s" "$4"
297   declare -i LFBFL_i
298   local LFBFL_argument
299   for ((LFBFL_i = 6; LFBFL_i <= $#; ++LFBFL_i)); do
300     LFBFL_argument="${!LFBFL_i}"
301     if [[ "${LFBFL_argument}" == "$2" ]]; then
302       printf -v "$1" "%s" "$3"
303       break
304     fi
305     if [[ "${LFBFL_argument}" == "$2"=* ]]; then
306       printf -v "$1" "%s" "${LFBFL_argument#"$2"=}"
307       printf -v "$1" "%s" "${!1:-$3}"
308       break
309     fi
```

```
  done
  if [[ $5 -eq 1 ]]; then
    readonly "$1"
  fi
}

keep_options(){
  # $1=variable_name of array of options
  # Other arguments: the options to keep.
  declare -a LFBFL_array_of_filtered_options
  declare -n LFBFL_array_of_options=$1
  local LFBFL_some_argument
  local LFBFL_some_argument2
  for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
    for LFBFL_some_argument2 in "$@"; do
      if [[
        "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
        || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
      ]]; then
        LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
      fi
    done
  done
  LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1
# Or:
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory --trap-popd
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  get_some_option LFBFL_work_directory --work-directory "." "." 0 "$@"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  # Options:
  #   --trap-popd
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
  if [[ i_enhanced_pushd_result -eq 0 ]]; then
    declare -i LFBFL_i_trap_popd
    get_some_flag LFBFL_i_trap_popd --trap-popd 1 "$@"
```

```
366     if [[ LFBFL_i_trap_popd -eq 1 ]]; then
367       declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
368       local LFBFL_some_return_trap
369       LFBFL_some_return_trap="${LFBFL_function_depth}:"
370       LFBFL_some_return_trap+="popd_from_work_directory 4"
371       readonly LFBFL_some_return_trap
372       LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
373     fi
374   fi
375   # shellcheck disable=SC2248
376   return ${i_enhanced_pushd_result}
377 }
378
379 popd_from_work_directory(){
380   # $1=offset for where_was_i
381   enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
382 }
383
384 work_directory_is_top_dirstack_directory(){
385   is_top_dirstack_directory "${LFBFL_work_directory}"
386 }
387
388
389 get_offset_option(){
390   # This command is to be called in another one with same arguments on top
391   # of the default value.
392   # $1=option_default_value
393   # Options:
394   #   --offset=2
395   get_some_option LFBFL_i_offset --offset "$1" "$1" 1 "$@"
396 }
397
398 get_max_line_length_option(){
399   # This command is to be called in another one with same arguments.
400   # Its most useful feature is that it factorizes the default maximum line
401   # length of 70 characters.
402   # Options:
403   #   --max-line-length
404   get_some_option LFBFL_i_max_line_length --max-line-length 70 70 1 "$@"
405 }
406
407
408 # Functions relative to shell and bash options
409 # (See SHELLOPTS and BASHOPTS, set and shopt):
410 # - shell options usage, example with pipefail:
411 #     enhanced_set_shell_option pipefail\
412 #       && trap 'enhanced_unset_shell_option pipefail' RETURN
413 #   or maybe:
414 #     enhanced_unset_shell_option pipefail\
415 #       && trap 'enhanced_set_shell_option pipefail' RETURN
416 #   if you need to unset it in some function.
417 #   or better if you use my RETURN traps handler:
418 #     enhanced_set_shell_option pipefail --trap-unset
419 #     enhanced_unset_shell_option pipefail --trap-set
420 # - bash options usage, example with globstar:
421 #     enhanced_set_bash_option globstar\
```

```
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_unset
  get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
  if [[ LFBFL_i_trap_unset -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
```

```
478   set +o "$1"
479   declare -i LFBFL_i_offset=2
480   get_offset_option 2 "$@"
481   # shellcheck disable=SC2154,SC2309
482   if [[ LFBFL_i_verbose -eq 1 ]]; then
483     local LFBFL_where_was_i
484     # shellcheck disable=SC2248
485     get_where_was_i ${LFBFL_i_offset}
486     printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
487   fi
488   declare -i LFBFL_i_trap_set
489   get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
490   if [[ LFBFL_i_trap_set -eq 1 ]]; then
491     declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
492     local LFBFL_some_return_trap
493     LFBFL_some_return_trap="${LFBFL_function_depth}:"
494     LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
495     readonly LFBFL_some_return_trap
496     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
497   fi
498   # shellcheck disable=SC2086
499   return ${!LFBFL_result_name}
500 }
501
502 enhanced_set_bash_option(){
503   # $1=optname some bash option for shopt
504   # Options:
505   #   --offset=2
506   #   --trap-unset
507   declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
508   declare -gi "${LFBFL_result_name}"=0
509   # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
510   # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
511   if shopt -q "$1"; then
512     printf -v "${LFBFL_result_name}" "%d" 1
513     # shellcheck disable=SC2086
514     return ${!LFBFL_result_name}
515   fi
516   shopt -s "$1"
517   declare -i LFBFL_i_offset=2
518   get_offset_option 2 "$@"
519   # shellcheck disable=SC2154,SC2309
520   if [[ LFBFL_i_verbose -eq 1 ]]; then
521     local LFBFL_where_was_i
522     # shellcheck disable=SC2248
523     get_where_was_i ${LFBFL_i_offset}
524     printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
525   fi
526   declare -i LFBFL_i_trap_unset
527   get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
528   if [[ LFBFL_i_trap_unset -eq 1 ]]; then
529     declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
530     local LFBFL_some_return_trap
531     LFBFL_some_return_trap="${LFBFL_function_depth}:"
532     LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
533     readonly LFBFL_some_return_trap
```

```
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_set
  get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
  if [[ LFBFL_i_trap_set -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

3.13 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
```

```
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_argument}"
      LFBFL_i_first=0
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_argument}" ]]; then
      return 0
    fi
  done
  return 1
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j = LFBFL_i + 1; LFBFL_j <= LFBFL_ij_max; ++LFBFL_j)); do
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
```

```
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq --repeated
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}
```

```
126
127 some_equal(){
128   ! all_distinct "$@"
129 }
130
131 some_equal2(){
132   ! all_distinct2 "$@"
133 }
134
135 some_equal3(){
136   ! all_distinct3 "$@"
137 }
138
139 max(){
140   # Returns the maximum string among the arguments.
141   # Renvoie la chaîne la plus grande parmi celles proposées.
142   # $1=sort_command with the flags you want to be applied.
143   # $ max sort a b c
144   # c
145   if [[ $# -eq 0 ]]; then
146     return 1
147   fi
148   enhanced_set_shell_option pipefail\
149     && trap 'enhanced_unset_shell_option pipefail' RETURN
150   printf "%s\n" "${@:2}" \
151     | eval -- "$1 --reverse"\
152     | head --lines=1
153 }
154
155 min(){
156   # Returns the minimum string among the arguments.
157   # Renvoie la chaîne la plus petite parmi celles proposées.
158   # $1=sort_command with the flags you want to be applied.
159   # $ min sort a b c
160   # a
161   if [[ $# -eq 0 ]]; then
162     return 1
163   fi
164   enhanced_set_shell_option pipefail\
165     && trap 'enhanced_unset_shell_option pipefail' RETURN
166   printf "%s\n" "${@:2}" \
167     | eval -- "$1"\
168     | head --lines=1
169 }
170
171 is_substring(){
172   # $1=string
173   # $2=substring
174   declare -r LFBFL_var_1=$(
175     printf "%s" "$1"\
176     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
177   )
178   declare -r LFBFL_var_2=$(
179     printf "%s" "$2"\
180     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
181   )
```

```
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php --run="${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php --run="${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.14 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
```

```
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_path="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_path}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_web_url=""
  grep_variable "${LFBFL_data_file_path}" repository_web_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_git_url=""
  grep_variable "${LFBFL_data_file_path}" repository_git_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_abstract=""
  grep_variable "${LFBFL_data_file_path}" abstract\
    --result-variable-prefix="LFBFL_"

  local LFBFL_acknowledgments=""
  grep_variable "${LFBFL_data_file_path}" acknowledgments\
    --result-variable-prefix="LFBFL_"

  local LFBFL_author_email=""
  grep_variable "${LFBFL_data_file_path}" author_email\
    --result-variable-prefix="LFBFL_"\
```

```
    --replace-line-returns-by=""

  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_path}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  local LFBFL_author_website=""
  grep_variable "${LFBFL_data_file_path}" author_website\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_path}" max_line_length\
    --result-variable-prefix="LFBFL_i_"

  declare -r LFBFL_current_date=$(date --iso-8601="date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_i_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc --lines
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "${LFBFL_common_options[@]}"
    code_lines_count_not_empty "${LFBFL_common_options[@]}"
    code_lines_count_empty "${LFBFL_common_options[@]}"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
```

```
144       > "${LFBFL_temp_path}/current_tree_light.txt"
145
146     tree -a -Fh --timefmt='%F %R' --gitignore\
147       -I "node_modules/"\
148       -I "__pycache__/"\
149       -I ".mypy_cache/"\
150       -I ".ruff_cache/"\
151       -I ".git/"\
152       | replace_non_ascii_spaces\
153       > "${LFBFL_temp_path}/current_tree.txt.temp"
154
155     local LFBFL_s_files_paths
156     # Remove line returns that are here to keep lines short.
157     LFBFL_s_files_paths=$(
158       sed --regexp-extended --null-data\
159         --expression='s/(.)\\\n/\1/Mg'\
160         -- "${LFBFL_subdir2}/files_names_listing.txt"\
161       | grep --invert-match '^//'
162     )
163     readonly LFBFL_s_files_paths
164     declare -a LFBFL_arr_files_paths
165     if [[ -n "${LFBFL_s_files_paths}" ]]; then
166       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
167     fi
168     readonly LFBFL_arr_files_paths
169
170     local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
171     LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
172     readonly LFBFL_temp_files_listing
173     : > "${LFBFL_temp_files_listing}"
174     local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
175     LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
176     readonly LFBFL_temp_files_listing2
177     : > "${LFBFL_temp_files_listing2}"
178     # HTML <li> elements, hence "lis".
179     local LFBFL_temp_files_lis
180     LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
181     readonly LFBFL_temp_files_lis
182     : > "${LFBFL_temp_files_lis}"
183     # shellcheck disable=SC2248
184     get_split_score_simple 1 ${LFBFL_i_max_line_length} /
185     declare -r LFBFL_score_command="${get_split_score_result}"
186     declare -i LFBFL_i_score_command_properties
187     LFBFL_i_score_command_properties=${i_get_split_score_result2}
188     readonly LFBFL_i_score_command_properties
189     # shellcheck disable=SC2248
190     get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
191     declare -r LFBFL_score_command2="${get_split_score_result}"
192     declare -i LFBFL_i_score_command_properties2
193     LFBFL_i_score_command_properties2=${i_get_split_score_result2}
194     readonly LFBFL_i_score_command_properties2
195     declare -r LFBFL_suffix='%'
196     declare -i LFBFL_i=0
197     local LFBFL_cleaned_path1
198     local LFBFL_cleaned_path2
199     local LFBFL_new_lines
```

```
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  local LFBFL_s_format
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed --expression='s/\//:/g;s/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"
```

```
256
257       # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
258       ((++LFBFL_i))
259       LFBFL_new_lines="${LFBFL_file_path}"
260       if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
261         # shellcheck disable=SC2248
262         repeated_split_last_line "${LFBFL_new_lines}"\
263           "-->"\
264           ${LFBFL_i_max_line_length}\
265           "<!--"\
266           "${LFBFL_score_command}"\
267           ${LFBFL_i_score_command_properties}
268         LFBFL_new_lines=${repeated_split_last_line_result}
269       fi
270       {
271         LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
272         LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
273         # shellcheck disable=SC2059
274         printf "${LFBFL_s_format}"\
275           "${LFBFL_i}"\
276           "${LFBFL_i}"\
277           "${LFBFL_new_lines}"
278         sed --expression='s/&/\&amp;/g;s/</\&lt;/g'\
279           < "${LFBFL_file_path}"
280         printf "</pre>\n\n\n"
281       } >> "${LFBFL_temp_files_listing2}"
282       LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
283       LFBFL_s_format+="      </a></li>\n"
284       # shellcheck disable=SC2059
285       printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
286         >> "${LFBFL_temp_files_lis}"
287     done
288     overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
289       "${LFBFL_temp_files_listing}"
290     # Example of generated content of files_listing.tex.sub:
291     # ------------------------------------------------------------------------
292     # \subsection{
293     #   build\_and\_checks\_dependencies/licenses\_templates/%
294     # build\_licenses\_templates.exec.sh
295     # }
296     # \label{
297     #   build_and_checks_dependencies:licenses_templates:%
298     # build_licenses_templatesexecsh
299     # }
300     #
301     # \VerbatimInput[numbers=left,xleftmargin=-5mm]{
302     # build_and_checks_dependencies/licenses_templates/%
303     # build_licenses_templates.exec.sh
304     # }
305     # ------------------------------------------------------------------------
306     sed --in-place --expression="s|\\n</pre>|</pre>|g"\
307       -- "${LFBFL_temp_files_listing2}"
308     overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
309       "${LFBFL_temp_files_listing2}"
310     # Example of generated content of files_listing.html.sub:
311     # ------------------------------------------------------------------------
```

```
312   # <h3 id="subsection3.15">3.15
313   # build_and_checks_dependencies/licenses_templates/<!--
314   # -->build_licenses_templates.exec.sh
315   # </h3>
316   # ...
317   # <h3 id="subsection3.33">3.33
318   # build_and_checks_variables/.gitignore
319   # </h3>
320   # <pre class="numbered_lines">
321   # temp/
322   # upgrade_venvs_ts
323   # </pre>
324   # ------------------------------------------------------------------------
325   overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
326     "${LFBFL_temp_files_lis}"
327   # Example of generated content of files_lis.html.sub:
328   # ------------------------------------------------------------------------
329   #       <li><a href="#subsection3.15">
330   # build_and_checks_dependencies/licenses_templates/<!--
331   # -->build_licenses_templates.exec.sh
332   #       </a></li>
333   # ------------------------------------------------------------------------
334
335   # shellcheck disable=SC2248
336   get_overlength_regexp ${LFBFL_i_max_line_length}
337
338   # We verify if some lines are beyond max_line_length characters
339   # in current_tree_light.txt and current_tree.txt.
340   declare -ar LFBFL_trees=(
341     "current_tree_light.txt"
342     "current_tree.txt.temp"
343   )
344   local LFBFL_tree
345   local LFBFL_tree_path
346   local LFBFL_line
347   local LFBFL_prefix
348   declare -i LFBFL_i_prefix_last_position
349   local LFBFL_char
350   local LFBFL_line_start
351   local LFBFL_file_name
352   local LFBFL_s_lines
353   declare -a LFBFL_arr_lines
354   declare -i LFBFL_i_keep_length
355   for LFBFL_tree in "${LFBFL_trees[@]}"; do
356     LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
357     # shellcheck disable=SC2248
358     LFBFL_s_lines=$(
359       grep --regexp="${get_overlength_regexp_result}" --\
360         "${LFBFL_tree_path}"
361     )
362     if [[ -z "${LFBFL_s_lines}" ]]; then
363       continue
364     fi
365     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
366     for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
367       # printf "LFBFL_line: %s\n" "${LFBFL_line}"
```

```
368       LFBFL_prefix=$(
369         printf "%s" "${LFBFL_line}"\
370         | sed --regexp-extended --expression='s/(.*)──[^─]+$/\1/g'
371       )
372       # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
373       LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
374       LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
375       # printf "LFBFL_char: %s\n" "${LFBFL_char}"
376       LFBFL_prefix=$(
377         printf "%s" "${LFBFL_prefix}"\
378         | sed --regexp-extended --expression='s/[^ ]+$//g'
379       )
380       if [[ "${LFBFL_char}" == "└" ]]; then
381         LFBFL_prefix+="  "
382       else
383         LFBFL_prefix+="│ "
384       fi
385       # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
386       LFBFL_file_name=$(
387         printf "%s" "${LFBFL_line}"\
388         | sed --regexp-extended --expression='s|.* ([^ ]+)$|\1|g'
389       )
390       # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
391       LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
392       LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
393       LFBFL_line_start=$(
394         printf "%s" "${LFBFL_line_start}"\
395         | sed --expression='s/ *$//g'
396       )
397       # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
398       LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
399       if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
400         # shellcheck disable=SC2248
401         repeated_split_last_line "${LFBFL_new_lines}"\
402           "${LFBFL_prefix}"\
403           ${LFBFL_i_max_line_length}\
404           ""\
405           ""\
406           ""
407         LFBFL_new_lines=${repeated_split_last_line_result}
408       fi
409       printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
410         > "${LFBFL_tree_path}.some_line.temp"
411       insert_file_at_token "${LFBFL_tree_path}"\
412         "${LFBFL_line}"\
413         "${LFBFL_tree_path}.some_line.temp"\
414         ""\
415         --quiet
416     done
417     rm --force -- "${LFBFL_tree_path}.some_line.temp"
418   done
419
420   overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
421     "${LFBFL_temp_path}/current_tree.txt.temp" --keep-temp --tree-mode
422
423   local LFBFL_tex_path_start
```

```
424   LFBFL_tex_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
425   readonly LFBFL_tex_path_start
426   local LFBFL_html_path_start
427   LFBFL_html_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.html"
428   readonly LFBFL_html_path_start
429
430   if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
431     # If there is a template, we init the process from it.
432     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
433       "${LFBFL_tex_path_start}.1"
434   elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
435     # Otherwise if there is a tex file, we init the process from it.
436     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
437       "${LFBFL_tex_path_start}.1"
438   else
439     printf "Neither .tex.tpl, nor .tex in ./%s/.\n" "${LFBFL_subdir2}"
440   fi
441
442   # Same logic with repository HTML file.
443   if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
444     cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
445       "${LFBFL_html_path_start}.1"
446   elif [[ -f "${LFBFL_repository_name}.html" ]]; then
447     cp -- "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
448   else
449     printf "Neither .html.tpl in ./%s/, nor .html in ./.\n"\
450       "${LFBFL_subdir2}"
451   fi
452
453   # shellcheck disable=SC2001
454   printf "%s\n" "${LFBFL_abstract}"\
455     | sed --expression='s/\\n/\n/g'\
456     > "${LFBFL_temp_path}/abstract_temp"
457
458   # shellcheck disable=SC2001
459   printf "%s\n" "${LFBFL_acknowledgments}"\
460     | sed --expression='s/\\n/\n/g'\
461     > "${LFBFL_temp_path}/acknowledgments_temp"
462
463   # But the filling still occurs, in case the dev want to refill
464   # part of the tex/html file that he modified with @token@
465   # using some computed results.
466   # Tex filling:
467   if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
468     sed --in-place\
469       --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
470       --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
471       --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
472       --expression="s|@author_email@|${LFBFL_author_email}|g"\
473       --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
474       --expression="s|@author_website@|${LFBFL_author_website}|g"\
475       --expression="s|@current_date@|${LFBFL_current_date}|g"\
476       --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
477       --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
478       --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
479       -- "${LFBFL_tex_path_start}.1"
```

```
480
481       insert_file_at_token "${LFBFL_tex_path_start}.1"\
482         @abstract@\
483         "${LFBFL_temp_path}/abstract_temp"\
484         "${LFBFL_tex_path_start}.2"
485
486       insert_file_at_token "${LFBFL_tex_path_start}.2"\
487         @acknowledgments@\
488         "${LFBFL_temp_path}/acknowledgments_temp"\
489         "${LFBFL_tex_path_start}.3"
490
491       insert_file_at_token "${LFBFL_tex_path_start}.3"\
492         @current_tree_light@\
493         "${LFBFL_temp_path}/current_tree_light.txt"\
494         "${LFBFL_tex_path_start}.4"
495
496       insert_file_at_token "${LFBFL_tex_path_start}.4"\
497         @current_tree@\
498         "${LFBFL_temp_path}/current_tree.txt"\
499         "${LFBFL_tex_path_start}.5"
500
501       insert_file_at_token "${LFBFL_tex_path_start}.5"\
502         @dependencies_notes@\
503         "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
504         "${LFBFL_tex_path_start}.6"
505
506       insert_file_at_token "${LFBFL_tex_path_start}.6"\
507         @files_listing_VerbatimInput@\
508         "${LFBFL_temp_path}/files_listing.tex.sub"\
509         "${LFBFL_tex_path_start}.7"
510
511       overwrite_if_not_equal\
512         "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
513         "${LFBFL_tex_path_start}.7" --keep-temp
514     fi
515
516     # HTML filling:
517     declare -i LFBFL_i_HTML_updated=0
518     if [[ -f "${LFBFL_html_path_start}.1" ]]; then
519       sed --in-place\
520         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
521         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
522         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
523         --expression="s|@author_email@|${LFBFL_author_email}|g"\
524         --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
525         --expression="s|@author_website@|${LFBFL_author_website}|g"\
526         --expression="s|@current_date@|${LFBFL_current_date}|g"\
527         --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
528         --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
529         --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
530         -- "${LFBFL_html_path_start}.1"
531
532       insert_file_at_token "${LFBFL_html_path_start}.1"\
533         @abstract@\
534         "${LFBFL_temp_path}/abstract_temp"\
535         "${LFBFL_html_path_start}.2"
```

```
536
537       insert_file_at_token "${LFBFL_html_path_start}.2"\
538         @acknowledgments@\
539         "${LFBFL_temp_path}/acknowledgments_temp"\
540         "${LFBFL_html_path_start}.3"
541
542       insert_file_at_token "${LFBFL_html_path_start}.3"\
543         @files_lis@\
544         "${LFBFL_temp_path}/files_lis.html.sub"\
545         "${LFBFL_html_path_start}.4"
546
547       insert_file_at_token "${LFBFL_html_path_start}.4"\
548         @current_tree_light@\
549         "${LFBFL_temp_path}/current_tree_light.txt"\
550         "${LFBFL_html_path_start}.5"
551
552       insert_file_at_token "${LFBFL_html_path_start}.5"\
553         @current_tree@\
554         "${LFBFL_temp_path}/current_tree.txt"\
555         "${LFBFL_html_path_start}.6"
556
557       insert_file_at_token "${LFBFL_html_path_start}.6"\
558         @dependencies_notes@\
559         "${LFBFL_temp_path}/dependencies_notes.html.sub"\
560         "${LFBFL_html_path_start}.7"
561
562       insert_file_at_token "${LFBFL_html_path_start}.7"\
563         @files_listing_HTMLPreInput@\
564         "${LFBFL_temp_path}/files_listing.html.sub"\
565         "${LFBFL_html_path_start}.8"
566
567       overwrite_if_not_equal "${LFBFL_repository_name}.html"\
568         "${LFBFL_html_path_start}.8" --keep-temp
569       LFBFL_i_HTML_updated=$?
570     fi
571
572     # The HTML contains everything including listings with source code.
573     # If it doesn't get updated,
574     # then it is useless to recompute the PDF.
575     if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
576       return
577     fi
578
579     local LFBFL_latex_command=""
580     grep_variable "${LFBFL_data_file_path}" latex_command\
581       --result-variable-prefix="LFBFL_"\
582       --replace-line-returns-by=""
583
584     case ${LFBFL_latex_command} in
585       pdflatex) : ;;
586       lualatex) : ;;
587       *) printf "Unsupported latex command" ; return 1 ;;
588     esac
589
590     declare -a LFBFL_arr_commands
591     local LFBFL_command
```

```
592
593   local LFBFL_latex_targets=""
594   grep_variable "${LFBFL_data_file_path}" latex_targets\
595     --result-variable-prefix="LFBFL_"
596   declare -a LFBFL_arr_latex_targets
597   if [[ -z "${LFBFL_latex_targets}" ]]; then
598     LFBFL_command="${LFBFL_latex_command}"
599     LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
600     if [[ LFBFL_i_verbose -eq 0 ]]; then
601       LFBFL_command+=" > /dev/null"
602     fi
603     LFBFL_arr_commands+=("${LFBFL_command}")
604   else
605     mapfile -t LFBFL_arr_latex_targets <<< "${LFBFL_latex_targets}"
606     local LFBFL_latex_target
607     local LFBFL_jobname_suffix
608     local LFBFL_buildtarget
609     for LFBFL_latex_target in "${LFBFL_arr_latex_targets[@]}"; do
610       LFBFL_jobname_suffix=${LFBFL_latex_target%:*}
611       LFBFL_buildtarget=${LFBFL_latex_target##*:}
612       LFBFL_command="${LFBFL_latex_command}"
613       LFBFL_command+=" --jobname="
614       LFBFL_command+="'${LFBFL_repository_name}-${LFBFL_jobname_suffix}'"
615       LFBFL_command+=" '\def\buildtarget{${LFBFL_buildtarget}}"
616       LFBFL_command+=" \input{"
617       LFBFL_command+="${LFBFL_subdir2}/${LFBFL_repository_name}.tex}'"
618       if [[ LFBFL_i_verbose -eq 0 ]]; then
619         LFBFL_command+=" > /dev/null"
620       fi
621       LFBFL_arr_commands+=("${LFBFL_command}")
622     done
623   fi
624
625   for LFBFL_command in "${LFBFL_arr_commands[@]}"; do
626     for ((LFBFL_i = 0; LFBFL_i < 3; ++LFBFL_i)); do
627       eval -- "${LFBFL_command}"
628     done
629   done
630
631   # LFBFL_files_to_temp=(\
632   #   "${LFBFL_repository_name}.aux"\
633   #   "${LFBFL_repository_name}.log"\
634   #   "${LFBFL_repository_name}.out"\
635   #   "${LFBFL_repository_name}.toc"
636   # )
637   # for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
638   #   mv -- "${LFBFL_file_name}" "${LFBFL_temp_path}/"
639   # done
640
641   LFBFL_files_suffix_to_temp=(\
642     ".aux"\
643     ".log"\
644     ".out"\
645     ".toc"
646   )
647   for LFBFL_file_suffix in "${LFBFL_files_suffix_to_temp[@]}"; do
```

```
    mv -- "${LFBFL_repository_name}"*"${LFBFL_file_suffix}"\
      "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.15 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat -- "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

```

```
generate_from_template_with_line_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    printf "%s\n" "$5"\
      > "${LFBFL_temp}"
  fi
  sed --expression="s~^~$3~g" -- "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval -- "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_part1_of_file
  # $4=path_of_part2_of_file
  # We cannot add --line-regexp with --fixed-strings because of the
  # possible indentation in file,
  # instead we use grep_fixed_string_with_anchor.
  declare -ir LFBFL_i_line_number=$(
    grep_fixed_string_with_anchor "$1" "$2"\
      --enforce-line-ends-with-fixed-string --line-number\
    | head --lines=1\
    | cut --delimiter=':' --fields=1
  )
  declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_i_lines_after=$((
    LFBFL_i_line_count - LFBFL_i_line_number
  ))
  head --lines="$((LFBFL_i_line_number - 1))" -- "$1"\
    > "$3"
  tail --lines="${LFBFL_i_lines_after}" -- "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_file_to_insert
  # $4=path_of_result_file
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_path="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_path="$1.insert_file_at_token2.temp"
```

```
  local LFBFL_result_file_path
  if [[ -n "$4" ]]; then
    LFBFL_result_file_path="$4"
  else
    LFBFL_result_file_path="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_path
  split_file_in_two "$1" "$2" "${LFBFL_start_file_path}"\
    "${LFBFL_end_file_path}"
  cat -- "${LFBFL_start_file_path}" "$3" "${LFBFL_end_file_path}"\
    > "${LFBFL_result_file_path}"
  rm -- "${LFBFL_start_file_path}" "${LFBFL_end_file_path}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_path}" "$5"
  fi
}
```

3.16 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
```

```
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.neon'
    '**/*.php'
    '**/*.py'
    '**/*.rs'
    '**/*.rlib'
    '**/*.sh'
    '**/*.sql'
    '**/*.sub'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.xslt'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.neon'
    '*.php'
    '*.py'
    '*.rs'
    '*.rlib'
    '*.sh'
    '*.sql'
    '*.sub'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
```

```
    '*.txt'
    '*.xml'
    '*.xslt'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.17 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
```

```
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: No command in configuration.\n"
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval -- "${LFBFL_eval_string}"
    done
  done
}
```

3.18 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
56    declare -a LFBFL_return_traps_stack
57    init_return_trap
58
59    enhanced_set_shell_option pipefail --trap-unset
60
61    local LFBFL_s_format
62
63    # Source
64    local LFBFL_license_source_subdir
65    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
66    readonly LFBFL_license_source_subdir
67    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
68    LFBFL_source_license_prefix+="license_file_header_"
69    readonly LFBFL_source_license_prefix
70    # Target
71    local LFBFL_license_target_subdir
72    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
73    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
74    readonly LFBFL_license_target_subdir
75    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
76    LFBFL_target_license_prefix+="license_file_header_"
77    readonly LFBFL_target_license_prefix
78
79    declare -ar LFBFL_licenses=(
80      "AGPLv3+"
81      "GPLv3+"
82      "LGPLv3+"
83    )
84
85    # ----------------------------------------------------------------
86    # Various language definitions for text file formats
87    declare -ar LFBFL_block_comment_languages=(
88      "c"
89      "py"
90      "xml"
91    )
92    declare -ar LFBFL_block_comment_enters=(
93      '/*'
94      '"""'
95      '<!--'
96    )
97    declare -ar LFBFL_block_comment_exits=(
98      '*/'
99      '"""'
100     '-->'
101   )
102   equal "${#LFBFL_block_comment_languages[@]}"\
103         "${#LFBFL_block_comment_enters[@]}"\
104         "${#LFBFL_block_comment_exits[@]}"
105   declare -i LFBFL_i_equal_result=$?
106   if [[ LFBFL_i_equal_result -eq 0 ]]; then
107     printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
108     return 1
109   fi
110
111   declare -ar LFBFL_line_comment_languages=(
```

```
    "rs"
    "sh"
    "sql"
    "tex"
  )
  declare -ar LFBFL_line_comment_prefixes=(
    '// '
    '# '
    '-- '
    '% '
  )
  equal "${#LFBFL_line_comment_languages[@]}"\
        "${#LFBFL_line_comment_prefixes[@]}"
  LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ----------------------------------------------------------------------

  # ----------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
  local LFBFL_extension
  local LFBFL_license_file_path
  local LFBFL_enter_string
  local LFBFL_exit_string
  local LFBFL_temp2
  local LFBFL_prefix_string
  local LFBFL_intermediate_file_path
  local LFBFL_file_prefix
  declare -i LFBFL_i_generate_from_template_result
  declare -i LFBFL_i
  for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
    LFBFL_license_prefix="${LFBFL_source_license_prefix}"
    LFBFL_license_prefix+="${LFBFL_some_license}"
    LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
    LFBFL_license_prefix2+="${LFBFL_some_license}"
    for ((
      LFBFL_i = 0;
      LFBFL_i < ${#LFBFL_block_comment_languages[@]};
      ++LFBFL_i
    )); do
      LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
      LFBFL_license_file_path="${LFBFL_license_prefix2}"
      LFBFL_license_file_path+=".${LFBFL_extension}"
      LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
      LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
      generate_from_template_with_block_comments\
        "${LFBFL_license_prefix}.tpl"\
```

```
        "${LFBFL_license_file_path}.tpl"\
        "${LFBFL_enter_string}" "${LFBFL_exit_string}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_path}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_path}"
        fi
      fi
    done
    LFBFL_temp2=".generate_from_template_with_line_comments.temp"
    for ((
      LFBFL_i = 0;
      LFBFL_i < ${#LFBFL_line_comment_languages[@]};
      ++LFBFL_i
    )); do
      LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
      LFBFL_license_file_path="${LFBFL_license_prefix2}"
      LFBFL_license_file_path+=".${LFBFL_extension}"
      LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
      LFBFL_intermediate_file_path="${LFBFL_license_file_path}.tpl"
      LFBFL_intermediate_file_path+="${LFBFL_temp2}"
      LFBFL_file_prefix=""
      if [[ "${LFBFL_extension}" == "sh" ]]; then
        LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
      fi
      # The sed command is here to remove trailing spaces of previously
      # empty lines after comment prefix is added to the line.
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_path}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_path}'"\
        "${LFBFL_file_prefix}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_path}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_path}"
        fi
      fi
    done
  done
  # ----------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in work directory, we can go there.
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

```

```
224   local LFBFL_data_file_path="build_and_checks_variables/"
225   LFBFL_data_file_path+="repository_data.txt"
226   readonly LFBFL_data_file_path
227   local LFBFL_repository_name=""
228   grep_variable "${LFBFL_data_file_path}" repository_name\
229     --result-variable-prefix="LFBFL_"\
230     --replace-line-returns-by=""
231   local LFBFL_license=""
232   grep_variable "${LFBFL_data_file_path}" license\
233     --result-variable-prefix="LFBFL_"
234   local LFBFL_license2=""
235   grep_variable "${LFBFL_data_file_path}" license2\
236     --result-variable-prefix="LFBFL_"
237   local LFBFL_author_full_name=""
238   grep_variable "${LFBFL_data_file_path}" author_full_name\
239     --result-variable-prefix="LFBFL_"\
240     --replace-line-returns-by=" "
241
242   LFBFL_license_prefix="build_and_checks_variables/temp/"
243   LFBFL_license_prefix+="license_file_header_"
244   readonly LFBFL_license_prefix
245   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
246   readonly LFBFL_license_prefix2
247   local LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
248   readonly LFBFL_license_prefix3
249
250   # First year according to current state of git repository.
251   declare -ir LFBFL_i_first_year=$(
252     git log\
253     | grep 'Date:'\
254     | tail --lines=1\
255     | cut --delimiter=' ' --fields=8
256   )
257   # Last year according to current state of git repository.
258   declare -ir LFBFL_i_last_year=$(
259     git log\
260     | grep 'Date:'\
261     | head --lines=1\
262     | cut --delimiter=' ' --fields=8
263   )
264   local LFBFL_copyright_string
265   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
266   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
267   readonly LFBFL_copyright_string
268
269   LFBFL_files_without_license_header=""
270   grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
271     files_without_license_header\
272     --result-variable-prefix="LFBFL_"
273   declare -a LFBFL_arr_files_without_license_header
274   if [[ -n "${LFBFL_files_without_license_header}" ]]; then
275     mapfile -t LFBFL_arr_files_without_license_header\
276       <<< "${LFBFL_files_without_license_header}"
277   fi
278
279   prepare_filled_license_file_block(){
```

```
    # $1=license_file_path
    sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
        --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
        -- "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf "License template %s updated.\n" "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
        printf "License template %s created.\n" "$1"
      fi
    fi
  }

  prepare_filled_license_file_block_post(){
    # $1=license_file_path
    head --lines=-1 -- "$1"\
      | tail --lines=+2\
      > "$1.temp"
    # Strange world: tail --lines=-1 was working
    # to exclude first line.
    # I tested it months ago, and saw with my eyes the start and end
    # block comments removed with:
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["css"]="c"
    ["html"]="xml"
    ["php"]="c"
    ["py"]="py"
    ["js"]="c"
    ["ts"]="c"
    ["xml"]="xml"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_path2
  declare -i LFBFL_i_not_subfile
  declare -i LFBFL_i_not_subfile2
  local LFBFL_file_path
  local LFBFL_file_path2
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  declare -i LFBFL_i_is_whitelisted
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_path}"
```

```
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_path2="${LFBFL_license_prefix3}"
      LFBFL_license_file_path2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_path2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    prepare_filled_license_file_block_post "${LFBFL_license_file_path}"
    if [[ -n "${LFBFL_license2}" ]]; then
      prepare_filled_license_file_block_post "${LFBFL_license_file_path2}"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_i_is_whitelisted=0
      for LFBFL_file_path2 in\
        "${LFBFL_arr_files_without_license_header[@]}";
      do
        if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
          LFBFL_i_is_whitelisted=1
        fi
      done
      if [[ LFBFL_i_is_whitelisted -eq 1 ]]; then
        continue
      fi
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}.temp"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}.temp"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm -- "${LFBFL_license_file_path}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm -- "${LFBFL_license_file_path2}.temp"
    fi
  done

```

```
392   prepare_filled_license_file_line(){
393     # $1=LFBFL_license_file_path
394     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
395         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
396         -- "$1.tpl"\
397       > "$1.temp"
398     overwrite_if_not_equal "$1" "$1.temp"
399     declare -ir LFBFL_i_result=$?
400     if [[ LFBFL_i_verbose -eq 1 ]]; then
401       if [[ LFBFL_i_result -eq 1 ]]; then
402         printf "License template %s updated.\n" "$1"
403       elif [[ LFBFL_i_result -eq 2 ]]; then
404         printf "License template %s created.\n" "$1"
405       fi
406     fi
407   }
408
409   declare -Ar LFBFL_all_line_comment_languages=(
410     ["rlib"]="rs"
411     ["rs"]="rs"
412     ["sh"]="sh"
413     ["sql"]="sql"
414     ["tex"]="tex"
415     ["toml"]="sh"
416     ["yaml"]="sh"
417   )
418
419   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
420     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
421     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
422     prepare_filled_license_file_line "${LFBFL_license_file_path}"
423     if [[ -n "${LFBFL_license2}" ]]; then
424       LFBFL_license_file_path2="${LFBFL_license_prefix3}"
425       LFBFL_license_file_path2+=".${LFBFL_dest}"
426       prepare_filled_license_file_line "${LFBFL_license_file_path2}"
427     fi
428
429     LFBFL_s_files_paths=$(
430       find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
431         -printf '%P\n'\
432       | relevant_find
433     )
434     if [[ -z "${LFBFL_s_files_paths}" ]]; then
435       if [[ LFBFL_i_verbose -eq 1 ]]; then
436         LFBFL_s_format="build_licenses_templates: "
437         LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
438         # shellcheck disable=SC2059
439         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
440       fi
441       continue
442     fi
443     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
444     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
445       LFBFL_i_is_whitelisted=0
446       for LFBFL_file_path2 in\
447         "${LFBFL_arr_files_without_license_header[@]}";
```

```
      do
        if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
          LFBFL_i_is_whitelisted=1
        fi
      done
      if [[ LFBFL_i_is_whitelisted -eq 1 ]]; then
        continue
      fi
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
This file is part of @repository_name@.

@repository_name@ is free software:
you can redistribute it and/or modify it
under the terms of the GNU Affero General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

@repository_name@ is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Affero General Public License for more details.

You should have received a copy of
the GNU Affero General Public License
along with @repository_name@.
If not, see <https://www.gnu.org/licenses/>.

©Copyright @copyright_string@
```

3.20 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

```
This file is part of @repository_name@.

@repository_name@ is free software:
you can redistribute it and/or modify it
```

```
under the terms of the GNU General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

@repository_name@ is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU General Public License for more details.

You should have received a copy of
the GNU General Public License
along with @repository_name@.
If not, see <https://www.gnu.org/licenses/>.

©Copyright @copyright_string@
```

3.21 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```
This file is part of @repository_name@.

@repository_name@ is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

@repository_name@ is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with @repository_name@.
If not, see <https://www.gnu.org/licenses/>.

©Copyright @copyright_string@
```

3.22 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
```

```
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "all_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | tr '\n' '\0'\
      | xargs -0 grep --invert-match --perl-regexp --with-filename\
        -- 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
```

```
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  # below and another lines, see SC2145 false positive
  all_code_lines "${LFBFL_common_options[@]}" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
```

```
124   pushd_to_work_directory --trap-popd
125   can_continue_after_enhanced_pushd || return 1
126
127   enhanced_set_shell_option pipefail --trap-unset
128
129   declare -a LFBFL_common_options=("$@")
130   keep_options LFBFL_common_options --verbose
131
132   all_self_code_lines "${LFBFL_common_options[@]}" \
133     | not_empty_lines_after_file_name
134 }
135
136 code_lines_count_all(){
137   # Options:
138   #   --verbose
139   #   --work-directory=""
140   declare -i LFBFL_i_verbose=0
141   get_verbose_option "$@"
142   local LFBFL_work_directory=""
143   get_work_directory_option "$@"
144   declare -a LFBFL_return_traps_stack
145   init_return_trap
146   pushd_to_work_directory --trap-popd
147   can_continue_after_enhanced_pushd || return 1
148
149   enhanced_set_shell_option pipefail --trap-unset
150
151   declare -gi i_code_lines_count_all_result
152   i_code_lines_count_all_result=$(
153     all_self_code_lines\
154     | wc --lines
155   )
156   # The printf is for command line use.
157   # A script should use the result value instead.
158   # shellcheck disable=SC2248
159   printf "%s\n" ${i_code_lines_count_all_result}
160 }
161
162 code_lines_count_empty(){
163   # Options:
164   #   --verbose
165   #   --work-directory=""
166   declare -i LFBFL_i_verbose=0
167   get_verbose_option "$@"
168   local LFBFL_work_directory=""
169   get_work_directory_option "$@"
170   declare -a LFBFL_return_traps_stack
171   init_return_trap
172   pushd_to_work_directory --trap-popd
173   can_continue_after_enhanced_pushd || return 1
174
175   enhanced_set_shell_option pipefail --trap-unset
176
177   declare -gi i_code_lines_count_empty_result
178   # shellcheck disable=SC2119
179   i_code_lines_count_empty_result=$(
```

```
    all_self_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.23 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
```

```
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
```

```
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat -- "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent -- "${LFBFL_temp}" "${!#}"; then
    rm -- "${LFBFL_temp}"
```

```
    return
  fi
  mv -- "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below,
  #   you don't need to supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-starts-with-fixed-string"
    ]]; then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-ends-with-fixed-string"
    ]]; then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    LFBFL_grep_options+=("${LFBFL_argument}")
    if [[
      "${LFBFL_argument}" == "-q"
      || "${LFBFL_argument}" == "--quiet"
    ]]; then
      LFBFL_i_quiet=1
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_argument}")
  done

  if [[
    LFBFL_i_enforce_line_starts -eq 0
    && LFBFL_i_enforce_line_ends -eq 0
  ]]; then
    grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
```

```
188       return
189     fi
190     declare -r LFBFL_s_lines=$(
191       grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
192         "$2" "$1"
193     )
194     declare -i LFBFL_i_result=1
195     if [[ -z "${LFBFL_s_lines}" ]]; then
196       # shellcheck disable=SC2248
197       return ${LFBFL_i_result}
198     fi
199     declare -a LFBFL_arr_lines
200     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
201     readonly LFBFL_arr_lines
202
203     declare -i LFBFL_i_fixed_string_length=${#2}
204     local LFBFL_some_line
205     local LFBFL_line_part
206     for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
207       if [[ "${LFBFL_some_line}" == "$2" ]]; then
208         LFBFL_i_result=0
209         if [[ LFBFL_i_quiet -eq 1 ]]; then
210           break
211         fi
212         printf "%s\n" "${LFBFL_some_line}"
213         continue
214       fi
215       if [[
216         LFBFL_i_enforce_line_starts -eq 1
217         && LFBFL_i_enforce_line_ends -eq 1
218       ]]; then
219         continue
220       fi
221       if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
222         LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
223       fi
224       if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
225         LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
226       fi
227       if [[ "${LFBFL_line_part}" == "$2" ]]; then
228         LFBFL_i_result=0
229         if [[ LFBFL_i_quiet -eq 1 ]]; then
230           break
231         fi
232         printf "%s\n" "${LFBFL_some_line}"
233       fi
234     done
235     # shellcheck disable=SC2248
236     return ${LFBFL_i_result}
237   }
238
239   grep_variable(){
240     # $1=file_path
241     # $2=variable_name
242     # Options :
243     #   --replace-line-returns-by=""
```

```
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  # printf "%s\n" "${LFBFL_regexp}"
  local LFBFL_variable_value
  local LFBFL_s_replace
  get_some_option LFBFL_s_replace --replace-line-returns-by '' $'\n' 1 "$@"
  if [[ "${LFBFL_s_replace}" != $'\n' ]]; then
    # sed -Ez -e...
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"\
      | sed --null-data --regexp-extended\
        --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"
    )
  fi
  local LFBFL_prefix=""
  get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n" "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}

```

```
not_spampty_lines(){
  grep '^[^ ]'
  # grep --extended-regexp --invert-match '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep --invert-match '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep --invert-match '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep --invert-match '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep --extended-regexp '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}
```

```
356
357 not_all_space_lines_after_file_name(){
358   grep --invert-match '^[^:]\+:[ ]\+$'
359 }
360
361 aspampty_lines_after_file_name(){
362   grep '^[^:]\+:[ ]*$'
363 }
364
365 not_aspampty_lines_after_file_name(){
366   grep --invert-match '^[^:]\+:[ ]*$'
367 }
368
369 # If you need to test the previous 20 functions,
370 # look at file spampty_data.txt in miscellaneous.
371 # And give a try at something like:
372 # cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
373 # -E = --show-ends :). Nice command :).
374
375 not_JS_dependencies_find(){
376   grep --extended-regexp --invert-match\
377     --regexp="(^|/)node_modules/"\
378     --regexp="(^|/)package-lock\.json$"
379 }
380
381 not_JS_dependencies_grep(){
382   grep --extended-regexp --invert-match\
383     --regexp="^([^:]+/)?node_modules/"\
384     --regexp="^([^:]+/)?package-lock\.json:"
385 }
386
387 not_dependencies_find(){
388   not_JS_dependencies_find
389 }
390
391 not_dependencies_grep(){
392   not_JS_dependencies_grep
393 }
394
395 not_php_cache_find(){
396   grep --extended-regexp --invert-match\
397     --regexp="(^|/)\.phpdoc/cache/"
398 }
399
400 not_php_cache_grep(){
401   grep --extended-regexp --invert-match\
402     --regexp="^([^:]+/)?\.phpdoc/cache/"
403 }
404
405 not_python_cache_find(){
406   grep --extended-regexp --invert-match\
407     --regexp="(^|/)__pycache__/"\
408     --regexp="(^|/)\.mypy_cache/"\
409     --regexp="(^|/)\.ruff_cache/"
410 }
411
```

```
not_python_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?__pycache__/"\
    --regexp="^([^:]+/)?\.mypy_cache/"\
    --regexp="^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_php_cache_find\
    | not_python_cache_find
}

not_cache_grep(){
  not_php_cache_grep\
    | not_python_cache_grep
}

not_git_find(){
  grep --extended-regexp --invert-match "(^|/)\.git/"
}

not_git_grep(){
  grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

```

```
not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)build_and_checks_variables/temp/"
}

not_temp_file_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

not_unchecked_builds_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)\.phpdoc/build/"
}

not_unchecked_builds_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?\.phpdoc/build/"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
```

```
    | not_archive_find\
    | not_temp_file_find\
    | not_unchecked_builds_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep\
    | not_unchecked_builds_grep
}
```

3.24 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed\
    --expression='s/\xc2\xa0/ /g'\
```

```
    --expression='s/\xe2\x80\x80/ /g'\
    --expression='s/\xe2\x80\x81/ /g'\
    --expression='s/\xe2\x80\x82/ /g'\
    --expression='s/\xe2\x80\x83/ /g'\
    --expression='s/\xe2\x80\x84/ /g'\
    --expression='s/\xe2\x80\x85/ /g'\
    --expression='s/\xe2\x80\x86/ /g'\
    --expression='s/\xe2\x80\x87/ /g'\
    --expression='s/\xe2\x80\x88/ /g'\
    --expression='s/\xe2\x80\x89/ /g'\
    --expression='s/\xe2\x80\x8a/ /g'\
    --expression='s/\xe2\x80\x8b/ /g'\
    --expression='s/\xe2\x80\x8c/ /g'\
    --expression='s/\xe2\x80\x8d/ /g'\
    --expression='s/\xe2\x80\x8e/ /g'\
    --expression='s/\xe2\x80\x8f/ /g'\
    --expression='s/\xe2\x80\xa8/ /g'\
    --expression='s/\xe2\x80\xa9/ /g'\
    --expression='s/\xe2\x80\xaa/ /g'\
    --expression='s/\xe2\x80\xab/ /g'\
    --expression='s/\xe2\x80\xac/ /g'\
    --expression='s/\xe2\x80\xad/ /g'\
    --expression='s/\xe2\x80\xae/ /g'\
    --expression='s/\xe2\x80\xaf/ /g'\
    --expression='s/\xe2\x81\x9f/ /g'\
    --expression='s/\xe2\x81\xa0/ /g'\
    --expression='s/\xe2\x81\xa1/ /g'\
    --expression='s/\xe2\x81\xa2/ /g'\
    --expression='s/\xe2\x81\xa3/ /g'\
    --expression='s/\xe2\x81\xa4/ /g'\
    --expression='s/\xe2\x81\xa5/ /g'\
    --expression='s/\xe2\x81\xa6/ /g'\
    --expression='s/\xe2\x81\xa7/ /g'\
    --expression='s/\xe2\x81\xa8/ /g'\
    --expression='s/\xe2\x81\xa9/ /g'\
    --expression='s/\xe2\x81\xaa/ /g'\
    --expression='s/\xe2\x81\xab/ /g'\
    --expression='s/\xe2\x81\xac/ /g'\
    --expression='s/\xe2\x81\xad/ /g'\
    --expression='s/\xe2\x81\xae/ /g'\
    --expression='s/\xe2\x81\xaf/ /g'
}
```

3.25 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
```

```
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # Options:
  #   --keep-temp : keep the temporary file
  #   --quiet : we even don't want the message of diff saying that
  #             files are distinct
  #   --tree-mode : when the two files are the output of tree command,
  #   we ignore if all the differences are only on directories timestamps.
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  declare -i LFBFL_i_keep_temp
  get_some_flag LFBFL_i_keep_temp --keep-temp 1 "$@"
  declare -i LFBFL_i_quiet
  get_some_flag LFBFL_i_quiet --quiet 1 "$@"
  declare -i LFBFL_i_tree_mode
  get_some_flag LFBFL_i_tree_mode --tree-mode 1 "$@"

  if [[ ! -f "$1" ]]; then
    if [[ LFBFL_i_keep_temp -eq 1 ]]; then
      cp --preserve=mode,ownership,timestamps -- "$2" "$1"
    else
      mv -- "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_i_is_equal
  if [[ LFBFL_i_tree_mode -eq 1 ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff -- "$1" "$2"\
      | grep --extended-regexp "^(>|<)"\
      | grep --invert-match "/$"
    LFBFL_i_is_equal=${PIPESTATUS[2]}
    if [[ LFBFL_i_is_equal -eq 1 ]]; then
      grep --only-matching --regexp="[^ ]*/$" -- "$1"\
        > "overwrite_if_not_equal.file1.temp"
```

```
      grep --only-matching --regexp="[^ ]*/$" -- "$2"\
        > "overwrite_if_not_equal.file2.temp"
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        diff overwrite_if_not_equal.file1.temp\
          overwrite_if_not_equal.file2.temp\
          > /dev/null
      else
        diff overwrite_if_not_equal.file1.temp\
          overwrite_if_not_equal.file2.temp
      fi
      LFBFL_i_is_equal=1-$?
      rm overwrite_if_not_equal.file1.temp\
        overwrite_if_not_equal.file2.temp
    fi
  else
    if [[ LFBFL_i_quiet -eq 1 ]]; then
      diff --brief -- "$1" "$2" > /dev/null
    else
      diff --brief -- "$1" "$2"
    fi
    LFBFL_i_is_equal=1-$?
  fi
  if [[ LFBFL_i_is_equal -eq 1 ]]; then
    if [[ LFBFL_i_keep_temp -eq 0 ]]; then
      rm -- "$2"
    fi
    return 0
  fi
  if [[ LFBFL_i_keep_temp -eq 1 ]]; then
    cp --preserve=mode,ownership,timestamps -- "$2" "$1"
  else
    mv -- "$2" "$1"
  fi
  return 1
}
```

3.26 build_and_checks_dependencies/phpmd_baseline.xslt

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
```

```
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
    version="1.1">
  <xsl:output method="xml"/>
  <xsl:template match="phpmd-baseline">
    <xsl:copy>
      <xsl:text>
</xsl:text>
      <xsl:apply-templates select="node()">
        <xsl:sort select="@file"/>
        <xsl:sort select="@rule"/>
        <xsl:sort select="@method"/>
      </xsl:apply-templates>
    </xsl:copy>
  </xsl:template>
  <xsl:template match="violation">
    <xsl:text>  </xsl:text>
    <xsl:copy>
      <xsl:copy-of select="@*"/>
    </xsl:copy>
    <xsl:text>
</xsl:text>
  </xsl:template>
  <!--
  Remove all indentation and line returns in text nodes,
  because otherwise it will be output first after sorting above.
  -->
  <xsl:template match="text()"></xsl:template>
</xsl:stylesheet>
```

3.27 build_and_checks_dependencies/python_black_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
```

```
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep --multiline --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
    "${LFBFL_common_options[@]}"
}
```

3.28 build_and_checks_dependencies/python_isort_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # This function should be useless in a few months:
  # https://github.com/PyCQA/isort/issues/2250
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
      "Checking that import of _collections_abc is at the right place.\n"

  enhanced_set_shell_option pipefail --trap-unset

```

```
  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.py" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_collections_abc_place: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    sed --regexp-extended --null-data\
      --expression='s/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      -- "${LFBFL_file_path}"\
      > "${LFBFL_file_path}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_path}"\
      "${LFBFL_file_path}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

3.29 build_and_checks_dependencies/repository_data.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
```

```
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}

grep_variable_with_multiple_files(){
  # $1=data_file_path
  # $2=variable_name
  # Options :
```

```
  #   --result-variable-prefix="LFBFL_" for example
  local LFBFL_prefix=""
  get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
  grep_variable "$1" "$2" --result-variable-prefix="${LFBFL_prefix}"
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  declare -a LFBFL_arr
  if [[ -z "${!LFBFL_variable_name}" ]]; then
    return
  fi
  mapfile -t LFBFL_arr <<< "${!LFBFL_variable_name}"
  local LFBFL_s=""
  local LFBFL_file_path
  local LFBFL_previous_file_path
  for LFBFL_file_path in "${LFBFL_arr[@]}"; do
    if [[
      -n "${LFBFL_previous_file_path}"
      && "${LFBFL_previous_file_path}" != */
    ]]; then
      LFBFL_s+=$'\n'
    fi
    LFBFL_s+="${LFBFL_file_path}"
    LFBFL_previous_file_path="${LFBFL_file_path}"
  done
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_s}"
}
```

3.30 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

```

```
check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion.\n"
  # shellcheck disable=SC2059
  [[ LFBFL_i_verbose -eq 1 ]] && printf "${LFBFL_s}"

  pcre2grep --multiline -- $'\\$\\{#(|.*_)arr_(?!.*\\[@\\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
  get_max_line_length_option "$@"
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];[^\n]{0,${LFBFL_i_max_previous_line_length}})"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"
  # pcre2grep --multiline --
  # '^[^\n]{1,70}(?<=\]\];[^\n]{0,70})'
  # '(?<!^\s{0,70}\]\];)\n\s*then'
```

```
  pcre2grep --multiline -- "${LFBFL_regexp}" **/*.sh
}

check_no_negation_before_bash_test(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking that shell scripts do not contain ! before [[.\n"

  pcre2grep --multiline -- $'!\\s*\\[\\[' **/*.sh
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose --max-line-length

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
  check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
}
```

3.31 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
```

```
11 either version 3 of the License,
12 or (at your option) any later version.
13
14 DevOrSysAdminScripts is distributed
15 in the hope that it will be useful,
16 but WITHOUT ANY WARRANTY;
17 without even the implied warranty of
18 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
19 See the GNU Lesser General Public License for more details.
20
21 You should have received a copy of
22 the GNU Lesser General Public License
23 along with DevOrSysAdminScripts.
24 If not, see <https://www.gnu.org/licenses/>.
25
26 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
27 2026-01-04T13:38+01:00: This file was renamed from
28 "split_line_at_most.exec.php"
29 to
30 "split_score.exec.php".
31 This file was renamed from
32 "call_split_score_after_before_or_before_after.exec.php" to
33 "split_line_at_most.exec.php".
34 This file was renamed from
35 "call_split_score_after_before_or_before_after.php" to
36 "call_split_score_after_before_or_before_after.exec.php".
37
38 @category Executable
39 @package DevOrSysAdminScripts
40 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
41 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
42 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
43 */
44
45 declare(strict_types=1);
46 declare(encoding='UTF-8');
47
48 require_once './build_and_checks_dependencies/split_score.libr.php';
49
50 if(PHP_SAPI !== 'cli'){
51   fwrite(
52     STDERR,
53     "split_score.exec.php must be run in a shell.\n",
54   );
55   die(1);
56 }
57
58 if(count($argv) !== 7){
59   fwrite(
60     STDERR,
61     "split_score.exec.php must be given exactly 6 arguments.\n",
62   );
63   die(1);
64 }
65
66 /*
```

```
php -r\
$'require_once \'./build_and_checks_dependencies/split_score.libr.php\';'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'

php -r \
$'require_once \'./build_and_checks_dependencies/split_score.libr.php\';'\
$'echo(\SplitScore\generate_split_score(true, 70, [\'/\'])'\
$'(\'/\', 60, true));'

php -r \
$'require_once \'./build_and_checks_dependencies/split_score.libr.php\';'\
$'$o_split_score_closure ='\
$'  new \SplitScore\SplitScoreClosure(true, 70, [\'/\']);'\
$'echo($o_split_score_closure(\'/\', 60, true));'
*/

/*
// The first cast "(int) $func..." is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1],
    (int) $argv[2],
    explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
//*/
// AHAHAH PHPCS...
/* AHAHAH PHPCS XD */

$o_split_score_closure = new \SplitScore\SplitScoreClosure(
  (bool) $argv[1],
  (int) $argv[2],
  explode(',', $argv[3]),
);
echo($o_split_score_closure($argv[4], (int) $argv[5], (bool) $argv[6]));

?>
```

3.32 build_and_checks_dependencies/split_score.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
```

```
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');



namespace SplitScore;

/**
Given the 3 following arguments, this function returns a closure that can
be used to compute the split score given the current string
(hopefully matching a delimiter), the current cut/split position,
and if that split position is after or before the current string.

@param bool          $b_larger_after
                     Score is larger when splitting after,
                     previous name of this parameter was $b_after_before.
@param int           $i_max_length
                     The maximum length allowed (goal length).
@param array<string> $arr_delimiter_strings_domain
                     An array of "delimiters".

@return \Closure {
  The "split_score" closure to use repeatedly afterward.

  @CLparam string $s_delimiter_string The string tested to be a delimiter.
  @CLparam int    $i_cut_position     The offset of the cut in the string.
  @CLparam bool   $b_is_cut_after     Is the offset taken after the string?

  @CLreturn int The split score.
}
*/
function generate_split_score(
```

```
72    bool $b_larger_after,
73    int $i_max_length,
74    array $arr_delimiter_strings_domain,
75  ) : \Closure {
76    return function (
77      string $s_delimiter_string,
78      int $i_cut_position,
79      bool $b_is_cut_after,
80    ) use (
81      $b_larger_after,
82      $i_max_length,
83      $arr_delimiter_strings_domain,
84    ) : int {
85      if(
86        !in_array(
87          $s_delimiter_string,
88          $arr_delimiter_strings_domain,
89          true,
90        )
91      ){
92        // I do not handle regexps for the moment.
93        // I'm too brainfucked. I just want to finish this feature fast
94        // for HTML and Tex/PDF listings generation with HTML and Tex on
95        // at most 70 characters.
96        return 0;
97      }
98      if($b_is_cut_after === $b_larger_after){
99        // When $b_larger_after,
100       // always way better to cut after '/' than before it.
101       // When !$b_larger_after (aka larger_before),
102       // always way better to cut before '/' than after it.
103       return 1 + $i_max_length + $i_cut_position;
104     }
105     return 1 + $i_cut_position;
106   };
107 }//end generate_split_score()
108
109
110
111
112 /**
113 Given the type system of PHP, a better way to have a closure computing
114 the split score .
115
116 @category Library
117 @package DevOrSysAdminScripts
118 @class SplitScoreClosure
119 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
120 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
121 */
122 final class SplitScoreClosure {
123   /**
124   Score is larger when splitting after.
125
126   @var bool $b_larger_after
127   */
```

```
128     private bool $b_larger_after;
129
130     /**
131     The maximum length allowed (goal length).
132
133     @var int $i_max_length
134     */
135     private int $i_max_length;
136
137     /**
138     An array of "delimiters".
139
140     @var array<string> $arr_delimiter_strings_domain
141     */
142     private array $arr_delimiter_strings_domain;
143
144
145
146     /**
147     Given the 3 following arguments,
148     this constructor creates an object that will behave like a Closure
149     that can be used to compute the split score given the current string
150     (hopefully matching a delimiter), the current cut/split position,
151     and if that split position is after or before the current string.
152
153     @param bool          $b_larger_after
154                          Score is larger when splitting after,
155                          previous name of this parameter was $b_after_before.
156     @param int           $i_max_length
157                          The maximum length allowed (goal length).
158     @param array<string> $arr_delimiter_strings_domain
159                          An array of "delimiters".
160
161     @return void
162     */
163     public function __construct(
164       bool $b_larger_after,
165       int $i_max_length,
166       array $arr_delimiter_strings_domain,
167     ) {
168       $this->b_larger_after = $b_larger_after;
169       $this->i_max_length = $i_max_length;
170       $this->arr_delimiter_strings_domain = $arr_delimiter_strings_domain;
171     }//end __construct() SplitScoreClosure
172
173
174
175     /**
176     The "split_score" closure to use repeatedly afterward.
177
178     @param string $s_delimiter_string The string tested to be a delimiter.
179     @param int    $i_cut_position     The offset of the cut in the string.
180     @param bool   $b_is_cut_after     Is the offset taken after the string?
181
182     @return int The split score.
183     */
```

```
  public function __invoke(
    string $s_delimiter_string,
    int $i_cut_position,
    bool $b_is_cut_after,
  ) : int {
    if(
      !in_array(
        $s_delimiter_string,
        $this->arr_delimiter_strings_domain,
        true,
      )
    ){
      return 0;
    }
    if($b_is_cut_after === $this->b_larger_after){
      // When $b_larger_after,
      // always way better to cut after '/' than before it.
      // When !$b_larger_after (aka larger_before),
      // always way better to cut before '/' than after it.
      return 1 + $this->i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  }//end __invoke() SplitScoreClosure
}//end class SplitScoreClosure

?>
```

3.33 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
```

```
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#-------------------------------------------------------------------------
#-------------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#-------------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score for the result.
#    But if you look at split_line_at_most code,
#    you will see a micro-optimization where distinguishing < and <= could
#    have been interesting.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
```

```
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
```

```
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 - 1 and RP7 : p1 = p2 + 1, since it implies
#   some equalities between matched delimiters.
```

```
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that's not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case is not interesting (alone,
#     full analysis could combine it with some other case,
#     but it seems gratis combinatorial explosion without insight):
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1) that contains it. The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
```

```
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
```

```
309 #   whilst RS= is false.
310 #   Thus properties with RS<= can trivially be extended with
311 #   RMM-"allow-md1-0", not useful to explicit this for the moment.
312 #   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
313 #   nothing more.
314 #   Thus we can focus on: md1 and md2 are not NULL,
315 #   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
316 #   could be extended to a property RMM=or!=/RMMNotNULL.
317 #   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
318 #   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
319 #   or FALSE. Wrong ! Not pleasant ! XD
320 #   So for RIA, we keep either: empty RIAe (RIA8),
321 #   ia1 = ia2 RIA= (RIA0 or RIA4),
322 #   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
323 #   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
324 #   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
325 #   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
326 #   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
327 #   even less pleasant XD : 11 possibilities.
328 #   In total, we have now 1 (RS=) times 5 times 2 times 11
329 #   + 1 (RS<=) times 8 times 2 times 11 = 286
330 #   combinations (life is even shorter ;) :) ).
331 #   We only listed below properties of the third form.
332 # i) By the previous remark, there is a good chance that we missed some
333 #   possible property.
334 #   We hope we did found the properties that had a simple meaning,
335 #   in particular related to the algorithm where either you
336 #   loop on the string forward and keep last position with maximum score,
337 #   or you loop backward and abort as soon as possible as soon you know
338 #   that you found some position at the end that must have maximum score.
339 # j) These properties are "boolean" flags (integers that are powers of 2).
340 #   We first put the exact integers,
341 #   then wondered about switching everything to exponentiation 2**i,
342 #   then we took the solution that minimizes the character length between
343 #   [1-9][0-9]* and 2**i, then reverted to initial before committing.
344 #   Most efficient interpreted code is what we did first,
345 #   if there is no opcode cache for constants.
346 #   Most safe code is when the language hides the details and handles
347 #   families of flags,
348 #   but beware if some flag goes into the database.
349 #   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
350 #   or $((2<<0))...
351 #   What I would love to see in many languages is a flag type, as a subtype
352 #   of integers, flag32, flag64 for C for example.
353 #   I think there is no such thing because sooner or later you play with
354 #   pseudo-flags that are combination of flags, hence integers.
355 #   But just for the declaration to let the type checker check that the
356 #   integer is a power of 2: flag32 MY_FLAG = 4;
357 #   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
358 #   Another fact is that practice makes perfect, so working with the powers
359 #   of 2 written out in base 10 gets you accustomed to them.
360 #   Safe code should be favored since (pre-)compilation optimizes it,
361 #   but for this project, I prefer to keep the integers in decimal and
362 #   check that it doubles when reading my listing, because I'm familiar
363 #   with that and the powers of 2 doesn't go beyond what I can check fast
364 #   enough in my brain.
```

```
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#------------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
```

```
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

```

```
# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-------------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-------------------------------------------------------------------------
#-------------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
```

```
533       | SSP_DELIMITER_UNIFORM
534       | SSP_EQUAL_AFTER_OR_BEFORE
535       | SSP_LARGER_AFTER
536       | SSP_LARGER_BEFORE
537       | SSP_POSITION_NOT_DECREASING
538       | SSP_POSITION_NOT_DECREASING_AFTER
539       | SSP_POSITION_NOT_DECREASING_BEFORE
540       | SSP_POSITION_UNIFORM
541       | SSP_POSITION_UNIFORM_AFTER
542       | SSP_POSITION_UNIFORM_BEFORE
543       | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
544       | SSP_SAME_DELIMITER_LARGER_AFTER
545       | SSP_SAME_DELIMITER_LARGER_BEFORE
546       | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
547       | SSP_SAME_POSITION_LARGER_AFTER
548       | SSP_SAME_POSITION_LARGER_BEFORE
549       | SSP_UNIFORM
550     ))
551     until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
552       LFBFL_i_result_old=$((LFBFL_i_result))
553       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
554         LFBFL_i_result=$((
555           LFBFL_i_result
556           | SSP_LARGER_AFTER
557           | SSP_SAME_DELIMITER_LARGER_AFTER
558           | SSP_SAME_POSITION_LARGER_AFTER
559         ))
560       fi
561       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
562         LFBFL_i_result=$((
563           LFBFL_i_result
564           | SSP_LARGER_BEFORE
565           | SSP_SAME_DELIMITER_LARGER_BEFORE
566           | SSP_SAME_POSITION_LARGER_BEFORE
567         ))
568       fi
569       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
570         LFBFL_i_result=$((
571           LFBFL_i_result
572           | SSP_POSITION_NOT_DECREASING_AFTER
573           | SSP_POSITION_NOT_DECREASING_BEFORE
574         ))
575       fi
576       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
577         LFBFL_i_result=$((
578           LFBFL_i_result
579           | SSP_POSITION_UNIFORM_AFTER
580           | SSP_POSITION_UNIFORM_BEFORE
581           | SSP_POSITION_NOT_DECREASING
582           | SSP_POSITION_NOT_DECREASING_AFTER
583           | SSP_POSITION_NOT_DECREASING_BEFORE
584         ))
585       fi
586       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
587         LFBFL_i_result=$((
588           LFBFL_i_result
```

```
589         | SSP_POSITION_NOT_DECREASING_AFTER
590       ))
591     fi
592     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
593       LFBFL_i_result=$((
594         LFBFL_i_result
595         | SSP_POSITION_NOT_DECREASING_BEFORE
596       ))
597     fi
598     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
599       LFBFL_i_result=$((
600         LFBFL_i_result
601         | SSP_EQUAL_AFTER_OR_BEFORE
602         | SSP_LARGER_AFTER
603         | SSP_LARGER_BEFORE
604         | SSP_SAME_DELIMITER_LARGER_AFTER
605         | SSP_SAME_DELIMITER_LARGER_BEFORE
606       ))
607     fi
608     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
609       LFBFL_i_result=$((
610         LFBFL_i_result
611         | SSP_LARGER_AFTER
612       ))
613     fi
614     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
615       LFBFL_i_result=$((
616         LFBFL_i_result
617         | SSP_LARGER_BEFORE
618       ))
619     fi
620     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
621       LFBFL_i_result=$((
622         LFBFL_i_result
623         | SSP_DELIMITER_UNIFORM
624         | SSP_EQUAL_AFTER_OR_BEFORE
625         | SSP_LARGER_AFTER
626         | SSP_LARGER_BEFORE
627         | SSP_SAME_POSITION_LARGER_AFTER
628         | SSP_SAME_POSITION_LARGER_BEFORE
629       ))
630     fi
631     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
632       LFBFL_i_result=$((
633         LFBFL_i_result
634         | SSP_LARGER_AFTER
635       ))
636     fi
637     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
638       LFBFL_i_result=$((
639         LFBFL_i_result
640         | SSP_LARGER_BEFORE
641       ))
642     fi
643     if ((LFBFL_i_result & SSP_UNIFORM)); then
644       LFBFL_i_result=$((LFBFL_i_all_flags))
```

```
      break
    fi
    if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
    && ((
      LFBFL_i_result & (
        SSP_ALWAYS_LARGER_BEFORE
        | SSP_EQUAL_AFTER_OR_BEFORE
        | SSP_LARGER_BEFORE
        | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_DELIMITER_LARGER_BEFORE
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_POSITION_LARGER_BEFORE
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
    && ((
      LFBFL_i_result & (
        SSP_EQUAL_AFTER_OR_BEFORE
        | SSP_LARGER_AFTER
        | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_DELIMITER_LARGER_AFTER
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_POSITION_LARGER_AFTER
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
```

```
701       fi
702       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
703       && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
704         LFBFL_i_result=$((
705           LFBFL_i_result
706           | SSP_ALWAYS_LARGER_AFTER
707         ))
708       fi
709       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
710       && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
711         LFBFL_i_result=$((
712           LFBFL_i_result
713           | SSP_ALWAYS_LARGER_BEFORE
714         ))
715       fi
716       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
717       && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
718         LFBFL_i_result=$((
719           LFBFL_i_result
720           | SSP_EQUAL_AFTER_OR_BEFORE
721         ))
722       fi
723       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
724       && ((
725         LFBFL_i_result & (
726           SSP_POSITION_UNIFORM
727           | SSP_POSITION_UNIFORM_AFTER
728         )
729       )); then
730         LFBFL_i_result=$((
731           LFBFL_i_result
732           | SSP_SAME_DELIMITER_LARGER_AFTER
733         ))
734       fi
735       if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
736       && ((
737         LFBFL_i_result & (
738           SSP_POSITION_UNIFORM
739           | SSP_POSITION_UNIFORM_BEFORE
740         )
741       )); then
742         LFBFL_i_result=$((
743           LFBFL_i_result
744           | SSP_SAME_DELIMITER_LARGER_BEFORE
745         ))
746       fi
747       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
748       && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
749         LFBFL_i_result=$((
750           LFBFL_i_result
751           | SSP_POSITION_NOT_DECREASING
752         ))
753       fi
754       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
755       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
756         LFBFL_i_result=$((
```

```
        LFBFL_i_result
        | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
    && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_POSITION_UNIFORM
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
    && ((
      LFBFL_i_result & (
        SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_POSITION_LARGER_AFTER
        | SSP_SAME_POSITION_LARGER_BEFORE
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
```

```
    && ((
      LFBFL_i_result & (
        SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_POSITION_LARGER_BEFORE
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
    && ((
      LFBFL_i_result & (
        SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        |  SSP_SAME_POSITION_LARGER_AFTER
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
  done
  declare -gi i_split_score_properties_logical_closure_result=$((
    LFBFL_i_result
  ))
}

split_score_simple(){
  # $1=larger_after (The name was "after_before" previously,
  #                  the meaning was not modified.
  #                  Remark not repeated below.)
  # $2=delimiters_strings_domain concatenated characters/delimiters
  # $3=delimiter_string a single character
  # $4=is_cut_after
  declare -gi i_split_score_result
  if [[ "$2" != *$3* ]]; then
    i_split_score_result=0
```

```
869       return
870     fi
871     if [[ "$4" == "$1" ]]; then
872       i_split_score_result=2
873       return
874     fi
875     i_split_score_result=1
876   }
877
878   get_split_score_simple(){
879     # $1=larger_after
880     # $2=max_length
881     # $3=delimiters_strings_domain
882     declare -g get_split_score_result="split_score_simple $1 '$3'"
883     declare -gi i_get_split_score_result2
884     # i_get_split_score_result2=$((11-$1*4))
885     # if [[ $1 -eq 1 ]]; then
886     #   i_get_split_score_result2=7
887     # else
888     #   i_get_split_score_result2=11
889     # fi
890     i_get_split_score_result2=$((
891       SSP_DELIMITER_UNIFORM
892       + SSP_POSITION_UNIFORM
893       + SSP_LARGER_BEFORE
894       + (
895         $1 * (
896           SSP_LARGER_AFTER
897           - SSP_LARGER_BEFORE
898         )
899       )
900     ))
901     # shellcheck disable=SC2248
902     split_score_properties_logical_closure ${i_get_split_score_result2}
903     i_get_split_score_result2=$((
904       i_split_score_properties_logical_closure_result
905     ))
906   }
907
908   get_split_score_exec(){
909     declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
910     SPLIT_SCORE_EXEC+="/split_score.exec.php"
911   }
912
913   split_score(){
914     # $1=larger_after
915     # $2=max_length
916     # $3=delimiters_strings_domain concatenated characters/delimiters
917     # $4=delimiter_string a single character
918     # $5=cut_position
919     # $6=is_cut_after
920     get_split_score_exec
921     declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
922     declare -gi i_split_score_result
923     i_split_score_result=$(eval -- "${LFBFL_command}")
924   }
```

```
925
926 # This function is currently not called,
927 # hence we need this shellcheck disable.
928 # shellcheck disable=SC2034
929 get_split_score(){
930   # $1=larger_after
931   # $1=max_length
932   # $2=delimiters_strings_domain
933   declare -g get_split_score_result="split_score $1 $2 '$3'"
934   declare -gi i_get_split_score_result2
935   # i_get_split_score_result2=$((9-$1*4))
936   i_get_split_score_result2=$((
937     SSP_DELIMITER_UNIFORM
938     + SSP_LARGER_BEFORE
939     + (
940       $1 * (
941         SSP_LARGER_AFTER
942         - SSP_LARGER_BEFORE
943       )
944     )
945   ))
946   # shellcheck disable=SC2248
947   split_score_properties_logical_closure ${i_get_split_score_result2}
948   i_get_split_score_result2=$((
949     i_split_score_properties_logical_closure_result
950   ))
951 }
952
953 split_line_at_most(){
954   # $1=line
955   # $2=max_length of beginning result string
956   # $3=split_score_command
957   #   negative score means do not consider as a possible split
958   # $4=split_score_command_properties :
959   #   These properties are flags.
960   #   If someday some property is not a flag,
961   #   then create split_score_command_properties2.
962   #   Old values to get a simple idea without reading everything at the
963   #   beginning of file:
964   #   - null or 0 no property;
965   #   - 1 split score doesn't depend on which delimiter matches;
966   #   - 2 split score doesn't depend on the current position;
967   #   - 4 any split score after is always larger
968   #     than any split score before;
969   #   - 8 any split score before is always larger
970   #     than any split score after;
971   # $5=forbidden_previous_character
972   # $6=split_score_command_delimiter_strings_domain
973   #   optimisation if small domain
974   #   - this argument can be "null", then we will use $3 only instead,
975   #     in that case, only mono-character strings are considered,
976   #   - this argument can be an array of characters,
977   #     optimize accordingly,
978   #   - this argument can be an array of strings,
979   #     optimize accordingly,
980   #   - this argument can be an array of regexps,
```

```
    #     optimize accordingly.
    #     Not sure an array of regexps makes more sense than a single
    #     one. Maybe they could be both cases with corresponding
    #     optimizations. Same for a single character or substring.
    #
    # Note that we're always looking for the last position to split
    # among the positions with maximum score;
    # if some position gets a score from 2 delimiters,
    # its score is always the maximum of the two corresponding scores.
    declare -g split_line_at_most_result_start
    declare -g split_line_at_most_result_end
    declare -A LFBFL_positions
    # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
    # For my use case in Bash scripts, I will need only an array of
    # characters. See get_split_score().
    if [[ -n "$6" ]]; then
      printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
    fi
    if [[ $2 -ge ${#1} ]]; then
      split_line_at_most_result_start=$1
      split_line_at_most_result_end=""
      return
    fi
    # We know that ${#1} > $2, hence ${#1} - 1 >= $2
    # Edge cases:
    # There is always at least one character that is overlength,
    # but we need to look only at the score before that character that is
    # overlength, the score after that character is invalid,
    # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
    # That first character that is overlength has index $2.
    # The cut position after that character has index $2 + 1.
    # Looping from character position going from 0 to $2, we need to drop
    # score before character 0 and drop score after character $2.
    # We do this with forbidden score -1.
    declare -ir LFBFL_i_max=$(($2))
    declare -i LFBFL_i
    declare -i LFBFL_j
    local LFBFL_current_char

    if [[ $2 -lt 2 ]]; then
      printf "split_line_at_most: max_length should be at least 2...\n"
      return
    fi
    if [[ -n "$5" && "$1" =~ ^"$5"*$ ]]; then
      for ((LFBFL_i = 2; LFBFL_i <= $2; LFBFL_i <<= 1)); do
        :
      done
      ((LFBFL_i >>= 1))
      split_line_at_most_result_start="${1:0:LFBFL_i}"
      split_line_at_most_result_end="${1:LFBFL_i}"
      return
    fi

    # At the beginning, each position has score 0.
    LFBFL_positions["0"]="-1"
    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
```

```
1037      LFBFL_positions["${LFBFL_i}"]="0"
1038    done
1039    LFBFL_positions["${LFBFL_i}"]="-1"
1040
1041    # If we have a forbidden previous character,
1042    # we mark the forbidden positions with -1.
1043    if [[ -n "$5" ]]; then
1044      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1045        LFBFL_current_char="${1:${LFBFL_i}:1}"
1046        ((++LFBFL_i))
1047        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1048          LFBFL_positions["${LFBFL_i}"]="-1"
1049        fi
1050      done
1051    fi
1052
1053    declare -r LFBFL_sort_command="sort --numeric-sort"
1054
1055    update_position_score(){
1056      # $1=position
1057      # Returns 0 if updated, 1 otherwise
1058      # printf "%s|%s\n" "${i_split_score_result}" "$1"
1059      if [[ i_split_score_result -ge 1 ]]; then
1060        # printf "valid result"
1061        if [[ ${LFBFL_positions["$1"]} != "-1" ]]; then
1062          LFBFL_positions["$1"]=$(
1063            max "${LFBFL_sort_command}"\
1064              "${LFBFL_positions["$1"]}"\
1065              "${i_split_score_result}"
1066          )
1067          return 0
1068        fi
1069      fi
1070      return 1
1071    }
1072
1073    local LFBFL_split_score_command=$3
1074    declare -i LFBFL_split_score_command_properties=$4
1075
1076    normal_or_position_uniform_call(){
1077      # $1=cut_position
1078      # $2=is_cut_after
1079      if ((LFBFL_split_score_command_properties & SSP_POSITION_UNIFORM));
1080      then
1081        eval --\
1082          "${LFBFL_split_score_command} '${LFBFL_current_char}' $2"
1083      else
1084        eval --\
1085          "${LFBFL_split_score_command} '${LFBFL_current_char}' $1 $2"
1086      fi
1087    }
1088
1089    if (($4 & SSP_DELIMITER_UNIFORM))\
1090    && (($4 & SSP_LARGER_AFTER))\
1091    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1092      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
```

```
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      LFBFL_j=$((LFBFL_i + 1))
      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_j} 1
      # shellcheck disable=SC2248
      if update_position_score ${LFBFL_j}
      then
        break
      fi

      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_i} 0
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
    done
  elif (($4 & SSP_DELIMITER_UNIFORM))\
  && (($4 & SSP_LARGER_BEFORE))\
  && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      # We still need to break in second if, after computing score after,
      # because SSP_LARGER_BEFORE implies large inequality instead of
      # strict inequality.
      # And since the algorithm is to take the largest position with
      # maximum score, there is a possibility that we'll break on a found
      # score before but for splitting on a score after.
      LFBFL_j=$((LFBFL_i + 1))
      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_j} 1
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_j}

      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_i} 0
      # shellcheck disable=SC2248
      if update_position_score ${LFBFL_i}
      then
        break
      fi
    done
  else
    for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
      ((++LFBFL_i))
      eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
    done
  fi
  declare -ir LFBFL_i_max_score=$(
    max 'sort --numeric-sort' "${LFBFL_positions[@]}"
  )
  local LFBFL_value
```

```
  declare -i LFBFL_i_best_position
  for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
    if [[ LFBFL_i -eq 0 ]]; then
      printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
    fi
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
      LFBFL_i_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_i_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=new_lines
  # $2=prefix
  # $3=max_length
  # $4=suffix
  # $5=split_score_command
  # $6=split_score_command_properties
  # $7=forbidden_previous_character ('\' usually)
  # Options:
  #   --verbose
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap

```

```
  enhanced_set_shell_option pipefail --trap-unset

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_i_length2=$(($3 - ${#4}))
  get_overlength_regexp "$3"

  # Testing if some line is too long.
  printf "%s" "$1"\
    | grep --quiet "${get_overlength_regexp_result}"
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    printf "%s" "$1"\
    | head --lines=-1
  )
  # printf "start: %s\n" "${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    printf "%s" "$1"\
    | tail --lines=1
  )
  # printf "last_line: %s\n" "${LFBFL_last_line}"
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    # shellcheck disable=SC2248
    split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
```

```
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.34 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  # $1=length
```

```
  # Option:
  #   --length-is-overlength
  declare -i LFBFL_i_overlength=$1
  declare -i LFBFL_i_length_is_overlength
  get_some_flag LFBFL_i_length_is_overlength --length-is-overlength 1 "$@"
  if [[ LFBFL_i_length_is_overlength -eq 0 ]]; then
    ((++LFBFL_i_overlength))
  fi
  readonly LFBFL_i_overlength
  declare -g get_overlength_regexp_result=".\{${LFBFL_i_overlength}\}"
  # declare -g get_overlength_regexp_result="^.\{${LFBFL_i_overlength}\}"
  # declare -g get_overlength_regexp_result=".\{${LFBFL_i_overlength}\}\$"
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
  get_max_line_length_option "$@"

  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_max_line_length}

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  local LFBFL_long_line
  local LFBFL_file_path
  local LFBFL_line
  local LFBFL_extension
  local LFBFL_base_name
  local LFBFL_s_long_lines
  declare -a LFBFL_arr_long_lines
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "too_long_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
```

```
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | tr '\n' '\0'\
      | xargs -0 grep --line-number --with-filename\
        --regexp="${get_overlength_regexp_result}" --
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "too_long_code_lines: No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      # Remove file_path and line_number
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.sub"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      elif [[ "${LFBFL_extension}" == "neon" ]]; then
        if [[ ! -f "${LFBFL_base_name}.neon.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

3.35 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

cd .. || exit
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./build_and_checks_dependencies/overwrite_if_not_equal.libr.sh"
cd build_and_checks_dependencies || exit

update_common_build_and_checks(){
  # Option:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp -- "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_files_names=(
    "build_dependencies_notes.exec.php"
    "build_dependencies_notes.libr.php"
    "build_file_from_printable_file__standard.exec.sh"
    "build_md_from_printable_md.exec.sh"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spelling_check.libr.sh"
```

```
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "overwrite_if_not_equal.libr.sh"
    "phpmd_baseline.xslt"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "shell_checks_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut --delimiter=' ' --fields=1
    )
    LFBFL_file_sha512_all="LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_name}")
    sed --in-place\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut --delimiter=' ' --fields=1
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
```

```
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed --in-place\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.36 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
```

```
    #   --verbose
    #   --work-directory=""
    declare -i LFBFL_i_verbose=0
    get_verbose_option "$@"
    local LFBFL_work_directory=""
    get_work_directory_option "$@"
    declare -a LFBFL_return_traps_stack
    init_return_trap
    pushd_to_work_directory --trap-popd
    can_continue_after_enhanced_pushd || return 1

    declare -i LFBFL_i_write
    get_some_flag LFBFL_i_write --write 1 "$@"
    declare -i LFBFL_i_append
    get_some_flag LFBFL_i_append --append 1 "$@"

    enhanced_set_shell_option pipefail --trap-unset

    LFBFL_subdir2="build_and_checks_variables"
    declare -r LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
    declare -r LFBFL_data_file_path="./${LFBFL_subdir2}/repository_data.txt"

    declare -i LFBFL_i_max_line_length
    grep_variable "${LFBFL_data_file_path}" max_line_length\
      --result-variable-prefix="LFBFL_i_"

    if [[ LFBFL_i_write -eq 1 ]]; then
      : > "${LFBFL_listing}"
    fi
    # Remove line returns that are here to keep lines short.
    sed --regexp-extended --null-data --expression='s/(.)\\\n/\1/Mg'\
      -- "${LFBFL_listing}"\
      > "${LFBFL_listing}.temp"
    # shopt -s dotglob was needed at some point but I don't see why now.
    # shellcheck disable=SC2248
    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
    declare -r LFBFL_split_score_command="${get_split_score_result}"
    declare -i LFBFL_i_split_score_command_properties
    LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
    readonly LFBFL_i_split_score_command_properties
    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
    readonly LFBFL_suffix
    # There is always a $'\n' final suffix from >>.
    # But when origin string ends with the suffix we must add something.
    declare -r LFBFL_final_suffix=$'\n\\'

    local LFBFL_s_files_paths
    LFBFL_s_files_paths=$(
      find . -type f -printf '%P\n'\
      | relevant_find\
      | sort
    )
    # printf "%s\n" "${LFBFL_s_files_paths}"
    declare -a LFBFL_arr_files_paths
    if [[ -n "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
      printf\
        "update_or_check_files_names_listing: Some files to process.\n"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  # printf "%s\n" "${LFBFL_arr_files_paths[@]}"

  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    # if [[ LFBFL_i_verbose -eq 1 ]]; then
    #   printf\
    #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
    #     "${LFBFL_file_path}"
    # fi
    git check-ignore --quiet -- "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_i_write -eq 1 ]]; then
      # shellcheck disable=SC2001
      printf "%s\n" "${repeated_split_last_line_result}"\
        >> "${LFBFL_listing}"
      continue
    fi
    if ! grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      printf "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
      if [[ LFBFL_i_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done
```

```
156
157   LFBFL_s_files_paths=$(cat -- "${LFBFL_listing}.temp")
158   if [[ -n "${LFBFL_s_files_paths}" ]]; then
159     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
160   fi
161
162   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
163     [[ "${LFBFL_file_path}" != '//'* ]] || continue
164     if [[ ! -f "${LFBFL_file_path}" ]]; then
165       printf "The non-file %s is listed in %s.\n"\
166         "${LFBFL_file_path}"\
167         "${LFBFL_listing}"
168     fi
169   done
170
171   rm -- "${LFBFL_listing}.temp"
172
173   if [[ LFBFL_i_verbose -eq 1 ]]; then
174     cat -- "${LFBFL_listing}"
175   fi
176 }
177
178 update_or_check_files_names_listing "$@"
```

3.37 build_and_checks_variables/.gitignore

```
1 temp/
2 upgrade_venvs_ts
```

3.38 build_and_checks_variables/bandit.ini

```
1 # FILE: .bandit
2 [bandit]
```

3.39 build_and_checks_variables/bandit_baseline.json

```
1 {"results":[]}
```

3.40 build_and_checks_variables/black.toml

```
1 [tool.black]
2 line-length = 75
```

3.41 build_and_checks_variables/dependencies_data.json

```
1  [
2    { "dependency_name": "Bash", "URLs": [
3        "https://www.gnu.org/software/bash/",
4        "https://tiswww.case.edu/php/chet/bash/bashtop.html",
5        "https://savannah.gnu.org/projects/bash/"
6    ] },
7    { "dependency_name": "GNU coreutils", "URLs": [
8        "https://www.gnu.org/software/coreutils/",
9        "https://savannah.gnu.org/projects/coreutils/"
10   ] },
11   { "dependency_name": "uutils coreutils", "URLs": [
12       "https://uutils.github.io/coreutils/",
```

```
13          "https://github.com/uutils/coreutils"
14      ] },
15      { "dependency_name": "diff (GNU diffutils)", "URLs": [
16          "https://www.gnu.org/software/diffutils/",
17          "https://savannah.gnu.org/projects/diffutils/"
18      ] },
19      { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
20          "https://www.gnu.org/software/findutils/",
21          "https://savannah.gnu.org/projects/findutils/"
22      ] },
23      { "dependency_name": "grep (GNU grep)", "URLs": [
24          "https://www.gnu.org/software/grep/",
25          "https://savannah.gnu.org/projects/grep/"
26      ] },
27      { "dependency_name": "sed (GNU sed)", "URLs": [
28          "https://www.gnu.org/software/sed/",
29          "https://savannah.gnu.org/projects/sed/"
30      ] },
31      { "dependency_name": "ShellCheck", "URLs": [
32          "https://www.shellcheck.net/",
33          "https://github.com/koalaman/shellcheck"
34      ] },
35      { "dependency_name": "GNU Wget", "URLs": [
36          "https://www.gnu.org/software/wget/",
37          "https://savannah.gnu.org/projects/wget/"
38      ] },
39      { "dependency_name": "more of util-linux", "URLs": [
40          "https://www.kernel.org/pub/linux/utils/util-linux/",
41          "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42          {
43            "URL": "https://github.com/util-linux/util-linux",
44            "comment": "official?"
45          }
46      ] },
47      { "dependency_name": "pandoc", "URLs": [
48          "https://pandoc.org/",
49          "https://github.com/jgm/pandoc"
50      ] },
51      { "dependency_name": "pdftex/pdflatex", "URLs": [
52          "https://www.tug.org/applications/pdftex/",
53          "https://tug.org/texlive/svn/"
54      ] },
55      { "dependency_name": "tree", "URLs": [
56          "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57          "https://gitlab.com/OldManProgrammer/unix-tree",
58          "https://github.com/Old-Man-Programmer/tree"
59      ] },
60      { "dependency_name": "git", "URLs": [
61          "https://git-scm.com/about",
62          "https://github.com/git/git"
63      ] },
64      { "dependency_name": "pylint", "URLs": [
65          "https://www.pylint.org/",
66          "https://pylint.readthedocs.io/en/stable/",
67          "https://github.com/pylint-dev/pylint",
68          "https://pypi.org/project/pylint/"
```

```
69      ] },
70      { "dependency_name": "notify-send", "URLs": [
71          "https://gitlab.gnome.org/GNOME/libnotify/",
72          "https://gnome.pages.gitlab.gnome.org/libnotify/"
73      ] },
74      { "dependency_name": "black", "URLs": [
75          "https://github.com/psf/black",
76          "https://black.readthedocs.io/en/stable/",
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
91          "https://phpdoc.org/",
92          "https://docs.phpdoc.org/",
93          "https://github.com/phpdocumentor/phpdocumentor"
94      ] },
95      { "dependency_name": "pip", "URLs": [
96          "https://pip.pypa.io/en/stable/",
97          "https://github.com/pypa/pip",
98          {
99            "URL": "https://pypi.org/project/pip/",
100           "comment": "(egg and chicken ;) )"
101         }
102     ] },
103     { "dependency_name": "ruff", "URLs": [
104         "https://docs.astral.sh/ruff/",
105         "https://github.com/astral-sh/ruff",
106         "https://pypi.org/project/ruff/"
107     ] },
108     { "dependency_name": "xmllint (libxml2)", "URLs": [
109         "https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home",
110         "https://gitlab.gnome.org/GNOME/libxml2"
111     ] },
112     { "dependency_name": "bandit", "URLs": [
113         "https://bandit.readthedocs.io/en/latest/",
114         "https://github.com/PyCQA/bandit",
115         "https://pypi.org/project/bandit/"
116     ] },
117     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
118         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
119         "https://packagist.org/packages/squizlabs/php_codesniffer"
120     ] },
121     { "dependency_name": "phpmd", "URLs": [
122         "https://phpmd.org/",
123         "https://github.com/phpmd/phpmd",
124         "https://packagist.org/packages/phpmd/phpmd"
```

```
  ] },
  { "dependency_name": "php", "URLs": [
      "https://www.php.net/",
      "https://github.com/php/php-src"
  ] },
  { "dependency_name": "tex", "URLs": [
      "https://ctan.org/tex-archive/systems/knuth/dist/tex"
  ] },
  { "dependency_name": "pcregrep, pcre2grep", "URLs": [
      "https://www.pcre.org/",
      "https://pcre2project.github.io/pcre2/",
      "https://github.com/PCRE2Project/pcre1",
      "https://github.com/PCRE2Project/pcre2",
      "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
  ] },
  { "dependency_name": "python3", "URLs": [
      "https://www.python.org/",
      "https://docs.python.org/3/"
  ] }
]
```

3.42 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
```

```
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/
https://docs.phpdoc.org/
https://github.com/phpdocumentor/phpdocumentor

```

```
pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

xmllint (libxml2) can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home
https://gitlab.gnome.org/GNOME/libxml2

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer
https://packagist.org/packages/squizlabs/php_codesniffer

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.43 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
```

```
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
```

```
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
```

```
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files trees</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files trees</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
```

```
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, phpdoc, pip, ruff, xmllint;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpcs, phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
```

```
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.44 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it
% under the terms of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed
% in the hope that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\if\buildtarget0% A4 paper
\documentclass{article}
\usepackage[a4paper,vmargin=20mm,left=27mm,right=27mm,nohead]{geometry}
\else\if\buildtarget1% US Letter paper
\documentclass{article}
\usepackage[letterpaper,vmargin=20mm,left=27mm,right=27mm,nohead]{geometry}
\else\if\buildtarget2% Book with US Letter paper
\documentclass[twoside]{article}
\usepackage[letterpaper,vmargin=20mm,outer=24mm,inner=24mm,nohead,
bindingoffset=8mm]{geometry}
\fi
\fi
\fi

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
```

```
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

Buildtarget: \buildtarget
\if\buildtarget0
A4
\else\if\buildtarget1% US Letter paper
USLetter
\else\if\buildtarget2% Book with US Letter paper
Book
```

```
\fi
\fi
\fi

\tableofcontents
\newpage

\section{Files trees}
\label{section:trees}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, phpdoc, pip, ruff, xmllint;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpcs, phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
```

```
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.45 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
archive_libre_office_documents.exec.php
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_dependencies_notes.libr.php
build_and_checks_dependencies/\
build_file_from_printable_file__standard.exec.sh
```

```
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/phpmd_baseline.xslt
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpcs_ruleset.xml
build_and_checks_variables/phpmd_baseline.xml
// build_and_checks_variables/phpstan_baseline.neon
build_and_checks_variables/phpstan_baseline.neon.tpl
// build_and_checks_variables/phpstan_config.neon
build_and_checks_variables/phpstan_config.neon.tpl
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
```

```
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
files.libr.php
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.exec.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/grep_benchmark.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
unicode_exceptions.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.46 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
```

```
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

3.47 build_and_checks_variables/phpcs_ruleset.xml

```
<?xml version="1.0"?>
<ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
  <description>A custom coding standard</description>
  <config name="tab_width" value="2"/>

  <file>../</file>

  <arg name="extensions" value="php"/>
  <arg name="report" value="summary"/>
  <arg name="colors"/>
  <arg value="sp"/>

  <!--
  I added all the standards, then excluded all the rules that gave me
  false positives or that seemed unappropriate when looking at the
  documentation.
  Some are bugs, some are conflicting with my own coding style.
  You may delete an exclude and see what pops up to maybe understand
  my choices.
  -->

  <rule ref="Generic">
    <!-- You have to allow long or short or both syntaxes. -->
    <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
    <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
    <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
    <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
    <!--
    Not standard in Unix/Linux world to keep files not ending with \n.
    This is always an option of the code editors.
    Since PHP doesn't output the final \n after ?>, it is ok.
    -->
    <exclude name="Generic.Files.EndFileNoNewline.Found"/>
    <exclude name="Generic.Files.ExecutableFile.Executable"/>
    <exclude name="Generic.Files.OneClassPerFile.MultipleFound"/>
```

```
    <exclude name="Generic.Files.OneObjectStructurePerFile.MultipleFound"/>
    <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
    <exclude
     name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
    />
    <exclude
      name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
    />
    <exclude
   name="Generic.NamingConventions.CamelCapsFunctionName.ScopeNotCamelCaps"
    />
    <!-- Applies only to true, false and null. -->
    <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
    <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
    <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
    <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
  </rule>
  <rule ref="PEAR">
    <exclude name="PEAR.Classes.ClassDeclaration.OpenBraceNewLine"/>
    <exclude name="PEAR.Commenting.ClassComment.MissingLinkTag"/>
    <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
    <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
    <!--
    Look at the comment before generate_split_score.
    PEAR and Squiz require to align the parameter names parts in the same
    comment, and also the parameter comments parts in the same comment,
    but with a long parameter name like $a_delimiter_strings_domain
    and a line length limit at 75 characters, parameter comments parts
    have only 24 characters left per line, which is ridiculous.
    Our choice to put the parameter comments parts starting on another line
    with same "indent" as parameter names parts is way better in those
    cases. Alignment has always been a trap with short lines.
    Since phpDocumentor handles the parameter comments parts correctly,
    it is enough for me.
    Worse is the handling of "nested doc block comment" for the Closure
    returned by generate_split_score().
    Our solution makes at least the source code readable.
    Maybe one day the dev tools will handle it correctly.
    -->
    <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
    <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
    <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
    <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
    <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
    <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
    <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
    <exclude
      name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
    />
    <exclude
     name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
    />
    <exclude
      name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
    />
    <!--
```

```
94      Don't laugh when looking at the false positives of the two exclude
95      below ;) XD.
96      They don't check for multiline function declarations apparently.
97      -->
98      <exclude
99        name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
100     />
101     <exclude
102       name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
103     />
104     <exclude
105       name="PEAR.Functions.FunctionDeclaration.SpaceBeforeOpenBrace"
106     />
107     <exclude
108       name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
109     />
110     <exclude
111       name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
112     />
113     <exclude
114       name="PEAR.NamingConventions.ValidFunctionName.ScopeNotCamelCaps"
115     />
116     <!--
117     You have to choose between the following exclude or another one
118     for PSR2.
119     -->
120     <exclude
121       name="PEAR.NamingConventions.ValidVariableName.PrivateNoUnderscore"
122     />
123   </rule>
124   <rule ref="PSR1">
125     <exclude name="PSR1.Classes.ClassDeclaration.MultipleClasses"/>
126     <exclude
127       name="PSR1.Methods.CamelCapsMethodName.NotCamelCaps"
128     />
129   </rule>
130   <rule ref="PSR2">
131     <exclude name="PSR2.Classes.ClassDeclaration.OpenBraceNewLine"/>
132     <!-- Back to my habits at Teliae with always a closing tag. -->
133     <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
134     <exclude
135 name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
136     />
137   </rule>
138   <rule ref="PSR12">
139     <exclude
140  name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
141     />
142     <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
143     <!--
144     3 false positives, see:
145     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
146     -->
147     <exclude
148       name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
149     />
```

```
150       <exclude
151         name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
152       />
153       <exclude
154         name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
155       />
156       <!-- and even a fourth and fifth false positives -->
157       <exclude
158         name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
159       />
160       <exclude
161         name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
162       />
163       <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
164       <exclude
165         name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
166       />
167       <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
168       <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
169       <!-- I agree except for string concatenation. -->
170       <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
171       <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
172     </rule>
173     <rule ref="Squiz">
174       <exclude name="Squiz.Arrays.ArrayBracketSpacing.SpaceAfterBracket"/>
175       <exclude name="Squiz.Arrays.ArrayBracketSpacing.SpaceBeforeBracket"/>
176       <exclude name="Squiz.Arrays.ArrayDeclaration.DoubleArrowNotAligned"/>
177       <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
178       <exclude name="Squiz.Classes.ClassDeclaration.MultipleClasses"/>
179       <exclude
180         name="Squiz.Classes.ClassDeclaration.NewlinesAfterCloseBrace"
181       />
182       <exclude name="Squiz.Classes.ClassDeclaration.OpenBraceNewLine"/>
183       <exclude name="Squiz.Classes.ClassDeclaration.SpaceBeforeCloseBrace"/>
184       <exclude name="Squiz.Classes.ClassFileName.NoMatch"/>
185       <exclude name="Squiz.Classes.ValidClassName.NotPascalCase"/>
186       <!--
187       It doesn't know the //*/ trick...
188       https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
189       -->
190       <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
191       <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
192       <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
193       <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
194       <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
195       <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
196       <exclude name="Squiz.Commenting.BlockComment.SingleLine"/>
197       <!-- Problem for doc block comment before namespace -->
198       <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
199       <exclude name="Squiz.Commenting.ClassComment.TagNotAllowed"/>
200       <exclude name="Squiz.Commenting.ClosingDeclarationComment.Incorrect"/>
201       <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
202       <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
203       <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
204       <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
205       <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
```

```
206     <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
207     <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
208     <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
209     <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
210     <exclude name="Squiz.Commenting.FunctionComment.EmptyThrows"/>
211     <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
212     <!-- asks for boolean instead of bool which is incorrect -->
213     <exclude
214       name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
215     />
216     <exclude name="Squiz.Commenting.FunctionComment.IncorrectTypeHint"/>
217     <!-- asks for boolean instead of bool which is incorrect -->
218     <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
219     <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
220     <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
221     <exclude
222       name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
223     />
224     <exclude
225       name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
226     />
227     <exclude name="Squiz.Commenting.FunctionCommentThrowTag.WrongNumber"/>
228     <!-- Problem for doc block comment before namespace -->
229     <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
230     <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
231     <!-- It doesn't know the //*/ trick... -->
232     <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
233     <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
234     <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
235     <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
236     <!-- asks for boolean instead of bool which is incorrect -->
237     <exclude name="Squiz.Commenting.VariableComment.IncorrectVarType"/>
238     <exclude
239       name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
240     />
241     <exclude
242   name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
243     />
244     <exclude
245       name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
246     />
247     <exclude
248       name="Squiz.ControlStructures.ForEachLoopDeclaration.SpacingBeforeAs"
249     />
250     <exclude
251       name="Squiz.ControlStructures.InlineIfDeclaration.NotSingleLine"
252     />
253     <exclude name="Squiz.Files.FileExtension.ClassFound"/>
254     <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
255     <exclude name="Squiz.Functions.GlobalFunction.Found"/>
256     <exclude
257       name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
258     />
259     <exclude
260   name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
261     />
```

```
262       <exclude
263      name="Squiz.Functions.MultiLineFunctionDeclaration.SpaceBeforeOpenBrace"
264       />
265       <exclude
266         name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
267       />
268       <exclude
269         name="Squiz.NamingConventions.ValidFunctionName.ScopeNotCamelCaps"
270       />
271       <exclude
272         name="Squiz.NamingConventions.ValidVariableName.MemberNotCamelCaps"
273       />
274       <exclude
275         name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
276       />
277       <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
278       <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
279       <exclude name="Squiz.PHP.CommentedOutCode"/>
280       <exclude name="Squiz.PHP.DisallowBooleanStatement.Found"/>
281       <exclude
282         name="Squiz.PHP.DisallowComparisonAssignment.AssignedComparison"
283       />
284       <exclude name="Squiz.PHP.DisallowInlineIf.Found"/>
285       <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
286       <exclude
287         name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
288       />
289       <exclude name="Squiz.WhiteSpace.FunctionSpacing"/>
290       <exclude name="Squiz.WhiteSpace.MemberVarSpacing.FirstIncorrect"/>
291       <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
292       <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
293       <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
294
295     </rule>
296     <rule ref="Zend">
297       <!-- Back to my habits at Teliae with always a closing tag. -->
298       <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
299       <exclude
300         name="Zend.NamingConventions.ValidVariableName.MemberVarNotCamelCaps"
301       />
302       <exclude
303         name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
304       />
305       <exclude
306         name="Zend.NamingConventions.ValidVariableName.PrivateNoUnderscore"
307       />
308     </rule>
309
310     <rule ref="Generic.Arrays.ArrayIndent">
311       <properties>
312         <property name="indent" value="2" />
313       </properties>
314     </rule>
315     <rule ref="Generic.Files.LineLength">
316       <properties>
317         <property name="lineLimit" value="75"/>
```

```
      <property name="absoluteLineLimit" value="75"/>
    </properties>
  </rule>
  <rule ref="Generic.Formatting.SpaceAfterNot">
    <properties>
      <property name="spacing" value="0" />
    </properties>
  </rule>
  <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
    <properties>
      <property name="checkFunctions" value="true"/>
      <property name="checkClosures" value="true"/>
    </properties>
  </rule>
  <rule ref="Generic.Metrics.CyclomaticComplexity">
    <properties>
      <property name="complexity" value="40" />
      <property name="absoluteComplexity" value="80" />
    </properties>
  </rule>
  <rule ref="Generic.Metrics.NestingLevel">
    <properties>
      <property name="nestingLevel" value="7" />
      <property name="absoluteNestingLevel" value="10" />
    </properties>
  </rule>
  <rule ref="Generic.Strings.UnnecessaryStringConcat">
    <properties>
      <property name="allowMultiline" value="true" />
    </properties>
  </rule>
  <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
    <properties>
      <property name="ignoreNewlines" value="true" />
    </properties>
  </rule>
  <rule ref="Generic.WhiteSpace.ScopeIndent">
    <properties>
      <property name="ignoreIndentationTokens" type="array">
        <element value="T_COMMENT"/>
        <element value="T_DOC_COMMENT_OPEN_TAG"/>
      </property>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="PEAR.Functions.FunctionCallSignature">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
  <rule ref="PEAR.Functions.FunctionDeclaration">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
```

```
  <rule ref="PEAR.WhiteSpace.ScopeIndent">
    <properties>
      <property name="ignoreIndentationTokens" type="array">
        <element value="T_COMMENT"/>
        <element value="T_DOC_COMMENT_OPEN_TAG"/>
      </property>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="PSR2.Methods.FunctionCallSignature">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="Squiz.Classes.ClassDeclaration">
      <properties>
          <property name="indent" value="2" />
      </properties>
  </rule>
  <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
  <rule ref="Squiz.Strings.ConcatenationSpacing">
    <properties>
      <property name="ignoreNewlines" value="true" />
    </properties>
  </rule>
  <rule ref="Squiz.WhiteSpace.FunctionSpacing">
    <properties>
      <property name="spacing" value="3" />
      <property name="spacingBeforeFirst" value="0" />
      <property name="spacingAfterLast" value="0" />
    </properties>
  </rule>
</ruleset>
```

3.48 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation
    rule="PHPMD\Rule\CleanCode\ElseExpression"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
```

```
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CyclomaticComplexity"
      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\LongMethod"
      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\NpathComplexity"
      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Naming\LongVariable"
      file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="build_and_checks_dependencies/split_score.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="build_and_checks_dependencies/split_score.libr.php"
      method="__construct"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="build_and_checks_dependencies/split_score.libr.php"
      method="__invoke"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
      file="build_and_checks_dependencies/split_score.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="build_and_checks_dependencies/split_score.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Naming\LongVariable"
      file="build_and_checks_dependencies/split_score.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CleanCode\BooleanArgumentFlag"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CleanCode\ElseExpression"
      file="files.libr.php"
    />
    <violation
```

```
66      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
67      file="files.libr.php"
68      method="add_file_data"
69    />
70    <violation
71      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
72      file="files.libr.php"
73      method="add_file_data_for_file_path"
74    />
75    <violation
76      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
77      file="files.libr.php"
78      method="get_s_content"
79    />
80    <violation
81      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
82      file="files.libr.php"
83    />
84    <violation
85      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
86      file="files.libr.php"
87      method="__construct"
88    />
89    <violation
90      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
91      file="files.libr.php"
92      method="add_file_data"
93    />
94    <violation
95      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
96      file="files.libr.php"
97      method="add_file_data_for_file_path"
98    />
99    <violation
100     rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
101     file="files.libr.php"
102   />
103   <violation
104     rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
105     file="files.libr.php"
106   />
107   <violation
108     rule="PHPMD\Rule\CyclomaticComplexity"
109     file="files.libr.php"
110   />
111   <violation
112     rule="PHPMD\Rule\Design\LongMethod"
113     file="files.libr.php"
114   />
115   <violation
116     rule="PHPMD\Rule\Design\NpathComplexity"
117     file="files.libr.php"
118   />
119   <violation
120     rule="PHPMD\Rule\Naming\LongVariable"
121     file="files.libr.php"
```

```
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="string_escaping.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="string_escaping.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CleanCode\BooleanArgumentFlag"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CleanCode\ElseExpression"
      file="unicode.libr.php"
    />
    <!-- https://github.com/phpmd/phpmd/issues/910 -->
    <!-- <violation -->
    <!--   rule="PHPMD\Rule\CleanCode\MissingImport" -->
    <!--   file="unicode.libr.php" -->
    <!-- /> -->
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CyclomaticComplexity"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\LongMethod"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\LongParameterList"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\NpathComplexity"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Naming\LongVariable"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
      file="unicode_exceptions.libr.php"
      method="get_data"
    />
    <violation
```

```
    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode_exceptions.libr.php"
    method="__construct"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
    file="unicode_exceptions.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode_exceptions.libr.php"
  />
</phpmd-baseline>
```

3.49 build_and_checks_variables/phpstan_baseline.neon.tpl

```
parameters:
 ignoreErrors:
  # https://github.com/phpstan/phpstan/issues/14585
  -
   rawMessage: Variable $argv might not be defined.
   identifier: variable.undefined
   count: 1
   path: ../archive_libre_office_documents.exec.php

  -
   rawMessage: Variable $argv might not be defined.
   identifier: variable.undefined
   count: 1
   path: ../build_and_checks_dependencies/build_dependencies_notes.exec.php

  -
   rawMessage: Variable $argv might not be defined.
   identifier: variable.undefined
   count: 1
   path: ../build_and_checks_dependencies/split_score.exec.php

  -
   rawMessage: 'Comparison operation "<" between int<0, 1114111> \
               and 0 is always false.'
   identifier: smaller.alwaysFalse
   count: 1
   path: ../unicode.libr.php

  -
   rawMessage: 'Comparison operation ">" between int<-283305, 283305> \
               and 1114111 is always false.'
   identifier: greater.alwaysFalse
   count: 1
   path: ../unicode.libr.php

  -
   rawMessage: 'Parameter #1 $i_unicode_code_point of function \
               DOSAS_unicode\unicode_code_point_to_UTF8 expects \
               int<0, 1114111>, int<-1114095, 1114111> given.'
   identifier: argument.type
   count: 1
```

```
    path: ../unicode.libr.php

  -
    rawMessage: 'Parameter #1 $i_unicode_code_point of function \
                DOSAS_unicode\unicode_code_point_to_UTF8 expects \
                int<0, 1114111>, int<-283305, 283305> given.'
    identifier: argument.type
    count: 1
    path: ../unicode.libr.php

  -
    rawMessage: 'Parameter #2 $i_current_octet of function \
                DOSAS_unicode\get_message_and_data_array expects int, \
                int<0, 191>|int<194, 244>|int<246, 254>|null given.'
    identifier: argument.type
    count: 1
    path: ../unicode.libr.php
```

3.50 build_and_checks_variables/phpstan_config.neon.tpl

```
includes:
    - phpstan_baseline.neon
    - /home/laurent/.config/composer/vendor/\
      phpstan/phpstan/conf/bleedingEdge.neon

parameters:
    level: max

    paths:
        - ..

    checkTooWideReturnTypesInProtectedAndPublicMethods: true
```

3.51 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
```

```
[tool.pylint.reports]
output-format = "colorized"
```

3.52 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

You can build multiple targets of the source listing according to various
formats and sizes.
Specify a jobname suffix (that will become a filename suffix)
and an integer that will set a buildtarget macro, separated with ':'.
latex_targets=A4:0
latex_targets=USLetter:1
latex_targets=Book:2

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
```

Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- default behavior, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
it requires that all dependency constraints are declared;
- custom behavior, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.

```
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
```

Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

```
109
110 bandit_venv=../venv_bandit
111 black_venv=../venv_black
112 isort_venv=../venv_isort
113 mypy_venv=../venv_mypy
114 pylint_venv=../venv_pylint
115 ruff_venv=../venv_ruff
116
117
118 upgrade venvs can be auto or ask, any other value is interpreted
119 as ask.
120
121 upgrade_venvs=ask
122
123
124 time interval can be 0 to always upgrade or ask to upgrade,
125 or a number at most bash max int
126 https://superuser.com/questions/1030122/
127 what-is-the-maximum-value-of-a-numeric-bash-shell-variable
128 it should be enough to simulate never "upgrade or ask for upgrade"
129 unless you claim to live many billions years ;)
130 If it is wat, this is a random int XD (see RANDOM).
131 Note that the probability that it upgrades your dependencies is not
132 1/2, it increases over time, and is 1 after slightly
133 less than one day.
134 If it is watyouwant?, see SRANDOM XD.
135
136 upgrade_venvs_time_interval_in_seconds=wat
137
138
139 For the Nu HTML, CSS, and SVG checker.
140
141 vnu_jar_path=../jar_venv/vnu.jar
142
143 Not fully implemented yet, it will only warn that vnu jar was not
144 updated since some time.
145 "wat" works, but "watyouwant?" joke was not repeated ;).
146
147 upgrade_vnu_jar_time_interval_in_seconds=604800
148
149 use_vnu_for_xhtml=false
150
151 For xmllint_files_without_DTD and files_without_license_header,
152 each line is a file path, except if the line ends with a /,
153 then it is concatenated with the following line(s) to obtain a path.
154
155 xmllint_files_without_DTD=./build_and_checks_variables/phpcs_ruleset.xml
156 xmllint_files_without_DTD=./build_and_checks_variables/phpmd_baseline.xml
157
158 If you use Ubuntu/Debian, install w3c-sgml-lib instead of =true.
159 See https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
160
161 xmllint_use_downloaded_DTDs=false
162
163 files_without_license_header=build_and_checks_variables/phpcs_ruleset.xml
164 files_without_license_header=build_and_checks_variables/
```

```
files_without_license_header=phpmd_baseline.xml
files_without_license_header=README.html
files_without_license_header=miscellaneous/string_escaping.html
files_without_license_header=build_and_checks_variables/ruff.toml
files_without_license_header=build_and_checks_variables/black.toml
files_without_license_header=build_and_checks_variables/pylintrc.toml
```

3.53 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=

max_line_length=

latex_command=

latex_targets=
latex_targets=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=
```

```

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=

xmllint_files_without_DTD=
xmllint_files_without_DTD=

xmllint_use_downloaded_DTDs=

files_without_license_header=
files_without_license_header=
```

3.54 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  "PLR0917",  # too-many-positional-arguments AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.exec.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
```

```
]
```

3.55 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"
```

3.56 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2016 : Expressions don't expand in single quotes,
           use double quotes for that.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
           Usually because the format string is very long.
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
```

```
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
           https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2016$" -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$"\
  -e "=SC2120$" -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$"\
  -e "=SC2309$" -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.57 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
```

```
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir --parents "${LFBFL_subdir}/licenses_templates"
  local LFBFL_variables_directory
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  readonly LFBFL_variables_directory
  mkdir --parents -- "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  local LFBFL_script_download_URL
  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
  LFBFL_script_download_URL+="${LFBFL_common_file_name}"
  readonly LFBFL_script_download_URL
  declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='c090463990b393a10177e9d0db26c59d45711bfdb4e08'
```

```
  LFBFL_correct_sha512+='1c7b9da75a9d14275fe5edffe1a1cddd460b6d0c92a8'
  LFBFL_correct_sha512+='d6eb76450550797861badac33847d8b24d26c67'
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.58 COPYING

```
                    GNU GENERAL PUBLIC LICENSE
                       Version 3, 29 June 2007

 Copyright (C) 2007 Free Software Foundation, Inc. <https://fsf.org/>
 Everyone is permitted to copy and distribute verbatim copies
 of this license document, but changing it is not allowed.

                            Preamble

```

The GNU General Public License is a free, copyleft license for software and other kinds of works.

The licenses for most software and other practical works are designed to take away your freedom to share and change the works. By contrast, the GNU General Public License is intended to guarantee your freedom to share and change all versions of a program--to make sure it remains free software for all its users. We, the Free Software Foundation, use the GNU General Public License for most of our software; it applies also to any other work released this way by its authors. You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for them if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs, and that you know you can do these things.

To protect your rights, we need to prevent others from denying you these rights or asking you to surrender the rights. Therefore, you have certain responsibilities if you distribute copies of the software, or if you modify it: responsibilities to respect the freedom of others.

For example, if you distribute copies of such a program, whether gratis or for a fee, you must pass on to the recipients the same freedoms that you received. You must make sure that they, too, receive or can get the source code. And you must show them these terms so they know their rights.

Developers that use the GNU GPL protect your rights with two steps: (1) assert copyright on the software, and (2) offer you this License giving you legal permission to copy, distribute and/or modify it.

For the developers' and authors' protection, the GPL clearly explains that there is no warranty for this free software. For both users' and authors' sake, the GPL requires that modified versions be marked as changed, so that their problems will not be attributed erroneously to authors of previous versions.

Some devices are designed to deny users access to install or run modified versions of the software inside them, although the manufacturer can do so. This is fundamentally incompatible with the aim of protecting users' freedom to change the software. The systematic pattern of such abuse occurs in the area of products for individuals to use, which is precisely where it is most unacceptable. Therefore, we have designed this version of the GPL to prohibit the practice for those products. If such problems arise substantially in other domains, we stand ready to extend this provision to those domains in future versions of the GPL, as needed to protect the freedom of users.

Finally, every program is threatened constantly by software patents. States should not allow patents to restrict development and use of software on general-purpose computers, but in those that do, we wish to avoid the special danger that patents applied to a free program could make it effectively proprietary. To prevent this, the GPL assures that

patents cannot be used to render the program non-free.

  The precise terms and conditions for copying, distribution and
modification follow.

                       TERMS AND CONDITIONS

  0. Definitions.

  "This License" refers to version 3 of the GNU General Public License.

  "Copyright" also means copyright-like laws that apply to other kinds of
works, such as semiconductor masks.

  "The Program" refers to any copyrightable work licensed under this
License. Each licensee is addressed as "you". "Licensees" and
"recipients" may be individuals or organizations.

  To "modify" a work means to copy from or adapt all or part of the work
in a fashion requiring copyright permission, other than the making of an
exact copy. The resulting work is called a "modified version" of the
earlier work or a work "based on" the earlier work.

  A "covered work" means either the unmodified Program or a work based
on the Program.

  To "propagate" a work means to do anything with it that, without
permission, would make you directly or secondarily liable for
infringement under applicable copyright law, except executing it on a
computer or modifying a private copy. Propagation includes copying,
distribution (with or without modification), making available to the
public, and in some countries other activities as well.

  To "convey" a work means any kind of propagation that enables other
parties to make or receive copies. Mere interaction with a user through
a computer network, with no transfer of a copy, is not conveying.

  An interactive user interface displays "Appropriate Legal Notices"
to the extent that it includes a convenient and prominently visible
feature that (1) displays an appropriate copyright notice, and (2)
tells the user that there is no warranty for the work (except to the
extent that warranties are provided), that licensees may convey the
work under this License, and how to view a copy of this License. If
the interface presents a list of user commands or options, such as a
menu, a prominent item in the list meets this criterion.

  1. Source Code.

  The "source code" for a work means the preferred form of the work
for making modifications to it. "Object code" means any non-source
form of a work.

  A "Standard Interface" means an interface that either is an official
standard defined by a recognized standards body, or, in the case of
interfaces specified for a particular programming language, one that
is widely used among developers working in that language.

The "System Libraries" of an executable work include anything, other than the work as a whole, that (a) is included in the normal form of packaging a Major Component, but which is not part of that Major Component, and (b) serves only to enable use of the work with that Major Component, or to implement a Standard Interface for which an implementation is available to the public in source code form. A "Major Component", in this context, means a major essential component (kernel, window system, and so on) of the specific operating system (if any) on which the executable work runs, or a compiler used to produce the work, or an object code interpreter used to run it.

The "Corresponding Source" for a work in object code form means all the source code needed to generate, install, and (for an executable work) run the object code and to modify the work, including scripts to control those activities. However, it does not include the work's System Libraries, or general-purpose tools or generally available free programs which are used unmodified in performing those activities but which are not part of the work. For example, Corresponding Source includes interface definition files associated with source files for the work, and the source code for shared libraries and dynamically linked subprograms that the work is specifically designed to require, such as by intimate data communication or control flow between those subprograms and other parts of the work.

The Corresponding Source need not include anything that users can regenerate automatically from other parts of the Corresponding Source.

The Corresponding Source for a work in source code form is that same work.

2. Basic Permissions.

All rights granted under this License are granted for the term of copyright on the Program, and are irrevocable provided the stated conditions are met. This License explicitly affirms your unlimited permission to run the unmodified Program. The output from running a covered work is covered by this License only if the output, given its content, constitutes a covered work. This License acknowledges your rights of fair use or other equivalent, as provided by copyright law.

You may make, run and propagate covered works that you do not convey, without conditions so long as your license otherwise remains in force. You may convey covered works to others for the sole purpose of having them make modifications exclusively for you, or provide you with facilities for running those works, provided that you comply with the terms of this License in conveying all material for which you do not control copyright. Those thus making or running the covered works for you must do so exclusively on your behalf, under your direction and control, on terms that prohibit them from making any copies of your copyrighted material outside their relationship with you.

Conveying under any other circumstances is permitted solely under the conditions stated below. Sublicensing is not allowed; section 10 makes it unnecessary.

3. Protecting Users' Legal Rights From Anti-Circumvention Law.

No covered work shall be deemed part of an effective technological measure under any applicable law fulfilling obligations under article 11 of the WIPO copyright treaty adopted on 20 December 1996, or similar laws prohibiting or restricting circumvention of such measures.

When you convey a covered work, you waive any legal power to forbid circumvention of technological measures to the extent such circumvention is effected by exercising rights under this License with respect to the covered work, and you disclaim any intention to limit operation or modification of the work as a means of enforcing, against the work's users, your or third parties' legal rights to forbid circumvention of technological measures.

4. Conveying Verbatim Copies.

You may convey verbatim copies of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice; keep intact all notices stating that this License and any non-permissive terms added in accord with section 7 apply to the code; keep intact all notices of the absence of any warranty; and give all recipients a copy of this License along with the Program.

You may charge any price or no price for each copy that you convey, and you may offer support or warranty protection for a fee.

5. Conveying Modified Source Versions.

You may convey a work based on the Program, or the modifications to produce it from the Program, in the form of source code under the terms of section 4, provided that you also meet all of these conditions:

a) The work must carry prominent notices stating that you modified it, and giving a relevant date.

b) The work must carry prominent notices stating that it is released under this License and any conditions added under section 7. This requirement modifies the requirement in section 4 to "keep intact all notices".

c) You must license the entire work, as a whole, under this License to anyone who comes into possession of a copy. This License will therefore apply, along with any applicable section 7 additional terms, to the whole of the work, and all its parts, regardless of how they are packaged. This License gives no permission to license the work in any other way, but it does not invalidate such permission if you have separately received it.

d) If the work has interactive user interfaces, each must display Appropriate Legal Notices; however, if the Program has interactive interfaces that do not display Appropriate Legal Notices, your work need not make them do so.

A compilation of a covered work with other separate and independent works, which are not by their nature extensions of the covered work, and which are not combined with it such as to form a larger program, in or on a volume of a storage or distribution medium, is called an "aggregate" if the compilation and its resulting copyright are not used to limit the access or legal rights of the compilation's users beyond what the individual works permit. Inclusion of a covered work in an aggregate does not cause this License to apply to the other parts of the aggregate.

6. Conveying Non-Source Forms.

You may convey a covered work in object code form under the terms of sections 4 and 5, provided that you also convey the machine-readable Corresponding Source under the terms of this License, in one of these ways:

a) Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by the Corresponding Source fixed on a durable physical medium customarily used for software interchange.

b) Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by a written offer, valid for at least three years and valid for as long as you offer spare parts or customer support for that product model, to give anyone who possesses the object code either (1) a copy of the Corresponding Source for all the software in the product that is covered by this License, on a durable physical medium customarily used for software interchange, for a price no more than your reasonable cost of physically performing this conveying of source, or (2) access to copy the Corresponding Source from a network server at no charge.

c) Convey individual copies of the object code with a copy of the written offer to provide the Corresponding Source. This alternative is allowed only occasionally and noncommercially, and only if you received the object code with such an offer, in accord with subsection 6b.

d) Convey the object code by offering access from a designated place (gratis or for a charge), and offer equivalent access to the Corresponding Source in the same way through the same place at no further charge. You need not require recipients to copy the Corresponding Source along with the object code. If the place to copy the object code is a network server, the Corresponding Source may be on a different server (operated by you or a third party) that supports equivalent copying facilities, provided you maintain clear directions next to the object code saying where to find the Corresponding Source. Regardless of what server hosts the Corresponding Source, you remain obligated to ensure that it is available for as long as needed to satisfy these requirements.

e) Convey the object code using peer-to-peer transmission, provided you inform other peers where the object code and Corresponding

Source of the work are being offered to the general public at no charge under subsection 6d.

A separable portion of the object code, whose source code is excluded from the Corresponding Source as a System Library, need not be included in conveying the object code work.

A "User Product" is either (1) a "consumer product", which means any tangible personal property which is normally used for personal, family, or household purposes, or (2) anything designed or sold for incorporation into a dwelling. In determining whether a product is a consumer product, doubtful cases shall be resolved in favor of coverage. For a particular product received by a particular user, "normally used" refers to a typical or common use of that class of product, regardless of the status of the particular user or of the way in which the particular user actually uses, or expects or is expected to use, the product. A product is a consumer product regardless of whether the product has substantial commercial, industrial or non-consumer uses, unless such uses represent the only significant mode of use of the product.

"Installation Information" for a User Product means any methods, procedures, authorization keys, or other information required to install and execute modified versions of a covered work in that User Product from a modified version of its Corresponding Source. The information must suffice to ensure that the continued functioning of the modified object code is in no case prevented or interfered with solely because modification has been made.

If you convey an object code work under this section in, or with, or specifically for use in, a User Product, and the conveying occurs as part of a transaction in which the right of possession and use of the User Product is transferred to the recipient in perpetuity or for a fixed term (regardless of how the transaction is characterized), the Corresponding Source conveyed under this section must be accompanied by the Installation Information. But this requirement does not apply if neither you nor any third party retains the ability to install modified object code on the User Product (for example, the work has been installed in ROM).

The requirement to provide Installation Information does not include a requirement to continue to provide support service, warranty, or updates for a work that has been modified or installed by the recipient, or for the User Product in which it has been modified or installed. Access to a network may be denied when the modification itself materially and adversely affects the operation of the network or violates the rules and protocols for communication across the network.

Corresponding Source conveyed, and Installation Information provided, in accord with this section must be in a format that is publicly documented (and with an implementation available to the public in source code form), and must require no special password or key for unpacking, reading or copying.

7. Additional Terms.

"Additional permissions" are terms that supplement the terms of this

License by making exceptions from one or more of its conditions.
Additional permissions that are applicable to the entire Program shall
be treated as though they were included in this License, to the extent
that they are valid under applicable law. If additional permissions
apply only to part of the Program, that part may be used separately
under those permissions, but the entire Program remains governed by
this License without regard to the additional permissions.

  When you convey a copy of a covered work, you may at your option
remove any additional permissions from that copy, or from any part of
it. (Additional permissions may be written to require their own
removal in certain cases when you modify the work.) You may place
additional permissions on material, added by you to a covered work,
for which you have or can give appropriate copyright permission.

  Notwithstanding any other provision of this License, for material you
add to a covered work, you may (if authorized by the copyright holders of
that material) supplement the terms of this License with terms:

    a) Disclaiming warranty or limiting liability differently from the
    terms of sections 15 and 16 of this License; or

    b) Requiring preservation of specified reasonable legal notices or
    author attributions in that material or in the Appropriate Legal
    Notices displayed by works containing it; or

    c) Prohibiting misrepresentation of the origin of that material, or
    requiring that modified versions of such material be marked in
    reasonable ways as different from the original version; or

    d) Limiting the use for publicity purposes of names of licensors or
    authors of the material; or

    e) Declining to grant rights under trademark law for use of some
    trade names, trademarks, or service marks; or

    f) Requiring indemnification of licensors and authors of that
    material by anyone who conveys the material (or modified versions of
    it) with contractual assumptions of liability to the recipient, for
    any liability that these contractual assumptions directly impose on
    those licensors and authors.

  All other non-permissive additional terms are considered "further
restrictions" within the meaning of section 10. If the Program as you
received it, or any part of it, contains a notice stating that it is
governed by this License along with a term that is a further
restriction, you may remove that term. If a license document contains
a further restriction but permits relicensing or conveying under this
License, you may add to a covered work material governed by the terms
of that license document, provided that the further restriction does
not survive such relicensing or conveying.

  If you add terms to a covered work in accord with this section, you
must place, in the relevant source files, a statement of the
additional terms that apply to those files, or a notice indicating
where to find the applicable terms.

Additional terms, permissive or non-permissive, may be stated in the form of a separately written license, or stated as exceptions; the above requirements apply either way.

8. Termination.

You may not propagate or modify a covered work except as expressly provided under this License. Any attempt otherwise to propagate or modify it is void, and will automatically terminate your rights under this License (including any patent licenses granted under the third paragraph of section 11).

However, if you cease all violation of this License, then your license from a particular copyright holder is reinstated (a) provisionally, unless and until the copyright holder explicitly and finally terminates your license, and (b) permanently, if the copyright holder fails to notify you of the violation by some reasonable means prior to 60 days after the cessation.

Moreover, your license from a particular copyright holder is reinstated permanently if the copyright holder notifies you of the violation by some reasonable means, this is the first time you have received notice of violation of this License (for any work) from that copyright holder, and you cure the violation prior to 30 days after your receipt of the notice.

Termination of your rights under this section does not terminate the licenses of parties who have received copies or rights from you under this License. If your rights have been terminated and not permanently reinstated, you do not qualify to receive new licenses for the same material under section 10.

9. Acceptance Not Required for Having Copies.

You are not required to accept this License in order to receive or run a copy of the Program. Ancillary propagation of a covered work occurring solely as a consequence of using peer-to-peer transmission to receive a copy likewise does not require acceptance. However, nothing other than this License grants you permission to propagate or modify any covered work. These actions infringe copyright if you do not accept this License. Therefore, by modifying or propagating a covered work, you indicate your acceptance of this License to do so.

10. Automatic Licensing of Downstream Recipients.

Each time you convey a covered work, the recipient automatically receives a license from the original licensors, to run, modify and propagate that work, subject to this License. You are not responsible for enforcing compliance by third parties with this License.

An "entity transaction" is a transaction transferring control of an organization, or substantially all assets of one, or subdividing an organization, or merging organizations. If propagation of a covered work results from an entity transaction, each party to that transaction who receives a copy of the work also receives whatever

licenses to the work the party's predecessor in interest had or could give under the previous paragraph, plus a right to possession of the Corresponding Source of the work from the predecessor in interest, if the predecessor has it or can get it with reasonable efforts.

You may not impose any further restrictions on the exercise of the rights granted or affirmed under this License. For example, you may not impose a license fee, royalty, or other charge for exercise of rights granted under this License, and you may not initiate litigation (including a cross-claim or counterclaim in a lawsuit) alleging that any patent claim is infringed by making, using, selling, offering for sale, or importing the Program or any portion of it.

11. Patents.

A "contributor" is a copyright holder who authorizes use under this License of the Program or a work on which the Program is based. The work thus licensed is called the contributor's "contributor version".

A contributor's "essential patent claims" are all patent claims owned or controlled by the contributor, whether already acquired or hereafter acquired, that would be infringed by some manner, permitted by this License, of making, using, or selling its contributor version, but do not include claims that would be infringed only as a consequence of further modification of the contributor version. For purposes of this definition, "control" includes the right to grant patent sublicenses in a manner consistent with the requirements of this License.

Each contributor grants you a non-exclusive, worldwide, royalty-free patent license under the contributor's essential patent claims, to make, use, sell, offer for sale, import and otherwise run, modify and propagate the contents of its contributor version.

In the following three paragraphs, a "patent license" is any express agreement or commitment, however denominated, not to enforce a patent (such as an express permission to practice a patent or covenant not to sue for patent infringement). To "grant" such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party.

If you convey a covered work, knowingly relying on a patent license, and the Corresponding Source of the work is not available for anyone to copy, free of charge and under the terms of this License, through a publicly available network server or other readily accessible means, then you must either (1) cause the Corresponding Source to be so available, or (2) arrange to deprive yourself of the benefit of the patent license for this particular work, or (3) arrange, in a manner consistent with the requirements of this License, to extend the patent license to downstream recipients. "Knowingly relying" means you have actual knowledge that, but for the patent license, your conveying the covered work in a country, or your recipient's use of the covered work in a country, would infringe one or more identifiable patents in that country that you have reason to believe are valid.

If, pursuant to or in connection with a single transaction or

arrangement, you convey, or propagate by procuring conveyance of, a covered work, and grant a patent license to some of the parties receiving the covered work authorizing them to use, propagate, modify or convey a specific copy of the covered work, then the patent license you grant is automatically extended to all recipients of the covered work and works based on it.

A patent license is "discriminatory" if it does not include within the scope of its coverage, prohibits the exercise of, or is conditioned on the non-exercise of one or more of the rights that are specifically granted under this License. You may not convey a covered work if you are a party to an arrangement with a third party that is in the business of distributing software, under which you make payment to the third party based on the extent of your activity of conveying the work, and under which the third party grants, to any of the parties who would receive the covered work from you, a discriminatory patent license (a) in connection with copies of the covered work conveyed by you (or copies made from those copies), or (b) primarily for and in connection with specific products or compilations that contain the covered work, unless you entered into that arrangement, or that patent license was granted, prior to 28 March 2007.

Nothing in this License shall be construed as excluding or limiting any implied license or other defenses to infringement that may otherwise be available to you under applicable patent law.

12. No Surrender of Others' Freedom.

If conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot convey a covered work so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not convey it at all. For example, if you agree to terms that obligate you to collect a royalty for further conveying from those to whom you convey the Program, the only way you could satisfy both those terms and this License would be to refrain entirely from conveying the Program.

13. Use with the GNU Affero General Public License.

Notwithstanding any other provision of this License, you have permission to link or combine any covered work with a work licensed under version 3 of the GNU Affero General Public License into a single combined work, and to convey the resulting work. The terms of this License will continue to apply to the part which is the covered work, but the special requirements of the GNU Affero General Public License, section 13, concerning interaction through a network will apply to the combination as such.

14. Revised Versions of this License.

The Free Software Foundation may publish revised and/or new versions of the GNU General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Program specifies that a certain numbered version of the GNU General Public License "or any later version" applies to it, you have the option of following the terms and conditions either of that numbered version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of the GNU General Public License, you may choose any version ever published by the Free Software Foundation.

If the Program specifies that a proxy can decide which future versions of the GNU General Public License can be used, that proxy's public statement of acceptance of a version permanently authorizes you to choose that version for the Program.

Later license versions may give you additional or different permissions. However, no additional obligations are imposed on any author or copyright holder as a result of your choosing to follow a later version.

15. Disclaimer of Warranty.

THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.

16. Limitation of Liability.

IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MODIFIES AND/OR CONVEYS THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAMAGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

17. Interpretation of Sections 15 and 16.

If the disclaimer of warranty and limitation of liability provided above cannot be given local legal effect according to their terms, reviewing courts shall apply local law that most closely approximates an absolute waiver of all civil liability in connection with the Program, unless a warranty or assumption of liability accompanies a copy of the Program in return for a fee.

END OF TERMS AND CONDITIONS

How to Apply These Terms to Your New Programs

If you develop a new program, and you want it to be of the greatest

```
possible use to the public, the best way to achieve this is to make it
free software which everyone can redistribute and change under these terms.

  To do so, attach the following notices to the program.  It is safest
to attach them to the start of each source file to most effectively
state the exclusion of warranty; and each file should have at least
the "copyright" line and a pointer to where the full notice is found.

    <one line to give the program's name and a brief idea of what it does.>
    Copyright (C) <year>  <name of author>

    This program is free software: you can redistribute it and/or modify
    it under the terms of the GNU General Public License as published by
    the Free Software Foundation, either version 3 of the License, or
    (at your option) any later version.

    This program is distributed in the hope that it will be useful,
    but WITHOUT ANY WARRANTY; without even the implied warranty of
    MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
    GNU General Public License for more details.

    You should have received a copy of the GNU General Public License
    along with this program.  If not, see <https://www.gnu.org/licenses/>.

Also add information on how to contact you by electronic and paper mail.

  If the program does terminal interaction, make it output a short
notice like this when it starts in an interactive mode:

    <program>  Copyright (C) <year>  <name of author>
    This program comes with ABSOLUTELY NO WARRANTY; for details type 'show w'.
    This is free software, and you are welcome to redistribute it
    under certain conditions; type 'show c' for details.

The hypothetical commands 'show w' and 'show c' should show the appropriate
parts of the General Public License.  Of course, your program's commands
might be different; for a GUI interface, you would use an "about box".

  You should also get your employer (if you work as a programmer) or school,
if any, to sign a "copyright disclaimer" for the program, if necessary.
For more information on this, and how to apply and follow the GNU GPL, see
<https://www.gnu.org/licenses/>.

  The GNU General Public License does not permit incorporating your program
into proprietary programs.  If your program is a subroutine library, you
may consider it more useful to permit linking proprietary applications with
the library.  If this is what you want to do, use the GNU Lesser General
Public License instead of this License.  But first, please read
<https://www.gnu.org/licenses/why-not-lgpl.html>.
```

3.59 COPYING.LESSER

```
                   GNU LESSER GENERAL PUBLIC LICENSE
                       Version 3, 29 June 2007

 Copyright (C) 2007 Free Software Foundation, Inc. <https://fsf.org/>
 Everyone is permitted to copy and distribute verbatim copies
```

of this license document, but changing it is not allowed.

  This version of the GNU Lesser General Public License incorporates the terms and conditions of version 3 of the GNU General Public License, supplemented by the additional permissions listed below.

  0. Additional Definitions.

  As used herein, "this License" refers to version 3 of the GNU Lesser General Public License, and the "GNU GPL" refers to version 3 of the GNU General Public License.

  "The Library" refers to a covered work governed by this License, other than an Application or a Combined Work as defined below.

  An "Application" is any work that makes use of an interface provided by the Library, but which is not otherwise based on the Library. Defining a subclass of a class defined by the Library is deemed a mode of using an interface provided by the Library.

  A "Combined Work" is a work produced by combining or linking an Application with the Library. The particular version of the Library with which the Combined Work was made is also called the "Linked Version".

  The "Minimal Corresponding Source" for a Combined Work means the Corresponding Source for the Combined Work, excluding any source code for portions of the Combined Work that, considered in isolation, are based on the Application, and not on the Linked Version.

  The "Corresponding Application Code" for a Combined Work means the object code and/or source code for the Application, including any data and utility programs needed for reproducing the Combined Work from the Application, but excluding the System Libraries of the Combined Work.

  1. Exception to Section 3 of the GNU GPL.

  You may convey a covered work under sections 3 and 4 of this License without being bound by section 3 of the GNU GPL.

  2. Conveying Modified Versions.

  If you modify a copy of the Library, and, in your modifications, a facility refers to a function or data to be supplied by an Application that uses the facility (other than as an argument passed when the facility is invoked), then you may convey a copy of the modified version:

    a) under this License, provided that you make a good faith effort to ensure that, in the event an Application does not supply the function or data, the facility still operates, and performs whatever part of its purpose remains meaningful, or

    b) under the GNU GPL, with none of the additional permissions of this License applicable to that copy.

3. Object Code Incorporating Material from Library Header Files.

The object code form of an Application may incorporate material from a header file that is part of the Library. You may convey such object code under terms of your choice, provided that, if the incorporated material is not limited to numerical parameters, data structure layouts and accessors, or small macros, inline functions and templates (ten or fewer lines in length), you do both of the following:

a) Give prominent notice with each copy of the object code that the Library is used in it and that the Library and its use are covered by this License.

b) Accompany the object code with a copy of the GNU GPL and this license document.

4. Combined Works.

You may convey a Combined Work under terms of your choice that, taken together, effectively do not restrict modification of the portions of the Library contained in the Combined Work and reverse engineering for debugging such modifications, if you also do each of the following:

a) Give prominent notice with each copy of the Combined Work that the Library is used in it and that the Library and its use are covered by this License.

b) Accompany the Combined Work with a copy of the GNU GPL and this license document.

c) For a Combined Work that displays copyright notices during execution, include the copyright notice for the Library among these notices, as well as a reference directing the user to the copies of the GNU GPL and this license document.

d) Do one of the following:

0) Convey the Minimal Corresponding Source under the terms of this License, and the Corresponding Application Code in a form suitable for, and under terms that permit, the user to recombine or relink the Application with a modified version of the Linked Version to produce a modified Combined Work, in the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.

1) Use a suitable shared library mechanism for linking with the Library. A suitable mechanism is one that (a) uses at run time a copy of the Library already present on the user's computer system, and (b) will operate properly with a modified version of the Library that is interface-compatible with the Linked Version.

e) Provide Installation Information, but only if you would otherwise be required to provide such information under section 6 of the

GNU GPL, and only to the extent that such information is
necessary to install and execute a modified version of the
Combined Work produced by recombining or relinking the
Application with a modified version of the Linked Version. (If
you use option 4d0, the Installation Information must accompany
the Minimal Corresponding Source and Corresponding Application
Code. If you use option 4d1, you must provide the Installation
Information in the manner specified by section 6 of the GNU GPL
for conveying Corresponding Source.)

5. Combined Libraries.

You may place library facilities that are a work based on the
Library side by side in a single library together with other library
facilities that are not Applications and are not covered by this
License, and convey such a combined library under terms of your
choice, if you do both of the following:

a) Accompany the combined library with a copy of the same work based
on the Library, uncombined with any other library facilities,
conveyed under the terms of this License.

b) Give prominent notice with the combined library that part of it
is a work based on the Library, and explaining where to find the
accompanying uncombined form of the same work.

6. Revised Versions of the GNU Lesser General Public License.

The Free Software Foundation may publish revised and/or new versions
of the GNU Lesser General Public License from time to time. Such new
versions will be similar in spirit to the present version, but may
differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the
Library as you received it specifies that a certain numbered version
of the GNU Lesser General Public License "or any later version"
applies to it, you have the option of following the terms and
conditions either of that published version or of any later version
published by the Free Software Foundation. If the Library as you
received it does not specify a version number of the GNU Lesser
General Public License, you may choose any version of the GNU Lesser
General Public License ever published by the Free Software Foundation.

If the Library as you received it specifies that a proxy can decide
whether future versions of the GNU Lesser General Public License shall
apply, that proxy's public statement of acceptance of any version is
permanent authorization for you to choose that version for the
Library.

3.60 files.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
```

```
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
The namespace DOSASfiles contain additional classes and functions
to deal with files.

@package DevOrSysAdminScripts
@subpackage DOSASfiles
*/
namespace DOSASfiles {

  use Exception;




  /**
  Kind of a "dataclass" to keep the informations related with a file,
  slightly enhanced with:
  - a constructor that extract the distinct parts of the filepath;
  - a method/getter handling the file content loading.

  @category Library
  @package DevOrSysAdminScripts
  @subpackage DOSASfiles
  @class FileData
```

```
63    @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
64    @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
65    */
66    final class FileData {
67      /**
68      The hash function to use for file contents.
69
70      @var string $s_hash_function
71      */
72      public static string $s_hash_function = 'md5';
73
74      /**
75      The filepath of the file.
76      https://www.youtube.com/watch?v=2NRlZ1ZME5U
77      Why?
78      https://github.com/squizlabs/PHP_CodeSniffer/issues/258
79
80      @var string $s_filepath
81      */
82      public string $s_filepath;
83
84      /**
85      The dirname of the filepath.
86
87      @var string $s_dirname
88      */
89      public string $s_dirname;
90
91      /**
92      The basename of the filepath.
93
94      @var string $s_basename
95      */
96      public string $s_basename;
97
98      /**
99      The filename of the basename.
100
101     @var string $s_filename
102     */
103     public string $s_filename;
104
105     /**
106     The extension of the basename.
107
108     @var string $s_extension
109     */
110     public string $s_extension;
111
112     /**
113     The filesubpath starting from a reference directory.
114
115     @var string $s_path_from_reference_directory
116     */
117     public string $s_path_from_reference_directory;
118
```

```
119     /**
120     The size of the content of the file.
121
122     @var int $i_size
123     */
124     public int $i_size;
125
126     /**
127     The hash of the content of the file.
128
129     @var string $s_hash
130     */
131     public string $s_hash;
132
133     /**
134     The content of the file as a string.
135
136     @var string $s_content
137     */
138     public $s_content;
139
140     /**
141     Statuses relative to another directory : 1.
142
143     Does the file exists with same relative path and same content
144     At The Same Time (ATST)?
145
146     @var bool $b_exists_with_same__relative_path_and_content
147     */
148     public bool $b_exists_with_same__relative_path_and_content; // ATST
149
150     /**
151     Statuses relative to another directory : 2.
152
153     Does the file exists with same relative path?
154
155     @var bool $b_exists_with_same_relative_path
156     */
157     public bool $b_exists_with_same_relative_path;
158
159     /**
160     Statuses relative to another directory : 3.
161
162     Does the file exists with same content?
163
164     @var bool $b_exists_with_same_content
165     */
166     public bool $b_exists_with_same_content;
167
168     /**
169     Statuses relative to another directory : 4.
170
171     Does the file exists with suffixed relative path?
172     Only computed if some regular expression for valid suffixes
173     is supplied.
174     An empty suffix is always considered valid,
```

```
175       but it is handled if your regexp doesn't match the empty string.
176
177       @var bool $b_exists_with_suffixed_relative_path
178       */
179       public bool $b_exists_with_suffixed_relative_path;
180
181       /**
182       Statuses relative to another directory : 5.
183
184       Does the file exists with suffixed relative path and same content
185       At The Same Time (ATST)?
186       Only computed if some regular expression for valid suffixes
187       is supplied.
188       An empty suffix is always considered valid,
189       but it is handled if your regexp doesn't match the empty string.
190
191       @var bool $b_exists_with_suffixed_relative_path_and
192       */
193       public bool $b_exists_with_suffixed_relative_path_and; // ATST
194
195
196
197       /**
198       Constructor for FileData.
199
200       @param string $s_filepath The filepath of the file.
201       @param string $s_reference_directory_path
202                     The directorypath of the reference directory.
203
204       @return void
205       */
206       public function __construct(
207         string $s_filepath,
208         string $s_reference_directory_path,
209       ) {
210         $this->s_filepath = $s_filepath;
211         $path_parts = pathinfo($this->s_filepath);
212         $this->s_dirname = (
213           isset($path_parts['dirname']) ? $path_parts['dirname'] : ''
214         );
215         $this->s_basename = $path_parts['basename'];
216         $this->s_filename = $path_parts['filename'];
217         if(!isset($path_parts['extension'])){
218           $this->s_extension = '';
219         }
220         else{
221           $this->s_extension = $path_parts['extension'];
222           if($this->s_filename === ''){
223             // Hidden file
224             $this->s_filename = $this->s_basename;
225             $this->s_extension = '';
226           }
227         }
228         $this->s_path_from_reference_directory = substr(
229           $this->s_filepath,
230           strlen($s_reference_directory_path),
```

```
231       );
232       // To enforce a load attempt.
233       $this->i_size = 1;
234       $this->s_hash = '';
235       $this->s_content = '';
236       $this->get_s_content();
237       $this->b_exists_with_same__relative_path_and_content = false;
238       $this->b_exists_with_same_relative_path = false;
239       $this->b_exists_with_same_content = false;
240       $this->b_exists_with_suffixed_relative_path = false;
241       $this->b_exists_with_suffixed_relative_path_and = false;
242     }//end __construct() FileData
243
244
245
246     /**
247     Returns the content of the file as a string after fetching it from
248     the filesystem.
249
250     @throws \Exception When the content of the file cannot be retrieved
251                       from the filesystem.
252
253     @return string
254     */
255     public function get_s_content() : string {
256       if($this->i_size !== strlen($this->s_content)){
257         $s_content = file_get_contents($this->s_filepath);
258         if($s_content === false){
259           throw new Exception(
260             'Unexpected failure when fetching content of '
261             .$this->s_filepath
262           );
263         }
264         $this->s_content = $s_content;
265         if($this->s_hash === ''){
266           $this->i_size = strlen($this->s_content);
267           $this->s_hash = md5($this->s_content);
268         }
269       }
270       return $this->s_content;
271     }//end get_s_content()
272   }//end class FileData
273
274
275
276
277   /**
278   Kind of a "dataclass" to keep the informations related with the files
279   under a directory.
280
281   @category Library
282   @package DevOrSysAdminScripts
283   @subpackage DOSASfiles
284   @class DataOfFilesUnderDirectory
285   @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
286   @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
```

```
287   */
288   final class DataOfFilesUnderDirectory {
289     /**
290     The path of the reference directory.
291
292     @var string $s_reference_directory_path
293     */
294     public string $s_reference_directory_path;
295
296     /**
297     The data of the files under the reference directory
298     put in order by absolute path.
299
300     @var array<string, FileData> $arr_files_by_path
301     */
302     public array $arr_files_by_path;
303
304     /**
305     The data of the files under the reference directory
306     put in order by path from reference directory.
307
308     @var array<string, FileData> $arr_files_by_path_from
309     */
310     public array $arr_files_by_path_from;
311
312     /**
313     The data of the files under the reference directory
314     put in order by size, hash, and order of addition to this object.
315
316     @var array<int, array<string, array<int, FileData>>> $arr_files_by_size
317     */
318     public array $arr_files_by_size;
319
320
321
322     /**
323     Constructor for DataOfFilesUnderDirectory.
324
325     @param string $s_reference_directory_path
326                   The directorypath of the reference directory.
327
328     @return void
329     */
330     public function __construct(string $s_reference_directory_path) {
331       $this->s_reference_directory_path = $s_reference_directory_path;
332       $this->arr_files_by_path = [];
333       $this->arr_files_by_path_from = [];
334       $this->arr_files_by_size = [];
335     }//end __construct() DataOfFilesUnderDirectory
336
337
338
339     /**
340     Adds a FileData object in the arrays used for fast access by
341     different keys.
342
```

```
343       @param FileData $o_file_data The FileData object to add.
344
345       @return void
346       */
347       public function add_file_data(FileData $o_file_data) : void {
348         $this->arr_files_by_path[$o_file_data->s_filepath] = $o_file_data;
349         $this->arr_files_by_path_from[
350           $o_file_data->s_path_from_reference_directory
351         ] = $o_file_data;
352         if(!isset($this->arr_files_by_size[$o_file_data->i_size])){
353           $this->arr_files_by_size[$o_file_data->i_size] = [];
354         }
355         if(
356           !isset(
357             $this->arr_files_by_size[$o_file_data->i_size][
358               $o_file_data->s_hash
359             ]
360           )
361         ){
362           $this->arr_files_by_size[$o_file_data->i_size][
363             $o_file_data->s_hash
364           ] = [];
365         }
366         $this->arr_files_by_size[$o_file_data->i_size][
367           $o_file_data->s_hash
368         ] []= $o_file_data;
369       }//end add_file_data()
370
371
372
373       /**
374       Create a FileData object for the given FilePath.
375       And adds this FileData object in the arrays used for fast access by
376       different keys.
377
378       @param string $s_filepath The filepath required.
379
380       @return void
381       */
382       public function add_file_data_for_file_path(
383         string $s_filepath
384       ) : void {
385         $o_file_data = new FileData(
386           $s_filepath,
387           $this->s_reference_directory_path,
388         );
389         $o_file_data->s_content = '';
390         $this->add_file_data($o_file_data);
391       }//end add_file_data_for_file_path()
392     }//end class DataOfFilesUnderDirectory
393
394
395
396
397     /**
398     Loads the data of files under a given directory,
```

```
399   but doesn't keep the contents loaded.
400   Recusivity is not implemented yet.
401
402   @param string $s_dirpath The directory to search in.
403
404   @return \DOSASfiles\DataOfFilesUnderDirectory
405   */
406   function load_files_data_under_directory(
407     string $s_dirpath,
408   ) : DataOfFilesUnderDirectory {
409     $o_data_of_files_under_directory = new DataOfFilesUnderDirectory(
410       $s_dirpath
411     );
412     $arr_s_paths = glob($s_dirpath.'*');
413     if($arr_s_paths === false){
414       return $o_data_of_files_under_directory;
415     }
416     foreach($arr_s_paths as $s_filepath){
417       $o_data_of_files_under_directory->add_file_data_for_file_path(
418         $s_filepath
419       );
420     }//end foreach($arr_s_paths as $s_filepath)
421     return $o_data_of_files_under_directory;
422   }//end load_files_data_under_directory()
423
424
425
426   /**
427   Loads the data of files under two given directories,
428   but doesn't keep the contents loaded.
429   And then compares the files under both directories.
430   Recusivity is not implemented yet.
431
432   @param string $s_source_dirpath The first directory to search in.
433   @param string $s_dest_dirpath The second directory to search in.
434   @param bool   $b_compute_also_reverse_status
435                 If you want that the status of files under dest_dirpath
436                 must also be computed.
437   @param string $s_valid_suffixes_regexp A regexp for valid suffixes.
438
439   @return array{
440     'source_files_data': \DOSASfiles\DataOfFilesUnderDirectory,
441     'dest_files_data': \DOSASfiles\DataOfFilesUnderDirectory
442   }
443   */
444   function compare_files_under_directories(
445     string $s_source_dirpath,
446     string $s_dest_dirpath,
447     bool $b_compute_also_reverse_status = false,
448     string $s_valid_suffixes_regexp = '',
449   ) : array {
450     $o_files_data_source = load_files_data_under_directory(
451       $s_source_dirpath
452     );
453     $o_files_data_dest = load_files_data_under_directory(
454       $s_dest_dirpath
```

```
455       );
456       foreach($o_files_data_source->arr_files_by_path_from as $o_file_data){
457         if(
458           isset(
459             $o_files_data_dest->arr_files_by_path_from[
460               $o_file_data->s_path_from_reference_directory
461             ]
462           )
463         ){
464           $o_file_data->b_exists_with_same_relative_path = true;
465           $o_file_data2 = (
466             $o_files_data_dest->arr_files_by_path_from[
467               $o_file_data->s_path_from_reference_directory
468             ]
469           );
470           $o_file_data2->b_exists_with_same_relative_path = true;
471           $o_file_data->b_exists_with_suffixed_relative_path = true;
472           $o_file_data2->b_exists_with_suffixed_relative_path = true;
473           if(
474             $o_file_data2->i_size === $o_file_data->i_size
475             && $o_file_data2->s_hash === $o_file_data->s_hash
476             && $o_file_data2->get_s_content()
477             === $o_file_data->get_s_content()
478           ){
479             $o_file_data->b_exists_with_same__relative_path_and_content = (
480               true
481             );
482             $o_file_data->b_exists_with_same_content = true;
483             $o_file_data->b_exists_with_suffixed_relative_path_and = (
484               true
485             );
486             $o_file_data2->b_exists_with_same__relative_path_and_content = (
487               true
488             );
489             $o_file_data2->b_exists_with_same_content = true;
490             $o_file_data2->b_exists_with_suffixed_relative_path_and = (
491               true
492             );
493             $o_file_data->s_content = '';
494             $o_file_data2->s_content = '';
495             continue;
496           }
497         }//end if(isset($o_files_data_dest->arr_files_by_path_from[...])
498         if(
499           isset($o_files_data_dest->arr_files_by_size[$o_file_data->i_size])
500           && isset(
501             $o_files_data_dest->arr_files_by_size[$o_file_data->i_size][
502               $o_file_data->s_hash
503             ]
504           )
505         ){
506           foreach(
507              $o_files_data_dest->arr_files_by_size[$o_file_data->i_size][
508               $o_file_data->s_hash
509             ]
510             as $o_file_data2
```

```
511         ){
512           if(
513             $o_file_data2->get_s_content()
514             === $o_file_data->get_s_content()
515           ){
516             $o_file_data->b_exists_with_same_content = true;
517             $o_file_data2->b_exists_with_same_content = true;
518             $o_file_data->s_content = '';
519             $o_file_data2->s_content = '';
520             if($s_valid_suffixes_regexp !== ''){
521               if(
522                 preg_match(
523                   '/'
524                   .preg_quote(
525                     $o_file_data->s_path_from_reference_directory,
526                     '/',
527                   )
528                   .$s_valid_suffixes_regexp
529                   .'/',
530                   $o_file_data2->s_path_from_reference_directory,
531                 )
532               ){
533                 $o_file_data->b_exists_with_suffixed_relative_path = true;
534                 $o_file_data->b_exists_with_suffixed_relative_path_and = (
535                   true
536                 );
537               }
538               if(
539                 $b_compute_also_reverse_status
540                 && preg_match(
541                   '/'
542                   .preg_quote(
543                     $o_file_data2->s_path_from_reference_directory,
544                     '/',
545                   )
546                   .$s_valid_suffixes_regexp
547                   .'/',
548                   $o_file_data->s_path_from_reference_directory,
549                 )
550               ){
551                 $o_file_data2->b_exists_with_suffixed_relative_path = true;
552                 $o_file_data2->b_exists_with_suffixed_relative_path_and = (
553                   true
554                 );
555               }
556             }//end if($s_valid_suffixes_regexp !== '')
557           }//end if(file_data2->content == file_data->content)
558         }//end foreach(file with same size and same hash)
559       }//end if(there is some file with same size and same hash)
560     }//end foreach($arr_files_data_source[...] as $o_file_data)
561
562     if($b_compute_also_reverse_status){
563       foreach($o_files_data_dest->arr_files_by_path_from as $o_file_data){
564         if($o_file_data->b_exists_with_suffixed_relative_path){
565           continue;
566         }
```

```
        $arr_s_paths = glob(
          $s_source_dirpath.$o_file_data->s_basename.'*'
        );
        if($arr_s_paths === false){
          continue;
        }
        foreach($arr_s_paths as $s_filepath){
          if(!isset($o_files_data_source->arr_files_by_path[$s_filepath])){
            // A newly created file that wasn't previously loaded
            // is ignored.
            continue;
          }
          $o_file_data2 = (
            $o_files_data_source->arr_files_by_path[$s_filepath]
          );
          if(
            preg_match(
              '//'
              .preg_quote(
                $o_file_data->s_path_from_reference_directory,
                '//',
              )
              .$s_valid_suffixes_regexp
              .'//',
              $o_file_data2->s_path_from_reference_directory,
            )
          ){
            $o_file_data->b_exists_with_suffixed_relative_path = true;
          }
        }//end foreach($arr_s_paths as $s_filepath)
      }//end foreach($arr_files_data_dest[...] as $o_file_data)
    }//end if($b_compute_also_reverse_status)

    return [
      'source_files_data' => $o_files_data_source,
      'dest_files_data' => $o_files_data_dest,
    ];
  }//end compare_files_under_directories()



  /**
  Archive some directory files into another directory.

  @param string $s_source_dirpath The source directory to search in.
  @param string $s_dest_dirpath The archive directory to move files in.
  @param bool   $b_archive_if_missing_relative_filepath
                Archive if no destination file is found with a relative
                path that is identical.
  @param bool   $b_archive_if_missing_content
                Archive if no destination file is found with same content.
  @param bool   $b_archive_if_missing_same_relative_path_content
                Archive if no destination file is found with a relative
                path that is identical and with same content.
  @param bool   $b_archive_if_missing_suffixed_relative_path
                Archive if no destination file is found with a relative
```

```
623                 path that has a valid suffix
624                 after the source relative path.
625     @param bool   $b_archive_if_missing_suffixed_relative_path_content
626                 Archive if no destination file is found with a relative
627                 path that has a valid suffix
628                 after the source relative path
629                 and with same content.
630     @param string $s_valid_suffixes_regexp A regexp for valid suffixes.
631
632     @throws \Exception When $s_valid_suffixes_regexp is required
633                       and not supplied.
634
635     @return void
636     */
637     function archive_directory_into_another(
638       string $s_source_dirpath,
639       string $s_dest_dirpath,
640       bool $b_archive_if_missing_relative_filepath,
641       bool $b_archive_if_missing_content,
642       bool $b_archive_if_missing_same_relative_path_content,
643       bool $b_archive_if_missing_suffixed_relative_path,
644       bool $b_archive_if_missing_suffixed_relative_path_content,
645       string $s_valid_suffixes_regexp = '',
646     ) : void {
647       if(
648         (
649           $b_archive_if_missing_suffixed_relative_path
650           || $b_archive_if_missing_suffixed_relative_path_content
651         )
652         && $s_valid_suffixes_regexp === ''
653       ){
654         throw new Exception(
655           'archive_directory_into_another is missing an'
656           .' $s_valid_suffixes_regexp argument.'
657         );
658       }
659       $arr_files_data = compare_files_under_directories(
660         $s_source_dirpath,
661         $s_dest_dirpath,
662         false,
663         $s_valid_suffixes_regexp,
664       );
665       foreach(
666         $arr_files_data['source_files_data']->arr_files_by_path_from
667         as $o_file_data
668       ){
669         if(
670           (
671             $b_archive_if_missing_relative_filepath
672             && !$o_file_data->b_exists_with_same_relative_path
673           )
674           ||
675           (
676             $b_archive_if_missing_content
677             && !$o_file_data->b_exists_with_same_content
678           )
```

```
          ||
          (
            $b_archive_if_missing_same_relative_path_content
            && !$o_file_data->b_exists_with_same__relative_path_and_content
          )
          ||
          (
            $b_archive_if_missing_suffixed_relative_path
            && !$o_file_data->b_exists_with_suffixed_relative_path
          )
          ||
          (
            $b_archive_if_missing_suffixed_relative_path_content
            && !$o_file_data->b_exists_with_suffixed_relative_path_and
          )
        ){
          shell_exec(
            'cp --backup=numbered'
            ." '".preg_replace("/'/", "'\"'\"'", $o_file_data->s_filepath)
            ."'"
            ." '".preg_replace("/'/", "'\"'\"'", $s_dest_dirpath)."'"
          );
        }
      }//end foreach($arr_files_data['source_files_data'][] as $o_file_data)
    }//end archive_directory_into_another()
}//end namespace DOSASfiles



/*
$ php
<?php
require_once 'files.libr.php';
var_dump(
  DOSASfiles\compare_files_under_directories(
    '/home/laurent/.config/libreoffice/4/user/backup/',
    '/home/laurent/Documents/LibreOfficeArchives/',
  )
);
DOSASfiles\archive_directory_into_another(
  '/home/laurent/.config/libreoffice/4/user/backup/',
  '/home/laurent/Documents/LibreOfficeArchives/',
  false,
  true,
  false,
  false,
  false,
  '',
);
*/
?>
```

3.61 git_helpers/git_checks.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
```

```
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations."
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle."
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits : -> #Prefetch()."
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
```

```
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_no_abusive_trailing_comma(){
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
  readonly LFBFL_send_body_1
  declare -i LFBFL_i_error=0
  git diff --cached\
    > /tmp/DOSAS_django_git_check2.temp
  if grep --perl-regexp ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_s_files_paths=$(
    git diff --cached --name-only\
    | grep '\.py$'
  )
  printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  declare -i LFBFL_i_error=0
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s.\n" "${LFBFL_file_path}"
    if ! black --check --diff -- "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
```

```
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

3.62 git_helpers/post-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
```

```
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.63 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath -- "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname -- "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
```

```
pushd -- "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf\
    "post_commit_renaming_comment.libr.sh pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# This function will only handle the files that were only renamed in the
# preceding commit.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -i LFBFL_i_log_directory_change
  get_some_flag LFBFL_i_log_directory_change --log-directory-change 1 "$@"
  declare -i LFBFL_i_max_comment_line_length
  get_some_option LFBFL_i_max_comment_line_length\
    --max-comment-line-length 70 70 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  if [[ LFBFL_i_max_comment_line_length -lt 10 ]]; then
    declare -r LFBFL_send_summary="Max comment line length below 10."
    notify-send "${LFBFL_send_summary}" "${LFBFL_send_summary}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "Max comment line length below 10, aborting.\n"
    fi
    return
  fi
```

```
  local LFBFL_s_format
  #-----------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log --max-count=1 --patch\
    | grep --after-context=1 '^diff --git'
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  local LFBFL_split_score_command1
  declare -i LFBFL_i_split_score_command_properties1
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
  LFBFL_split_score_command1="${get_split_score_result}"
  LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
  readonly LFBFL_split_score_command1
  readonly LFBFL_i_split_score_command_properties1
  local LFBFL_split_score_command2
  declare -i LFBFL_i_split_score_command_properties2
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
  LFBFL_split_score_command2="${get_split_score_result}"
  LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
  readonly LFBFL_split_score_command2
  readonly LFBFL_i_split_score_command_properties2
  declare -i LFBFL_i_renaming_happened=0
  declare -i LFBFL_i
  declare -i LFBFL_j
  declare -r LFBFL_special_file_name="files_names_listing.txt"
  local LFBFL_diff_line
  local LFBFL_similarity_line
  local LFBFL_old_file_path
  local LFBFL_old_file_name
  local LFBFL_old_file_directory
  local LFBFL_new_file_path
  local LFBFL_new_file_name
  local LFBFL_new_file_directory
  local LFBFL_useful_file_name
  local LFBFL_extension
  local LFBFL_comment_prefix
  local LFBFL_comment_prefix2
  local LFBFL_old_file_name2
  local LFBFL_old_file_directory2
```

```
152     local LFBFL_new_file_name2
153     local LFBFL_new_file_directory2
154     local LFBFL_new_comment
155     declare -i LFBFL_i_copyright_line_number
156     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
157     declare -i LFBFL_i_line_count
158     declare -i LFBFL_i_lines_after
159     local LFBFL_temp_file_path
160     for ((LFBFL_i = 0; LFBFL_i < ${#LFBFL_renaming_lines_array[@]};)); do
161       LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
162       LFBFL_i=$((LFBFL_i + 1))
163       LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
164       LFBFL_i=$((LFBFL_i + 1))
165       # We need to increment 3 times since grep adds "--" lines between.
166       LFBFL_i=$((LFBFL_i + 1))
167       if [[ LFBFL_i_verbose -eq 1 ]]; then
168         printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
169           "${LFBFL_diff_line}"\
170           "${LFBFL_similarity_line}"
171       fi
172       if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
173         continue
174       fi
175
176       # Extract old file name from diff line. -------------------------
177       LFBFL_old_file_path=$(
178         printf "%s" "${LFBFL_diff_line}"\
179         | cut --delimiter=' ' --fields=3
180       )
181       # It starts by "a/".
182       LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
183       LFBFL_old_file_name=$(basename -- "${LFBFL_old_file_path}")
184       LFBFL_old_file_directory=$(dirname -- "${LFBFL_old_file_path}")
185       # Extract new file name from diff line. -------------------------
186       LFBFL_new_file_path=$(
187         printf "%s" "${LFBFL_diff_line}"\
188         | cut --delimiter=' ' --fields=4
189       )
190       # It starts by "b/".
191       LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
192       LFBFL_new_file_name=$(basename -- "${LFBFL_new_file_path}")
193       LFBFL_new_file_directory=$(dirname -- "${LFBFL_new_file_path}")
194       if [[ LFBFL_i_verbose -eq 1 ]]; then
195         LFBFL_s_format="LFBFL_old_file_path: %s\n"
196         LFBFL_s_format+="LFBFL_old_file_name: %s\n"
197         LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
198         LFBFL_s_format+="LFBFL_new_file_path: %s\n"
199         LFBFL_s_format+="LFBFL_new_file_name: %s\n"
200         LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
201         # shellcheck disable=SC2059
202         printf "${LFBFL_s_format}"\
203           "${LFBFL_old_file_path}"\
204           "${LFBFL_old_file_name}"\
205           "${LFBFL_old_file_directory}"\
206           "${LFBFL_new_file_path}"\
207           "${LFBFL_new_file_name}"\
```

```
208         "${LFBFL_new_file_directory}"
209       fi
210       # Extract "LFBFL_extension" with smart guessing for .tpl ---------
211       # and set prefix for line with ©Copyright. -----------------------
212       LFBFL_useful_file_name="${LFBFL_new_file_name}"
213       LFBFL_extension="${LFBFL_useful_file_name##*.}"
214       if [[ LFBFL_i_verbose -eq 1 ]]; then
215         printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
216           "${LFBFL_useful_file_name}"\
217           "${LFBFL_extension}"
218       fi
219       for ((LFBFL_j = 0; LFBFL_j < 3; ++LFBFL_j)); do
220         if [[ "${LFBFL_extension}" != "tpl" ]]; then
221           break
222         fi
223         LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
224         LFBFL_extension="${LFBFL_useful_file_name##*.}"
225         if [[ LFBFL_i_verbose -eq 1 ]]; then
226           printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
227             "${LFBFL_useful_file_name}"\
228             "${LFBFL_extension}"
229         fi
230       done
231       LFBFL_comment_prefix=""
232       if [[
233         "${LFBFL_extension}" == "sh"
234         || "${LFBFL_extension}" == "toml"
235         || "${LFBFL_extension}" == "yaml"
236       ]]; then
237         LFBFL_comment_prefix="# "
238       fi
239       if [[ "${LFBFL_extension}" == "sql" ]]; then
240         LFBFL_comment_prefix="-- "
241       fi
242       if [[ "${LFBFL_extension}" == "tex" ]]; then
243         LFBFL_comment_prefix="% "
244       fi
245       if [[
246         "${LFBFL_useful_file_name}" == "${LFBFL_special_file_name}"
247         || "${LFBFL_extension}" == "rs"
248         || "${LFBFL_extension}" == "rlib"
249       ]];
250       then
251         LFBFL_comment_prefix="// "
252       fi
253       if [[ LFBFL_i_verbose -eq 1 ]]; then
254         printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
255       fi
256       #-----------------------------------------------------------------
257       # Split if necessary old_file_name, new_file_name, etc. to have
258       # comment lines below max_comment_line_length.
259       LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
260       # -----
261       LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
262       LFBFL_old_file_name2+="${LFBFL_old_file_name}"
263       # shellcheck disable=SC2248
```

```
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. ---------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[
      LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
```

```
    ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. ------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep --line-number -- '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut --delimiter=':' --fields=1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. ------------------------------------
    LFBFL_i_line_count=$(
      wc --lines\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. --------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. --------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        -- "${LFBFL_new_file_path}"
```

```
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" -- "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv -- "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. -------------------------------------------
    git add -- "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit -------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.64 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
```

```
  printf "pre-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.65 IMAP_transfert.exec.py

```
"""
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
2026-05-12T12:06+02:00: This file was renamed from
"IMAP_transfert.py"
to
```

```
"IMAP_transfert.exec.py".

Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
```

```
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print('Connecting to "from account".')
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print('Creating new folder in "to account".')
        to_account.create(to_new_folder)

    if verbose:
        print('Fetching all emails ids in "from account".')
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
```

```
139         print(f'from_account.uid("search") result:{result} ids:{ids}')
140     ids = ids[0].decode().split()
141     if verbose:
142         print('Fetching and transfering each email in "from account".')
143     for my_id in ids:
144         result, raw_message = from_account.uid(
145             "fetch",
146             my_id,
147             "(RFC822)",
148         )
149         if verbose:
150             print(f'from_account.uid("fetch", {my_id}) result:{result}')
151         message = raw_message[0][1]
152         my_email = email.message_from_string(message.decode("utf-8"))
153         flags = imaplib.ParseFlags(raw_message[0][0])
154         # flags = " ".join(flags) mypy is unhappy but IMAP flags are
155         # ASCII strings, we give mypy what it wants.
156         flags_as_strs = [flag.decode("ascii") for flag in flags]
157         flags_str = " ".join(flags_as_strs)
158         # date = imaplib.Internaldate2tuple(raw_message[0][0])
159         date = '"' + my_email["date"] + '"'
160         if verbose:
161             print(
162                 f'from_account.uid("fetch", {my_id})'
163                 f" flags:{flags} date:{date}",
164             )
165         to_account.append(to_new_folder, flags_str, date, message)
166
167
168 def main() -> None:
169     """
170     Parse command-line arguments and
171     transfert all emails from some IMAP account
172     to another IMAP account.
173     """
174     parser = argparse.ArgumentParser(
175         description=(
176             "Transfert all emails from some IMAP account"
177             " to another IMAP account"
178         ),
179     )
180     parser.add_argument(
181         "from_login",
182         type=str,
183         help="Email account login of user one.",
184     )
185     parser.add_argument(
186         "from_password",
187         type=str,
188         help="Email account password of user one.",
189     )
190     parser.add_argument(
191         "to_login",
192         type=str,
193         help="Email account login of user two.",
194     )
```

```
195     parser.add_argument(
196         "to_password",
197         type=str,
198         help="Email account password of user two.",
199     )
200     parser.add_argument(
201         "--to_new_folder",
202         type=str,
203         default="",
204         help="Transfert emails into a new folder of this name.",
205     )
206     parser.add_argument(
207         "--from_host",
208         type=str,
209         default="imap.gmail.com",
210         help="Host of first email account",
211     )
212     parser.add_argument(
213         "--from_port",
214         type=int,
215         default=993,
216         help="Port on host of first email account",
217     )
218     parser.add_argument(
219         "--to_host",
220         type=str,
221         default="imap.gmail.com",
222         help="Host of second email account",
223     )
224     parser.add_argument(
225         "--to_port",
226         type=int,
227         default=993,
228         help="Port on host of second email account",
229     )
230     parser.add_argument(
231         "-v",
232         "--verbose",
233         action="store_true",
234         help="Print details of operations",
235     )
236
237     args = parser.parse_args()
238
239     if args.verbose:
240         print(
241             "-----------------------------------------------------\n"
242             "IMAP transfert:\n"
243             "-----------------------------------------------------\n"
244             "From account:\n"
245             f" - Login: {args.from_login}\n"
246             f" - Password: {args.from_password}\n"
247             f" - Host: {args.from_host}\n"
248             f" - Port: {args.from_port}\n"
249             "To account:\n"
250             f" - Login: {args.to_login}\n"
```

```
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

3.66 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
-----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
-----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
-----------------------------------------------------------------------
https://pubs.opengroup.org/onlinepubs/9699919799/
When looking `man sed` GNU sed says to use POSIX -E instead of
-r or --regexp-extended for portability.
But the link above says that still now only -e, -f, -n are POSIX options
for sed.
For this reason we replaced all short option names by long option names
when possible, even for sed.
```

3.67 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.68 miscellaneous/grep_benchmark.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

some_diff(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='<!-- '\
        --regexp='^[0-9]\+d[0-9]\+$'\
        --regexp='^[0-9]\+,[0-9]\+d[0-9]\+$'
  done
}

some_diff2(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='<!-- '\
        --regexp='^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$'
  done
}

some_diff3(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='\(<!-- \|^[0-9]\+d[0-9]\+$\|^[0-9]\+,[0-9]\+d[0-9]\+$\)'
  done
}

some_diff4(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
```

```
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
  done
}

#$ time some_diff
# real        0m1,663s
# user        0m0,795s
# sys        0m1,975s
#$ time some_diff2
# real        0m1,597s
# user        0m0,705s
# sys        0m2,009s
#$ time some_diff3
# real        0m1,619s
# user        0m0,758s
# sys        0m1,969s
#$ time some_diff4
# real        0m1,574s
# user        0m0,750s
# sys        0m1,932s
```

3.69 miscellaneous/spampty_data.txt

```


a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.70 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>'
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
```

```
This is some text
</pre>

<script>
let a = 'abcd & < > \\ \n  \"\' \u{00003c}/script>';
alert(a);
let b = "abcd & < > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.71 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.72 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
[![Security:bandit:badge]][Bandit-GitHub.com]
[![Code-style:black:badge]][Black-GitHub.com]
```

```
[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**                |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge
```

```
[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

[GitHub-top-language-badge]: https://img.shields.io/github/\
languages/top/llyaudet/DevOrSysAdminScripts

[GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.73 string_escaping.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
```

```
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for HTML (surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\HTML
*/
namespace string_escaping\HTML {
  /**
  When you want that the string appears as is in HTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
```

```
65        $s_string,
66      );
67    }//end escape_text()
68
69
70
71    /**
72    When you want that the string appears as is in an HTML tag attribute
73    that starts and ends with double quotes: my_attribute="something".
74    Example: a title attribute.
75    /!\ in double quotes /!\ my_attribute='something' is not supported.
76
77    @param string $s_string The string you want to escape.
78
79    @return string The escaped string.
80    */
81    function escape_attribute(string $s_string) : string {
82      return str_replace(
83        ['"'],
84        ['&quot;'],
85        $s_string,
86      );
87    }//end escape_attribute()
88
89
90
91    /**
92    When you want that the string appears verbatim inside HTML <pre> tags.
93
94    @param string $s_string The string you want to escape.
95
96    @return string The escaped string.
97    */
98    function escape_pre(string $s_string) : string {
99      return str_replace(
100       [
101         '&',
102         '<',
103       ],
104       [
105         '&amp;',
106         '&lt;',
107       ],
108       $s_string,
109     );
110   }//end escape_pre()
111 }//end namespace string_escaping\HTML
112
113
114
115
116 /**
117 The namespace string_escaping is splitted into subnamespaces for each
118 language.
119 This subnamespace is for XHTML (still surprised?)...
120
```

```
@package DevOrSysAdminScripts
@subpackage string_escaping\XHTML
*/
namespace string_escaping\XHTML {
  /**
  When you want that the string appears as is in XHTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an XHTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
        '"',
        "\n",
      ],
      [
        '&amp;',
        '&lt;',
        '&quot;',
        '&#10;',
      ],
      $s_string,
    );
  }//end escape_attribute()


```

```
177
178   /**
179   When you want that the string appears verbatim inside XHTML <pre> tags.
180
181   @param string $s_string The string you want to escape.
182
183   @return string The escaped string.
184   */
185   function escape_pre(string $s_string) : string {
186     return str_replace(
187       [
188         '&',
189         '<',
190       ],
191       [
192         '&amp;',
193         '&lt;',
194       ],
195       $s_string,
196     );
197   }//end escape_pre()
198 }//end namespace string_escaping\XHTML
199
200
201
202
203 /**
204 The namespace string_escaping is splitted into subnamespaces for each
205 language.
206 This subnamespace is for JS (you're hopeless?)...
207
208 @package DevOrSysAdminScripts
209 @subpackage string_escaping\JS
210 */
211 namespace string_escaping\JS {
212
213   use function string_escaping\XHTML\escape_text as XHTML_escape_text;
214
215
216
217   /**
218   When you want that the string can be put in a JS string definition.
219
220   @param string $s_string The string you want to escape.
221
222   @return string The escaped string.
223   */
224   function escape(string $s_string) : string {
225     return str_replace(
226       [
227         '\\',
228         "\n",
229         "'",
230         '"',
231       ],
232       [
```

```
          '\\\\',
          '\n',
          '\\\'',
          '\\"',
        ],
        $s_string,
      );
    }//end escape()



    /**
    When you want that the string can be put in a JS string definition
    to be included inside an HTML file <script> tag.

    @param string $s_string The string you want to escape.

    @return string The escaped string.
    */
    function escape_for_string_definition_inside_HTML(string $s_string)
    : string {
      return str_replace(
        ['</script>'],
        ['\u{00003c}/script>'],
        escape($s_string),
      );
    }//end escape_for_string_definition_inside_HTML()



    /**
    When you want that the string can be put in a JS string definition
    to be included inside an XHTML file <script> tag.

    @param string $s_string The string you want to escape.

    @return string The escaped string.
    */
    function escape_for_string_definition_inside_XHTML(string $s_string)
    : string {
      return XHTML_escape_text(escape($s_string));
    }//end escape_for_string_definition_inside_XHTML()
  }//end namespace string_escaping\JS



  /**
  The namespace string_escaping is splitted into subnamespaces for each
  language.
  This subnamespace is for TeX.

  @package DevOrSysAdminScripts
  @subpackage string_escaping\TeX
  */
  namespace string_escaping\TeX {
```

```
289     /**
290     When you want that the string can be written to a TeX file inside a
291     part where TeX is in text mode.
292
293     @param string $s_string The string you want to escape.
294
295     @return string The escaped string.
296     */
297     function escape_text(string $s_string) : string {
298       // https://tex.stackexchange.com/a/34586/270015
299       return str_replace(
300         [
301           '\\',
302           '&',
303           '%',
304           '$',
305           '#',
306           '_',
307           '{',
308           '}',
309           '~',
310           '^',
311         ],
312         [
313           '\\textbackslash',
314           '\\&',
315           '\\%',
316           '\\$',
317           '\\#',
318           '\\_',
319           '\\{',
320           '\\}',
321           '\\textasciitilde',
322           '\\textasciicircum',
323         ],
324         $s_string,
325       );
326     }//end escape_text()
327   }//end namespace string_escaping\TeX
328
329   /*
330   $ php
331   <?php
332   require_once 'string_escaping.libr.php';
333   var_dump(string_escaping\HTML\escape_text("<&\"\n'truc\\>"));
334   var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\>"));
335   var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\>"));
336   var_dump(string_escaping\XHTML\escape_text("<&\"\n'truc\\>"));
337   var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\>"));
338   var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\>"));
339   var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
340   var_dump(string_escaping\JS\escape_for_string_definition_inside_HTML(
341     "abcd & < > \\ \n  \"' </script>"
342   ));
343   var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
344     "abcd & < > \\ \n  \"'"
```

```
));
*/
?>
```

3.74 unicode.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
This file was renamed from "unicode.php" to "unicode.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

namespace DOSAS_unicode;

use Exception;

/*
Max value for Unicode is 10FFFF != 16*255*255 = 16*65025 = 1040400... XD
Max value for Unicode is 10FFFF = (16*256+255)*256+255
  = (2560+1536+255)*256+255 = (4096+255)*256+255 = 4351*256+255
  = 4351*1000/4 + 4351*6 + 255
  4351000 2175500 1087750
  = 1087750 + 26106 + 255 = 1114111
so there is 1114112 possible values in the "uncarved" range.
244,143,191,191 for UTF-8 yields max code point is ((4*64+15)*64+63)*64+63
  = ((256+15)*64+63)*64+63 = (271*64+63)*64+63 = (16260+1084+63)*64+63
```

```
  = 17407*64+63 = 1044420 + 69628 + 63 = 1114111
*/
const MAX_UNICODE_CODE_POINT = 1114111;



/**
Converts an int to the byte-string corresponding to the UTF-8 encoding
of the Unicode code point of same integer value.
Note that not all integers between 0 and 1114111 are valid code points
for UTF-8.

@param int<0, 1114111> $i_unicode_code_point The code point.

@throws \Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function unicode_code_point_to_UTF8(int $i_unicode_code_point) : string {
  // var_dump($i_unicode_code_point);
  if($i_unicode_code_point < 0){
    throw new Exception('A unicode code point must not be negative.');
  }
  // 0xxxxxxx ASCII
  if($i_unicode_code_point < 128){
    return chr($i_unicode_code_point);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_unicode_code_point < 2048){  // 2**11
    $i_first_byte_base_value = 192;
    /*
    $i_last_byte_significant_bits = $i_unicode_code_point % 64  // 2**6;
    $i_first_byte_significant_bits = intdiv(
      $i_unicode_code_point - $i_last_byte_significant_bits,
      64,  // 2**6
    );
    $s_result = chr(
      $i_first_byte_base_value
      + $i_first_byte_significant_bits
    );
    $s_result .= chr(
      $i_continuation_base_value
      + $i_last_byte_significant_bits
    );
    return $s_result;
    */
    return (
      chr($i_first_byte_base_value + ($i_unicode_code_point >> 6))
      .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
    );
  }//end if($i_unicode_code_point < 2048)
  // 1110xxxx 10xxxxxx 10xxxxxx
  if($i_unicode_code_point < 65536){  // 2**16
    $i_first_byte_base_value = 224;
    /*
```

```
    $arr_arr_data_per_byte_reverse = [
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_unicode_code_point;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      // J'aurais pu "roller" les 4 cas des ifs.
      // Mais pour les perfs, il faudrait tout unroller vu le nombre
      // d'itérations. Et j'aimerais bien comparer du code unrollé
      // aux 2 niveaux à la main avec celui unrollé par le compilateur
      // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
      // l'intérieur d'abord (pour ne pas refaire le même boulot)
      // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
      // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
    return $s_result;
    */
    return (
      chr($i_first_byte_base_value + ($i_unicode_code_point >> 12))
      .chr(
        $i_continuation_base_value
        + (($i_unicode_code_point >> 6) & 63)
      )
      .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
    );
  }//end if($i_unicode_code_point < 65536)
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  // if($i_unicode_code_point < 2097152){  // 2**21
  if($i_unicode_code_point < MAX_UNICODE_CODE_POINT){
    $i_first_byte_base_value = 240;
    /*
    $arr_arr_data_per_byte_reverse = [
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_unicode_code_point;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
```

```
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }
    return $s_result;
    */
    return (
      chr($i_first_byte_base_value + ($i_unicode_code_point >> 18))
      .chr(
        $i_continuation_base_value
        + (($i_unicode_code_point >> 12) & 63)
      )
      .chr(
        $i_continuation_base_value
        + (($i_unicode_code_point >> 6) & 63)
      )
      .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
    );
  }//end if($i_unicode_code_point < MAX_UNICODE_CODE_POINT)
  // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
    ." :)"
  );
  // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
}//end unicode_code_point_to_UTF8()



/**
Converts a string of hexadecimal digits to the byte-string corresponding
to the UTF-8 encoding of the Unicode code point of same integer value
that is denoted by the hexadecimal digits.

@param string $s_code_point_in_hexadecimal_notation The code point.

@throws \Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function hexa_code_point_to_UTF8(
  string $s_code_point_in_hexadecimal_notation
) : string {
```

```
  $s_code_point_in_hexadecimal_notation = strtoupper(
    $s_code_point_in_hexadecimal_notation
  );
  $i_unicode_code_point = 0;
  $i_ord_0 = ord('0');
  $i_ord_A = ord('A');
  $i_digit_offset = -$i_ord_0;
  $i_letter_offset = 10 - $i_ord_A;
  for(
    $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
    $i < $i_max;
    ++$i
  ){
    $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
    $i_unicode_code_point *= 16;
    if(ctype_digit($s_hexa_digit)){
      $i_unicode_code_point += (
        ord($s_hexa_digit) + $i_digit_offset
      );
      continue;
    }
    if(ctype_xdigit($s_hexa_digit)){
      $i_unicode_code_point += (
        ord($s_hexa_digit) + $i_letter_offset
      );
      continue;
    }
    throw new Exception('Not an hexadecimal digit.');
  }
  if($i_unicode_code_point > MAX_UNICODE_CODE_POINT){
    throw new Exception(
      'Hexadecimal integer is greater than UTF-8 maximum code point.'
    );
  }
  return unicode_code_point_to_UTF8($i_unicode_code_point);
}//end hexa_code_point_to_UTF8()



/**
Converts a string of decimal digits to the byte-string corresponding
to the UTF-8 encoding of the Unicode code point of same integer value
that is denoted by the decimal digits.

@param string $s_code_point_in_decimal_notation The code point.

@throws \Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function decimal_code_point_to_UTF8(
  string $s_code_point_in_decimal_notation
) : string {
  $i_unicode_code_point = 0;
  $i_ord_0 = ord('0');
  $i_digit_offset = -$i_ord_0;
```

```
276   for(
277     $i = 0, $i_max = strlen($s_code_point_in_decimal_notation);
278     $i < $i_max;
279     ++$i
280   ){
281     $s_digit = $s_code_point_in_decimal_notation[$i];
282     $i_unicode_code_point *= 10;
283     if(ctype_digit($s_digit)){
284       $i_unicode_code_point += (ord($s_digit) + $i_digit_offset);
285       continue;
286     }
287     throw new Exception('Not a decimal digit.');
288   }
289   if($i_unicode_code_point > MAX_UNICODE_CODE_POINT){
290     throw new Exception(
291       'Decimal integer is greater than UTF-8 maximum code point.'
292     );
293   }
294   return unicode_code_point_to_UTF8($i_unicode_code_point);
295 }//end decimal_code_point_to_UTF8()
296
297
298
299 /**
300 This function throws an exception with extended debug informations
301 if the input string is not valid ASCII.
302 Otherwise, it returns true.
303
304 @param string $s_string The input string.
305
306 @throws \DOSAS_unicode\InvalidEncodingException
307         When the input string is not valid ASCII.
308
309 @return true
310 */
311 function check_string_is_valid_ASCII(string $s_string) : bool {
312   //Fast-path
313   if(mb_check_encoding($s_string, 'ASCII')){
314     return true;
315   }
316
317   $i_offset_in_octets_from_string_start = 0;
318   $i_current_line_number = 1;
319   $i_offset_in_octets_from_line_start = -1;
320   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
321     $i_current_octet = ord($s_string[$i]);
322     $i_offset_in_octets_from_string_start = $i;
323     ++$i_offset_in_octets_from_line_start;
324
325     if($i_current_octet > 127){
326       throw new InvalidEncodingException(
327         'Non-ASCII character found on line '
328         .$i_current_line_number
329         .'; the octet/character '
330         .($i_offset_in_octets_from_line_start + 1)
331         .' has value '
```

```
332           .$i_current_octet
333           .' which is not ASCII.'
334           .' (Sequential position without line splitting:'
335           .' This is at octet/character '
336           .($i_offset_in_octets_from_string_start + 1)
337           .'.)',
338           [
339             'current_octet' => $i_current_octet,
340             'offset_from_string_start'
341               => $i_offset_in_octets_from_string_start,
342             'line' => $i_current_line_number,
343             'offset_from_line_start'
344               => $i_offset_in_octets_from_line_start,
345           ],
346         );
347       }//end if($i_current_octet > 127)
348
349       // if($s_string[$i] === "\n"){
350       if($i_current_octet === 10){
351         ++$i_current_line_number;
352         $i_offset_in_octets_from_line_start = -1;
353       }
354     }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
355     return true;
356   }//end check_string_is_valid_ASCII()
357
358
359
360   /**
361   This function throws an exception with extended debug informations
362   if the content of the file with given file_path is not valid ASCII.
363   Otherwise, it returns true.
364
365   @param string $s_file_path The input file_path.
366
367   @throws \Exception When the file is not found.
368   @throws \DOSAS_unicode\InvalidEncodingException
369           When the input string is not valid ASCII.
370
371   @return true
372   */
373   function check_file_is_valid_ASCII(string $s_file_path) : bool {
374     $s_string = file_get_contents($s_file_path);
375     if($s_string === false){
376       throw new Exception('File '.$s_file_path.' not found.');
377     }
378     return check_string_is_valid_ASCII($s_string);
379   }//end check_file_is_valid_ASCII()
380
381
382
383   /**
384   This function returns the message string and data array for
385   DOSAS_unicode\InvalidEncodingException inside check_string_is_valid_UTF8().
386
387   @param string $s_custom_message The custom part of the message.
```

```
388 @param int $i_current_octet The current octet in the string.
389 @param int $i_continuation_octet_needed The current number of continuation
390                                        octets needed.
391 @param int $i_offset_in_octets_from_string_start
392             The offset in octets from string start.
393 @param int $i_offset_in_characters_from_string_start
394             The offset in characters from string start.
395 @param int $i_character_start_position_from_string_start
396             The offset in octets from string start at which the current
397             character starts.
398 @param int $i_current_line_number The current line number.
399 @param int $i_offset_in_octets_from_line_start
400             The offset in octets from line start.
401 @param int $i_offset_in_characters_from_line_start
402             The offset in characters from line start.
403 @param int $i_character_start_position_from_line_start
404             The offset in octets from line start at which the current
405             character starts.
406 @param int $i_current_continuation_octet_minimum
407             The minimum value of the second continuation octet as specified
408             by the grammar.
409 @param int $i_current_continuation_octet_maximum
410             The maximum value of the second continuation octet as specified
411             by the grammar.
412
413 @return array{'message': string, 'data': array<mixed>}
414 */
415 function get_message_and_data_array(
416   string $s_custom_message,
417   int $i_current_octet,
418   int $i_continuation_octet_needed,
419   int $i_offset_in_octets_from_string_start,
420   int $i_offset_in_characters_from_string_start,
421   int $i_character_start_position_from_string_start,
422   int $i_current_line_number,
423   int $i_offset_in_octets_from_line_start,
424   int $i_offset_in_characters_from_line_start,
425   int $i_character_start_position_from_line_start,
426   int $i_current_continuation_octet_minimum,
427   int $i_current_continuation_octet_maximum,
428 ) : array {
429   return [
430     'message' => (
431       'Non-UTF8 character found on line '
432       .$i_current_line_number
433       .'; the octet '
434       .($i_offset_in_octets_from_line_start + 1)
435       .', part of the character '
436       .($i_offset_in_characters_from_line_start + 1)
437       .', has value '
438       .$i_current_octet
439       .$s_custom_message
440       .' This character starts at octet '
441       .($i_character_start_position_from_line_start + 1)
442       .' of the current line.'
443       .' (Sequential positions without line splitting:'
```

```
      .' This is at character '
      .($i_offset_in_characters_from_string_start + 1)
      .' and octet '
      .($i_offset_in_octets_from_string_start + 1)
      .'.'
      .' This character starts at octet '
      .($i_character_start_position_from_string_start + 1)
      .'.)'
    ),
    'data' => [
      'current_octet' => $i_current_octet,
      'continuation_octet_needed' => $i_continuation_octet_needed,
      'offset_in_octets_from_string_start'
        => $i_offset_in_octets_from_string_start,
      'offset_in_characters_from_string_start'
        => $i_offset_in_characters_from_string_start,
      'character_start_position_from_string_start'
        => $i_character_start_position_from_string_start,
      'line' => $i_current_line_number,
      'offset_in_octets_from_line_start'
        => $i_offset_in_octets_from_line_start,
      'offset_in_characters_from_line_start'
        => $i_offset_in_characters_from_line_start,
      'character_start_position_from_line_start'
        => $i_character_start_position_from_line_start,
      'current_continuation_octet_minimum'
        => $i_current_continuation_octet_minimum,
      'current_continuation_octet_maximum'
        => $i_current_continuation_octet_maximum,
    ],
  ];
}//end get_message_and_data_array()



/**
Checks that the version of PHP allows using the fast-path inside
check_string_is_valid_UTF8.

@return bool
*/
function get_use_fast_path() : bool {
  // Before PHP 5.4, Unicode support has bugs,
  // both in mb_check_encoding and preg_match.
  // See https://github.com/php/php-src/issues/22279
  return PHP_VERSION_ID >= 50400;
}//end get_use_fast_path()



/**
This function throws an exception with extended debug informations
if the input string is not valid UTF-8.
Otherwise, it returns true.

See https://datatracker.ietf.org/doc/html/rfc3629
```

```
See https://github.com/Seldaek/jsonlint/pull/98
for why it may be a good idea to have the last two parameters.
Note that a very clever compiler would merge both booleans into
a single integer parameter using binary flags.
If you want to use this function with fast-path and fallback on slow-path
in case of error, then set both booleans to true.
If, for testing/benchmarking purpose, you want to compare the results
of fast-path and slow-path, call this function twice with a single boolean
set, and store/compare the results.

@param string $s_string The input string.
@param bool $b_fast_path Use the fast-path or not to check quickly that
                         there is no error.
@param bool $b_slow_path Use the slow-path or not to check that
                         there is no error or give detailed error message.

@throws \Exception When called with $b_fast_path set to true but PHP
                   version is incompatible.
@throws \DOSAS_unicode\InvalidEncodingException
        When the input string is not valid UTF-8.

@return bool
*/
function check_string_is_valid_UTF8(
  string $s_string,
  bool $b_fast_path = true,
  bool $b_slow_path = true,
) : bool {
  // Fast-path
  // But before PHP 5.4 Unicode support has bugs.
  // See https://github.com/php/php-src/issues/22279
  if($b_fast_path && !get_use_fast_path()){
    throw new Exception(
      'Fast-path is not available with version '.PHP_VERSION_ID.' of PHP.'
    );
  }
  if($b_fast_path){
    if(function_exists('mb_check_encoding')){
      if(mb_check_encoding($s_string, 'UTF-8')){
        return true;
      }
    }
    else{
      if(preg_match('//u', $s_string) === 1){
        return true;
      }
    }
  }
  else{
    if(!$b_slow_path){
      throw new Exception(
        "Don't call check_string_is_valid_UTF8 without at least one of"
        .' fast-path or slow-path required.'
      );
    }
  }
```

```
556
557     if(!$b_slow_path){
558       return false;
559     }
560
561     $i_current_octet = null;
562     $i_continuation_octet_needed = 0;
563     $i_offset_in_octets_from_string_start = 0;
564     $i_offset_in_characters_from_string_start = -1;
565     $i_character_start_position_from_string_start = 0;
566     $i_current_line_number = 1;
567     $i_offset_in_octets_from_line_start = -1;
568     $i_offset_in_characters_from_line_start = -1;
569     $i_character_start_position_from_line_start = -1;
570
571     // For the second octet of a character, hence first continuation octet,
572     // further restriction may apply.
573     $I_CONTINUATION_OCTET_MINIMUM = 128;
574     $I_CONTINUATION_OCTET_MAXIMUM = 191;
575     $i_current_continuation_octet_minimum = $I_CONTINUATION_OCTET_MINIMUM;
576     $i_current_continuation_octet_maximum = $I_CONTINUATION_OCTET_MAXIMUM;
577     $s_message_for_continuation_octet_above_maximum = null;
578
579     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
580       $i_current_octet = ord($s_string[$i]);
581       $i_offset_in_octets_from_string_start = $i;
582       ++$i_offset_in_octets_from_line_start;
583
584       /*
585       The octet values C0, C1, F5 to FF never appear.
586       C0 = 12*16      = 192 = 11000000
587       C1 = 12*16 + 1  = 193 = 11000001
588       F5 = 15*16 + 5  = 245 = 11110101
589       FF = 15*16 + 15 = 255 = 11111111
590       */
591       if(
592         $i_current_octet === 192
593         || $i_current_octet === 193
594         || $i_current_octet === 245
595         || $i_current_octet === 255
596       ){
597         [
598           'message' => $s_message,
599           'data' => $arr_data,
600         ] = get_message_and_data_array(
601           ' which is one of the four forbidden values (C0, C1, F5, FF).',
602           $i_current_octet,
603           $i_continuation_octet_needed,
604           $i_offset_in_octets_from_string_start,
605           $i_offset_in_characters_from_string_start,
606           $i_character_start_position_from_string_start,
607           $i_current_line_number,
608           $i_offset_in_octets_from_line_start,
609           $i_offset_in_characters_from_line_start,
610           $i_character_start_position_from_line_start,
611           $i_current_continuation_octet_minimum,
```

```
612           $i_current_continuation_octet_maximum,
613         );
614         throw new InvalidEncodingException($s_message, $arr_data);
615       }
616
617       /*
618       UTF8-octets = *( UTF8-char )
619       UTF8-char   = UTF8-1 / UTF8-2 / UTF8-3 / UTF8-4
620       UTF8-1      = %x00-7F
621       UTF8-2      = %xC2-DF UTF8-tail
622         Notice that values C0 and C1 are forbidden, hence %xC2
623       UTF8-3      = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2( UTF8-tail ) /
624                     %xED %x80-9F UTF8-tail / %xEE-EF 2( UTF8-tail )
625         Notice that E0 = 224 adds a restriction on second octet.
626         Notice that ED = 237 adds a restriction on second octet.
627       UTF8-4      = %xF0 %x90-BF 2( UTF8-tail ) / %xF1-F3 3( UTF8-tail ) /
628                     %xF4 %x80-8F 2( UTF8-tail )
629         Notice that F0 = 240 adds a restriction on second octet.
630         Notice that F4 = 244 adds a restriction on second octet.
631       UTF8-tail   = %x80-BF
632       */
633       if($i_continuation_octet_needed > 0){
634         if(
635           $i_current_octet < $i_current_continuation_octet_minimum
636           || $i_current_octet > $i_current_continuation_octet_maximum
637         ){
638           [
639             'message' => $s_message,
640             'data' => $arr_data,
641           ] = get_message_and_data_array(
642             $s_message_for_continuation_octet_above_maximum !== null
643             && $i_current_octet > $i_current_continuation_octet_maximum
644             ? $s_message_for_continuation_octet_above_maximum
645             : ' which is not a continuation octet.',
646             $i_current_octet,
647             $i_continuation_octet_needed,
648             $i_offset_in_octets_from_string_start,
649             $i_offset_in_characters_from_string_start,
650             $i_character_start_position_from_string_start,
651             $i_current_line_number,
652             $i_offset_in_octets_from_line_start,
653             $i_offset_in_characters_from_line_start,
654             $i_character_start_position_from_line_start,
655             $i_current_continuation_octet_minimum,
656             $i_current_continuation_octet_maximum,
657           );
658           throw new InvalidEncodingException($s_message, $arr_data);
659         }//end if($i_current_octet < 128 || $i_current_octet >= 192)
660         --$i_continuation_octet_needed;
661         $i_current_continuation_octet_minimum = (
662           $I_CONTINUATION_OCTET_MINIMUM
663         );
664         $i_current_continuation_octet_maximum = (
665           $I_CONTINUATION_OCTET_MAXIMUM
666         );
667         continue;
```

```
668       }//end if($i_continuation_octet_needed > 0)
669
670       ++$i_offset_in_characters_from_string_start;
671       ++$i_offset_in_characters_from_line_start;
672       $i_character_start_position_from_string_start = (
673         $i_offset_in_octets_from_string_start
674       );
675       $i_character_start_position_from_line_start = (
676         $i_offset_in_octets_from_line_start
677       );
678
679       if($i_current_octet < 128){ // 0xxxxxxx ASCII
680         // if($s_string[$i] === "\n"){
681         if($i_current_octet === 10){
682           ++$i_current_line_number;
683           $i_offset_in_octets_from_line_start = -1;
684           $i_offset_in_characters_from_line_start = -1;
685         }
686         continue;
687       }
688
689       if(/*$i_current_octet >= 128 &&*/ $i_current_octet < 192){
690         [
691           'message' => $s_message,
692           'data' => $arr_data,
693         ] = get_message_and_data_array(
694           ' which is a continuation octet.',
695           $i_current_octet,
696           $i_continuation_octet_needed,
697           $i_offset_in_octets_from_string_start,
698           $i_offset_in_characters_from_string_start,
699           $i_character_start_position_from_string_start,
700           $i_current_line_number,
701           $i_offset_in_octets_from_line_start,
702           $i_offset_in_characters_from_line_start,
703           $i_character_start_position_from_line_start,
704           $i_current_continuation_octet_minimum,
705           $i_current_continuation_octet_maximum,
706         );
707         throw new InvalidEncodingException($s_message, $arr_data);
708       }//end if($i_current_octet >= 128 && $i_current_octet < 192)
709
710       if(/*$i_current_octet >= 192 &&*/ $i_current_octet < 224){
711         // 110xxxxx 10xxxxxx
712         $i_continuation_octet_needed = 1;
713         continue;
714       }
715
716       if(/*$i_current_octet >= 224 &&*/ $i_current_octet < 240){
717         // 1110xxxx 10xxxxxx 10xxxxxx
718         $i_continuation_octet_needed = 2;
719         /*
720         UTF8-3 = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2( UTF8-tail ) /
721                  %xED %x80-9F UTF8-tail / %xEE-EF 2( UTF8-tail )
722         Notice that E0 = 224 adds a restriction on second octet.
723         Notice that ED = 237 adds a restriction on second octet.
```

```
724
725         The definition of UTF-8 prohibits encoding character numbers between
726         U+D800 and U+DFFF.
727         D8 = 13*16 + 8  = 216 = 11010100
728         DF = 13*16 + 15 = 223 = 11010101
729         216*256 = 55296       = 1101 1000 0000 0000 = D800
730         to
731         223*256 + 255 = 57343 = 1101 1111 1111 1111 = DFFF
732
733         1110xxxx
734         11101101 = 237 = ED
735
736         237,160,128
737         237,191,191
738         Thus, the whole "continuation range" is forbidden if start
739         is 237 and second octet, first continuation octet, is >= 160.
740         */
741         if($i_current_octet === 224){
742           $i_current_continuation_octet_minimum = 160;
743           $i_current_continuation_octet_maximum = 191;  // Normal value
744         }
745         if($i_current_octet === 237){
746           $i_current_continuation_octet_minimum = 128;  // Normal value
747           $i_current_continuation_octet_maximum = 159;
748           $s_message_for_continuation_octet_above_maximum = (
749             ' which is into the forbidden range of surrogate pairs.'
750           );
751         }
752         continue;
753       }//end if(/*$i_current_octet >= 224 &&*/ $i_current_octet < 240)
754
755       if(/*$i_current_octet >= 240 &&*/ $i_current_octet < /*248*/ 245){
756         // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
757         $i_continuation_octet_needed = 3;
758         /*
759         UTF8-4 = %xF0 %x90-BF 2( UTF8-tail ) / %xF1-F3 3( UTF8-tail ) /
760                  %xF4 %x80-8F 2( UTF8-tail )
761         Notice that F0 = 240 adds a restriction on second octet.
762         Notice that F4 = 244 adds a restriction on second octet.
763         */
764         if($i_current_octet === 240){
765           $i_current_continuation_octet_minimum = 144;
766           $i_current_continuation_octet_maximum = 191;  // Normal value
767         }
768         if($i_current_octet === 244){
769           $i_current_continuation_octet_minimum = 128;  // Normal value
770           $i_current_continuation_octet_maximum = 143;
771         }
772         continue;
773       }
774
775       [
776         'message' => $s_message,
777         'data' => $arr_data,
778       ] = get_message_and_data_array(
779         ' which is invalid.',
```

```
780         $i_current_octet,
781         $i_continuation_octet_needed,
782         $i_offset_in_octets_from_string_start,
783         $i_offset_in_characters_from_string_start,
784         $i_character_start_position_from_string_start,
785         $i_current_line_number,
786         $i_offset_in_octets_from_line_start,
787         $i_offset_in_characters_from_line_start,
788         $i_character_start_position_from_line_start,
789         $i_current_continuation_octet_minimum,
790         $i_current_continuation_octet_maximum,
791       );
792       throw new InvalidEncodingException($s_message, $arr_data);
793     }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
794
795     if($i_continuation_octet_needed > 0){
796       [
797         'message' => $s_message,
798         'data' => $arr_data,
799       ] = get_message_and_data_array(
800         '; end of string was found instead of a continuation octet.',
801         $i_current_octet,
802         $i_continuation_octet_needed,
803         $i_offset_in_octets_from_string_start,
804         $i_offset_in_characters_from_string_start,
805         $i_character_start_position_from_string_start,
806         $i_current_line_number,
807         $i_offset_in_octets_from_line_start,
808         $i_offset_in_characters_from_line_start,
809         $i_character_start_position_from_line_start,
810         $i_current_continuation_octet_minimum,
811         $i_current_continuation_octet_maximum,
812       );
813       throw new InvalidEncodingException($s_message, $arr_data);
814     }
815
816     if($b_fast_path){
817       throw new InvalidEncodingException(
818         "Fast-path detected an error that this function couldn't found."
819         .'Please create an issue at '
820         .'"https://github.com/LLyaudet/DevOrSysAdminScripts/issues".',
821         [],
822       );
823     }
824
825     return true;
826   }//end check_string_is_valid_UTF8()
827
828
829
830   /**
831   This function throws an exception with extended debug informations
832   if the content of the file with given file_path is not valid UTF-8.
833   Otherwise, it returns true.
834
835   @param string $s_file_path The input file_path.
```

```
836 @param bool $b_fast_path Use the fast-path or not to check quickly that
837                           there is no error.
838
839 @throws \Exception When the file is not found OR when called with
840                    $b_fast_path set to true but PHP version is
841                    incompatible.
842 @throws \DOSAS_unicode\InvalidEncodingException
843         When the input string is not valid UTF-8.
844
845 @return bool
846 */
847 function check_file_is_valid_UTF8(
848   string $s_file_path,
849   bool $b_fast_path = true,
850 ) : bool {
851   $s_string = file_get_contents($s_file_path);
852   if($s_string === false){
853     throw new Exception('File '.$s_file_path.' not found.');
854   }
855   return check_string_is_valid_UTF8($s_string, $b_fast_path);
856 }//end check_file_is_valid_UTF8()
857
858
859
860 /*
861 ?>
862 <?php
863 require_once "./unicode_exceptions.libr.php";
864 require_once "./unicode.libr.php";
865 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("002B"));
866 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("00E6"));
867 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("1400"));
868 var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("10111"));
869 // https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
870 // Some invisible unicode characters that are a problem:
871 // 2000-200F 2028-202F 205F 2060-206F
872 $arr = [
873   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
874   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
875   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
876   '205F',
877   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
878   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
879 ];
880 $arr_s = [];
881 $arr_s2 = [];
882 $arr_s3 = [];
883 echo("sed\\\n");
884 foreach($arr as $s_hexa){
885   $s = DOSAS_unicode\hexa_code_point_to_UTF8($s_hexa);
886   $arr_s []= $s;
887   if(strlen($s) === 2){
888     $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
889   }
890   if(strlen($s) === 3){
891     $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
```

```
892         .'\x'.bin2hex($s[2]);
893     }
894     $arr_s2 []= $s2;
895     $s3 = "  --expression='s/".$s2."/ /g'\\";
896     $arr_s3 []= $s3;
897     echo($s3."\n");
898 }
899 
900 ?>
901 <?php
902 require_once "./unicode_exceptions.libr.php";
903 require_once "./unicode.libr.php";
904 // var_dump(DOSAS_unicode\check_file_is_valid_UTF8("validutf8.json"));
905 // var_dump(DOSAS_unicode\check_file_is_valid_UTF8("nonvalidutf8.json"));
906 var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé"));
907 var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé", true, false));
908 var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé", false, true));
909 var_dump(
910   DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233), true, false)
911 );
912 if(true){
913   var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233)));
914 }
915 else{
916   var_dump(
917     DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233), false, true)
918   );
919 }
920 //*/
921 ?>
```

3.75 unicode_exceptions.libr.php

```
1  <?php
2  
3  /**
4  This file is part of DevOrSysAdminScripts.
5  
6  DevOrSysAdminScripts is free software:
7  you can redistribute it and/or modify it
8  under the terms of the GNU Lesser General Public License
9  as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12 
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19 
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24 
```

```
©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
This file was renamed from "unicode.php" to "unicode.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

namespace DOSAS_unicode;

/**
A class to enhance standard PHP Exception with an array of data.

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/
class WithDataArrayException extends \Exception {
  /**
  A data array to store details related to an exception.

  @var array<mixed> $arr_data
  */
  protected $arr_data;



  /**
  Constructor for WithDataArrayException.

  @param string $message
  @param array<mixed> $arr_data

  @return void
  */
  public function __construct(string $message, array $arr_data = []) {
    $this->arr_data = $arr_data;
    parent::__construct($message);
  }//end __construct() WithDataArrayException



  /**
  Getter to obtain the data array with details of the exception.

  @return array<mixed>
  */
  public function get_data() : array {
    return $this->arr_data;
  }//end get_data()
```

```
}//end class WithDataArrayException




/**
A class to specialize WithDataArrayException for use with invalid
encoding detection (ASCII and UTF-8 validation) in unicode.libr.php.

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/
class InvalidEncodingException extends WithDataArrayException {
}//end class InvalidEncodingException

?>
```

3.76 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
  #   --add-test-files
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
```

```
  fi
  readonly LFBFL_verbose
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_fixed_point_build
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_add_test_files=0
  if [[ "$*" == *--add-test-files* ]]; then
    LFBFL_add_test_files=1
  fi
  readonly LFBFL_add_test_files
  declare -i LFBFL_remove_test_files=0
  if [[ "$*" == *--remove-test-files* ]]; then
    LFBFL_remove_test_files=1
  fi
  readonly LFBFL_remove_test_files
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )

  declare -i LFBFL_i
  local LFBFL_file_path="miscellaneous/12345678901234567890123456789012345678901234567890"
  declare -i LFBFL_digit
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    mkdir --parents -- --a/
    {
      printf "<?php\nfunction my_function(:;){\n\n}\n"
      # shellcheck disable=SC2016
      printf '$my_variable = "aaaa
```

```
      printf "def my_function3(my_argument : int) -> None:\n"
      printf "    \"\"\"Some comment.\"\"\"\n"
      printf "\n"
      printf "    print(my_argument)\n"
      printf "\n"
      printf "def my_function4(my_argument : int):\n"
      printf "    \"\"\"Some comment.\"\"\"\n"
      printf "    def my_sub_function():\n"
      printf "        print(my_argument)\n"
      printf "    return my_sub_function\n"
      printf "\n"
      printf "def my_function5(my_argument : int):\n"
      printf "    \"\"\"Some comment.\"\"\"\n"
      printf "\n"
      printf "    def my_sub_function():\n"
      printf "        print(my_argument)\n"
      printf "    return my_sub_function\n"
      printf "\n"
    } > --a.py
    cat -- --a.py
    cp -- --a.py --a/
    {
      printf "def my_function(:;):\n\n\n"
      printf 'my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
      printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaa";\n'
    } > --a2.py
    cat -- --a2.py
    cp -- --a2.py --a/
    {
      # Error 0 is missing shell header.
      printf "my_function(){\n"
      printf "   # Error 1: Indentation is not correct.\n"
      printf "  sed -e 's/a/b/g' \\\\\n"
      printf "      -e 's/a/b/g' \"\$1\" # Indentation is correct.\n"
      printf "  my_variable=\"%%s\\n\"\n"
      # shellcheck disable=SC2016
      printf '  printf "${my_variable}"  # Error 2: shellcheck warning\n'
      local LFBFL_ttp="ttp"
      local LFBFL_URL="h${LFBFL_ttp}://example.com"
      printf '  printf "%s" # Error 3: http -> https\n' "${LFBFL_URL}"
      # shellcheck disable=SC2016
      printf '  printf "${#some_'
      printf 'arr_1}" # Error 4: no_size_of_array_first_element\n'
      # shellcheck disable=SC2016
      printf '  printf "${#'
      printf 'arr_1}"  # Error 5: no_size_of_array_first_element\n'
      printf '  if !'
      # shellcheck disable=SC2016
      printf ' [[ $1 -eq 1 ]]; # Error 6: no_negation_before_bash_test\n'
      printf '  then  # Error 7: misplaced then\n'
      printf '    my_variable="a"\n'
      printf '  fi\n'
      printf "\n"
      # shellcheck disable=SC2016
      printf '  if [[ $1 -eq 1 ]];\n'
      printf '  then  # Error 8: another case of misplaced then\n'
```

```
      printf '    my_variable="a"\n'
      printf '  fi\n'
      printf '  my_variable="aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
      printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaa"  # Error 9 line too long.\n'
      printf '}\n'
    } > --a.sh
    cp -- --a.sh --a/

    # We add a file with space in it for tests.
    touch "a b.sh"

    # The following files cannot be commited,
    # because the backslash in them breaks CodeFactor.io.
    for ((LFBFL_i = 1; LFBFL_i < 90; ++LFBFL_i)); do
      LFBFL_digit=$((LFBFL_i % 10))
      LFBFL_file_path+="${LFBFL_digit}"
      touch "${LFBFL_file_path}"
      touch "${LFBFL_file_path}\\"
    done
  fi
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    cat -- --a.py
    cat -- --a2.py
  fi
  if [[ LFBFL_remove_test_files -eq 1 ]]; then
    rm --force --recursive -- --a/
    rm --force -- --a.php --a.py --a2.py --a.sh
    rm "a b.sh"
    for ((LFBFL_i = 1; LFBFL_i < 90; ++LFBFL_i)); do
      LFBFL_digit=$((LFBFL_i % 10))
      LFBFL_file_path+="${LFBFL_digit}"
      rm --force "${LFBFL_file_path}"
      rm --force "${LFBFL_file_path}\\"
    done
  fi
}

upgrade_build_and_checks "$@"
```

3.77 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
```

```
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded file)
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_i_error=0

  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} --output-document="$1"\
      -- "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum -- "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    LFBFL_i_error=1
  fi
  if [[ -n "${LFBFL_check_download}" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"}\
      --output-document="$1.wget_sha512.temp"\
      -- "$2"
    declare -r LFBFL_present_sha512_2=$(
      sha512sum -- "$1.wget_sha512.temp"\
      | cut --fields=1 --delimiter=' '
```

```
    )
    if [[ "${LFBFL_check_download}" != "keep" ]]; then
      rm -- "$1.wget_sha512.temp"
    fi
    if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
      printf\
        "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !